Unit II

Chapter 2

HUMAN SETTLEMENTS



Human Settlement means cluster of dwellings of any type or size where human beings live. For this purpose, people may erect houses and other structures and command some area or territory as their economic support-base. Thus, the process of settlement inherently involves grouping of people and apportioning of territory as their resource base.

Settlements vary in size and type. They range from a hamlet to metropolitan cities. With size, the economic character and social structure of settlements changes and so do its ecology and technology. Settlements could be small and sparsely spaced; they may also be large and closely spaced. The sparsely located small settlements are called villages, specialising in agriculture or other primary activities. On the other hand, there are fewer but larger settlements which are termed as urban settlements specialising in secondary and tertiary activities. The basic differences between rural and urban settlements are as follows :

- The rural settlements derive their life support or basic economic needs from land based primary economic activities, whereas, urban settlements, depend on processing of raw materials and manufacturing of finished goods on the one hand and a variety of services on the other.
- Cities act as nodes of economic growth, provide goods and services not only to urban dwellers but also to the people of the rural settlements in their hinterlands in return for food and raw materials. This functional relationship between the urban and rural settlements takes place through transport and communication network.
- Rural and urban settlements differ in terms of social relationship, attitude and outlook. Rural people are less mobile and therefore, social relations among them are intimate. In urban areas, on the other hand, way of life is complex and fast, and social relations are formal.

Types of Rural Settlement

Types of the settlement are determined by the extent of the built-up area and inter-house

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distance. In India compact or clustered village of a few hundred houses is a rather universal feature, particularly in the northern plains. However, there are several areas, which have other forms of rural settlements. There are various factors and conditions responsible for having different types of rural settlements in India. These include: (i) physical features – nature of terrain, altitude, climate and availability of water (ii) cultural and ethenic factors – social structure, caste and religion (iii) security factors – defence against thefts and robberies. Rural settlements in India can broadly be put into four types:

- Clustered, agglomerated or nucleated,
- Semi-clustered or fragmented,
- Hamleted, and
- Dispersed or isolated.

Clustered Settlements

The clustered rural settlement is a compact or closely built up area of houses. In this type of village the general living area is distinct and separated from the surrounding farms, barns and pastures. The closely built-up area and its



Fig. 2.1 : Clustered Settlements in the North-eastern states

intervening streets present some recognisable pattern or geometric shape, such as rectangular, radial, linear, etc. Such settlements are generally found in fertile alluvial plains and in the northeastern states. Sometimes, people live in compact village for security or defence reasons, such as in the Bundelkhand region of central India and in Nagaland. In Rajasthan, scarcity of water has necessitated compact settlement for maximum utilisation of available water resources.

Semi-Clustered Settlements

Semi-clustered or fragmented settlements may result from tendency of clustering in a restricted area of dispersed settlement. More often such a pattern may also result from segregation or fragmentation of a large compact village. In this case, one or more sections of the village society choose or is forced to live a little away from the main cluster or village. In such cases, generally, the land-owning and dominant community occupies the central part of the main village, whereas people of lower strata of society and menial workers settle on the outer flanks of the village. Such settlements are widespread in the Gujarat plain and some parts of Rajasthan.



Fig. 2.2 : Semi-clustered settlements

Hamleted Settlements

Sometimes settlement is fragmented into several units physically separated from each other bearing a common name. These units are locally called *panna, para, palli, nagla, dhani*, etc. in various parts of the country. This segmentation of a large village is often motivated by social and ethnic factors. Such villages are more frequently found in the middle and lower Ganga plain, Chhattisgarh and lower valleys of the Himalayas.

Dispersed Settlements

Dispersed or isolated settlement pattern in India appears in the form of isolated huts or hamlets of few huts in remote jungles, or on small hills



Fig. 2.3 : Dispersed settlements in Nagaland

with farms or pasture on the slopes. Extreme dispersion of settlement is often caused by extremely fragmented nature of the terrain and land resource base of habitable areas. Many areas of Meghalaya, Uttarakhand, Himachal Pradesh and Kerala have this type of settlement.

Urban Settlements

Unlike rural settlements, urban settlements are generally compact and larger in size. They are engaged in a variety of non-agricultural, economic and administrative functions. As mentioned earlier, cities are functionally linked to rural areas around them. Thus, exchange of goods and services is performed sometimes directly and sometimes through a series of market towns and cities. Thus, cities are connected directly as well as indirectly with the villages and also with each other.

> The census of India, 1991 defines urban settlements as "All places which have municipality, corporation, cantonment board or notified town area committee and have a minimum population of 5000 persons, at least 75 per cent of male workers are engaged in non-agriculture pursuits and a density of population of at least 400 persons per square kilometer are urban.

Evolution of Towns in India

Towns flourished since prehistoric times in India. Even at the time of Indus valley civilisation, towns like Harappa and Mohanjodaro were in existence. The following period has witnessed evolution of towns. It continued with periodic ups and downs until the arrival of Europeans in India in the eighteenth century. On the basis of their evolution in different periods, Indian towns may be classified as:

• Ancient towns, • Medieval towns, and • Modern towns.

Ancient Towns

There are number of towns in India having historical background spanning over 2000 years. Most of them developed as religious and cultural centres. Varanasi is one of the important towns among these. Prayag (Prayagraj), Pataliputra (Patna), Madurai are some other examples of ancient towns in the country.

Medieval Towns

About 100 of the existing towns have their roots in the medieval period. Most of them developed as headquarters of principalities and kingdoms. These are fort towns which came up on the ruins of ancient towns. Important among them are Delhi, Hyderabad, Jaipur, Lucknow, Agra and Nagpur.

Modern Towns

The British and other Europeans have developed a number of towns in India. Starting their foothold on coastal locations, they first developed some trading ports such as Surat, Daman, Goa, Pondicherry, etc. The British later consolidated their hold around three principal nodes – Mumbai (Bombay), Chennai (Madras), and Kolkata (Calcutta) – and built them in the British style. Rapidly



Fig. 2.4 : A view of the modern city

extending their domination either directly or through control over the princely states, they established their administrative centres, hill-towns as summer resorts, and added new civil,

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Table 2.1 : India – Trends of Urbanisation 1901-2011

Year	Number of Towns/UAs	Urban Population (in Thousands)	% of Total Population	Decennial Growth (%)
1901	1,827	25,851.9	10.84	—
1911	1,815	25,941.6	10.29	0.35
1921	1,949	28,086.2	11.18	8.27
1931	2,072	33,456.0	11.99	19.12
1941	2,250	44,153.3	13.86	31.97
1951	2,843	62,443.7	17.29	41.42
1961	2,365	78,936.6	17.97	26.41
1971	2,590	1,09,114	19.91	38.23
1981	3,378	1,59,463	23.34	46.14
1991	4,689	2,17,611	25.71	36.47
2001	5,161	2,85,355	27.78	31.13
2011*	6,171	3,77,000	31.16	31.08

**Source: Census of India, 2011 http://www.censusindia.gov.in (Provisional)*

administrative and military areas to them. Towns based on modern industries also evolved after 1850. Jamshedpur can be cited as an example.

After independence, a large number of towns have been developed as administrative headquarters, e.g., Chandigarh, Bhubaneswar, Gandhinagar, Dispur, etc., and industrial centres, such as Durgapur, Bhilai, Sindri, Barauni. Some old towns also developed as satellite towns around metropolitan cities, such as Ghaziabad, Rohtak, Gurugram around Delhi. With increasing investment in rural areas, a large number of medium and small towns have developed all over the country.

Urbanisation in India

The level of urbanisation is measured in terms of percentage of urban population to total population. The level of urbanisation in India in 2011 was 31.16 per cent, which is quite low in comparison to developed countries. Total urban population has increased eleven-fold during the twentieth century. Enlargement of urban centres and emergence of new towns have played a significant role in the growth of urban population and urbanisation in the country. (Table 2.1). But the growth rate of urbanisation has slowed down during last two decades.

Functional Classification of Towns

Apart from their role as central or nodal places, many towns and cities perform specialised services. Some towns and cities specialise in certain functions and they are known for some specific activities, products or services. However, each town performs a number of functions. On the basis of dominant or specialised functions, Indian cities and towns can be broadly classified as follows:

Administrative towns and cities

Towns supporting administrative headquarters of higher order are administrative towns, such as Chandigarh, New Delhi, Bhopal, Shillong, Guwahati, Imphal, Srinagar, Gandhinagar, Jaipur, Chennai, etc.

Industrial towns

Industries constitute prime motive force of these cities, such as Mumbai, Salem, Coimbatore, Modinagar, Jamshedpur, Hugli, Bhilai, etc.

Transport Cities

They may be ports primarily engaged in export and import activities such as Kandla, Cochin, Kozhikode, Vishakhapatnam, etc., or hubs of inland transport, such as Agra, Dhulia, Pandit Deendayal Upadhyay Nagar, Itarsi, Katni, etc.

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Commercial towns

Towns and cities specialising in trade and commerce are kept in this class. Kolkata, Saharanpur, Satna, etc., are some examples.

Mining towns

These towns have developed in mineral rich areas such as Raniganj, Jharia, Digboi, Ankaleshwar, Singrauli, etc.

Garrisson Cantonment towns

These towns emerged as garrisson towns such as Ambala, Jalandhar, Mhow, Babina, Udhampur, etc.

Smart Cities Mission

The objective of the *Smart Cities Mission* is to promote cities that provide core infrastructure, a clean and sustainable environment and give a decent quality of life to its citizens. One of the features of Smart Cities is to apply smart solutions to infrastructure and services in order to make them better. For example, making areas less vulnerable to disasters, using fewer resources and providing cheaper services. The focus is on sustainble and inclusive development and the idea is to look at compact areas, create a replicable model, which will act like a lighthouse to other aspiring cities.

List the urban agglomerations/cities state-wise and see the state-wise population under this category of cities.

Educational towns

Starting as centres of education, some of the towns have grown into major campus towns, such as Roorki, Varanasi, Aligarh, Pilani, Prayagraj, etc.

Religious and cultural towns

Varanasi, Mathura, Amritsar, Madurai, Puri, Ajmer, Pushkar, Tirupati, Kurukshetra, Haridwar, Ujjain came to prominence due to their religious/cultural significance.

Tourist towns

Nainital, Mussoorie, Shimla, Pachmarhi, Jodhpur, Jaisalmer, Udagamandalam (Ooty), Mount Abu are some of the tourist destinations.

The cities are not static in their function. The functions change due to their dynamic nature.

Even specialised cities, as they grow into metropolises become multifunctional wherein industry, business, administration, transport, etc., become important. The functions get so intertwined that the city can not be categorised in a particular functional class.



EXERCISES

1. Choose the right answers of the following from the given options.

(i) Which one of the following towns is NOT located on a river bank?

(a) Agra
(b) Bhopal
(c) Patna
(d) Kolkata

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(ii) Which one of the following is NOT the part of the definition of a town as per the census of India?

(a) Population density of 400 persons per sq km.

(b) Presence of municipality, corporation, etc.

(c) More than 75% of the population engaged in primary sector.

(d) Population size of more than 5,000 persons.

(iii) In which one of the following environments does one expect the presence of dispersed rural settlements?

(a) Alluvial plains of Ganga

(b) Arid and semi-arid regions of Rajasthan

(c) Lower valleys of Himalayas

(d) Forests and hills in north-east

2. Answer the following questions in about 30 words.

(i) What are garrisson towns? What is their function?

(ii) What are the main factors for the location of villages in desert regions?

3. Answer the following questions in about 150 words.

(i) Discuss the features of different types of rural settlements. What are the factors responsible for the settlement patterns in different physical environments?

(ii) Can one imagine the presence of only one-function town? Why do the cities become multi-functional?

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Table 3.1 : Composition of Total Cultivable Land

Agricultural Land-use Categories	*As a percentage of Reporting Area*		*As a percentage of total cultivable land*	
	1950-51	2019-20	1950-51	2019-20
Culturable Waste land	8.0	3.9	13.4	6.8
Fallow other than Current Fallow	6.1	3.7	10.2	6.4
Current Fallow	3.7	4.5	6.2	7.8
Net Area Sown	41.7	45.6	70.0	79.0
Total Cultivable Land	59.5	57.7	100.00	100.00

It is clear from the above discussion that the scope for bringing in additional land under net sown area in India is limited. There is, thus, an urgent need to evolve and adopt land-saving technologies. Such technologies can be classified under two heads – those which raise the yield of any particular crop per unit area of land and those which increase the total output per unit area of land from all crops grown over one agricultural year by increasing land-use intensity. The advantage of the latter kind of technology is that along with increasing output from limited land, it also increases the demand for labour significantly. For a land scarce but labour abundant country like India, a high cropping intensity is desirable not only for fuller utilisation of land resource, but also for reducing unemployment in the rural economy.

The ***cropping intensity (CI)*** is calculated as follows :

$$\text{Cropping Intensity in percentage} = \frac{\text{GCA}}{\text{NSA}} \times 100$$

Cropping Seasons in India

There are three distinct crop seasons in the northern and interior parts of country, namely ***kharif***, ***rabi*** and ***zaid***. The *kharif* season largely coincides with Southwest Monsoon under which the cultivation of tropical crops, such as rice, cotton, jute, jowar, bajra and tur is possible. The ***rabi*** season begins with the onset of winter in October-November and ends in March-April. The low temperature conditions during this season facilitate the cultivation of temperate and subtropical crops such as wheat, gram and mustard. ***Zaid*** is a short duration summer cropping season beginning after harvesting of rabi crops. The cultivation of watermelons, cucumbers, vegetables and fodder crops during this season is done on irrigated lands. However, this type of distinction in the cropping season does not exist in southern parts of the country. Here, the temperature is high enough to grow tropical crops during any period in the year provided the soil moisture is available. Therefore, in this region same crops can be grown thrice in an agricultural year provided there is sufficient soil moisture.

Types of Farming

On the basis of main source of moisture for crops, the farming can be classified as ***irrigated*** and ***rainfed (barani)***. There is difference in the nature of irrigated farming, as well as based on the objective of irrigation, i.e., protective or productive. The objective of protective irrigation is to protect the crops from adverse effects of

Table 3.2 : Cropping Seasons in India

Cropping Season	*Major Crops Cultivated*	
	Northern States	*Southern States*
Kharif June-September	Rice, Cotton, Bajra, Maize, Jowar, Tur	Rice, Maize, Ragi, Jowar, Groundnut
Rabi October – March	Wheat, Gram, Rapeseeds and Mustard, Barley	Rice, Maize, Ragi, Groundnut, Jowar
Zaid April–June	Vegetables, Fruits, Fodder	Rice, Vegetables, Fodder

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moisture deficiency which often means that irrigation acts as a supplementary source of water over and above the rainfall. The strategy of this kind of irrigation is to provide soil moisture to maximum possible area. Productive irrigation is meant to provide sufficient soil moisture in the cropping season to achieve high productivity. In such irrigation the water input per unit area of cultivated land is higher than protective irrigation. *Rainfed farming* is further classified on the basis of adequacy of soil moisture during cropping season into *dryland* and *wetland* farming. In India, the ***dryland farming*** is largely confined to the regions having annual rainfall less than 75 cm. These regions grow hardy and drought resistant crops such as *ragi*, *bajra*, *moong*, gram and *guar* (fodder crops) and practise various measures of soil moisture conservation and rain water harvesting. In ***wetland farming***, the rainfall is in excess of soil moisture requirement of plants during rainy season. Such regions may face flood and soil erosion hazards. These areas grow various water intensive crops such as rice, jute and sugarcane and practise aquaculture in the fresh water bodies.

Foodgrains

The importance of foodgrains in Indian agricultural economy may be gauged from the fact these crops occupy about two-third of total cropped area in the country. Foodgrains are dominant crops in all parts of the country whether they have subsistence or commercial agricultural economy. On the basis of the structure of grain the foodgrains are classified as cereals and pulses.

Cereals

The cereals occupy about 54 per cent of total cropped area in India. The country produces about 11 per cent cereals of the world and ranks third in production after China and U.S.A. India produces a variety of cereals, which are classified as fine grains (rice, wheat) and coarse grains (jowar, bajra, maize, ragi), etc. Account of important cereals has been given in the following paragraphs :

Rice

Rice is a staple food for the overwhelming majority of population in India. Though, it is considered to be a crop of tropical humid areas, it has many varieties which are grown in different agro-climatic regions. These are successfully grown from sea level to about 2,000 m altitude and from humid areas in eastern India to dry but irrigated areas of Punjab, Haryana, western U.P. and northern Rajasthan. In southern states and West Bengal the climatic conditions allow the cultivation of two or three crops of rice in an agricultural year. In West Bengal farmers grow three crops of rice called '*aus*', '*aman*' and '*boro*'. But in Himalayas and northwestern parts of the country, it is grown as a *kharif* crop during southwest Monsoon season.

India contributes 22.07 per cent of rice production in the world and ranked second after China in 2018. About one-fourth of the total cropped area in the country is under rice cultivation. West Bengal, Uttar Pradesh, and Punjab are the leading rice producing states in the country. The yield level of rice is high in Punjab, Tamil Nadu, Haryana, Andhra Pradesh, Telangana, West Bengal and Kerala. In the first four of these states almost the entire land under rice cultivation is irrigated. Punjab and Haryana are not traditional rice growing areas. Rice



Fig. 3.2 : Rice transplantation in southern parts of India

cultivation in the irrigated areas of Punjab and Haryana was introduced in 1970s following the Green Revolution. Genetically improved varieties of seed, relatively high usage of fertilizers and pesticides and lower levels of susceptibility of the crop to pests due to dry climatic conditions

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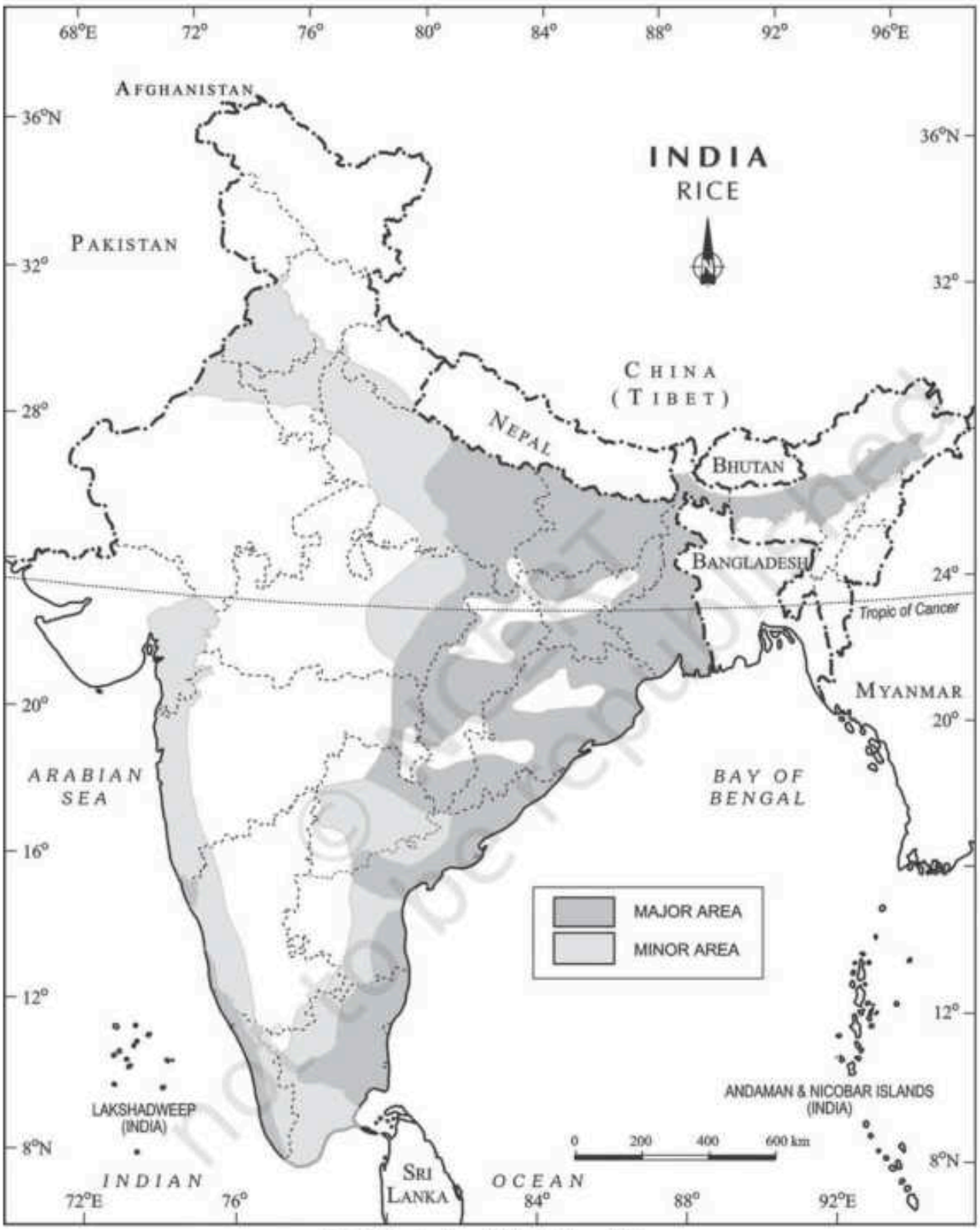


Fig. 3.3 : India – Distribution of Rice

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are responsible for higher yield of rice in this region. The yield of this crop is very low in rainfed areas of Madhya Pradesh, Chhattisgarh and Odisha.

Wheat

Wheat is the second most important cereal crop in India after rice. India produces about 12.8 per cent of total wheat production of world (2017). It is primarily a crop of temperate zone. Hence, its cultivation in India is done during winter i.e. *rabi* season. About 85 per cent of total area under this crop is concentrated in north and central regions of the country i.e. Indo-Gangetic Plain, Malwa Plateau and Himalayas up to 2,700 m altitude. Being a *rabi* crop, it is mostly grown under irrigated conditions. But it is a rainfed crop in Himalayan highlands and parts of Malwa plateau in Madhya Pradesh.

About 14 per cent of the total cropped area in the country is under wheat cultivation. Uttar Pradesh, Madhya Pradesh, Punjab, Haryana and Rajasthan are leading wheat producing states. The yield level of wheat is very high (above 4,000 k.g. per ha) in Punjab and Haryana whereas, Uttar Pradesh, Rajasthan and Bihar have moderate yields. The states like Madhya Pradesh, Himachal Pradesh and Jammu and Kashmir growing wheat under rainfed conditions have low yield.

Jowar

The coarse cereals together occupy about 16.50 per cent of total cropped area in the country. Among these, *jowar* or *sorghum* alone accounts for about 5.3 per cent of total cropped area. It is main food crop in semi-arid areas of central and southern India. Maharashtra alone produces more than half of the total *jowar* production of the country. Other leading producer states of *jowar* are Karnataka, Madhya Pradesh, Andhra Pradesh and Telangana. It is sown in both *kharif* and *rabi* seasons in southern states. But it is a *kharif* crop in northern India where it is mostly grown as a fodder crop. South of Vindhyachal it is a rainfed crop and its yield level is very low in this region.

Bajra

Bajra is sown in hot and dry climatic conditions in northwestern and western parts of the country. It is a hardy crop which resists frequent dry spells and drought in this region. It is cultivated alone as well as part of mixed cropping. This coarse cereal occupies about 5.2 per cent of total cropped area in the country. Leading producers of *bajra* are the states of Rajasthan, Maharashtra, Gujarat, Uttar Pradesh and Haryana. Yield of this crop has increased during recent years in Rajasthan, Haryana and Gujarat due to introduction of drought resistant varieties and expansion of irrigation under it.

Maize

Maize is a food as well as fodder crop grown under semi-arid climatic conditions and over inferior soils. This crop occupies only about 3.6 per cent of total cropped area. Maize cultivation is not concentrated in any specific region. It is sown all over India exceptPunjab and eastern and north-eastern regions. The leading producers of maize are the states of Karnataka, Madhya Pradesh, Bihar, Andhra Pradesh, Telangana, Rajasthan and Uttar Pradesh. Yield level of maize is higher than other coarse cereals. It is high in southern states and declines towards central parts.

Pulses

Pulses are a very important ingredient of vegetarian food as these are rich sources of proteins. These are legume crops which increase the natural fertility of soils through nitrogen fixation. India is a leading producer of pulses in the world. The cultivation of pulses in the country is largely concentrated in the drylands of Deccan and central plateaus and northwestern parts of the country. Pulses occupy about 11 per cent of the total cropped area in the country. Being the rainfed crops of drylands, the yields of pulses are low and fluctuate from year to year. Gram and *tur* are the main pulses cultivated in India.

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New seed varieties of wheat (Mexico) and rice (Philippines) known as high yielding varieties (HYVs) were available for cultivation by mid-1960s. India took advantage of this and introduced package technology comprising HYVs, along with chemical fertilisers in irrigated areas of Punjab, Haryana, Western Uttar Pradesh, Andhra Pradesh and Gujarat. Assured supply of soil moisture through irrigation was a basic pre-requisite for the success of this new agricultural technology. This strategy of agricultural development paid dividends instantly and increased the foodgrains production at very fast rate. This spurt of agricultural growth came to be known as '**Green Revolution**'. This also gave fillip to the development of a large number of agro-inputs, agro-processing industries and small-scale industries. This strategy of agricultural development made the country self-reliant in foodgrain production. But Green Revolution was initially confined to irrigated areas only. This led to regional disparities in agricultural development in the country till 1970s, after which the technology spread to the Eastern and Central parts of the country.

The Planning Commission of India focused its attention on the problems of agriculture in rainfed areas in 1980s. It initiated agro-climatic planning in 1988 to induce regionally balanced agricultural development in the country. It also emphasised on the need for diversification of agriculture and harnessing of resources for the development of dairy farming, poultry, horticulture, livestock rearing and aquaculture.

Initiation of the policy of liberalisation and free market economy in 1990s influenced the course of development of Indian agriculture.

National Mission for Sustainable Agriculture (NMSA)

National Mission for Sustainable Agriculture is to make agriculture more productive, sustainable, remunerative and climate resilient by promoting location specific integrated/composite farming systems and to conserve natural resources through appropriate soil and moisture conservation measures. The Government has been promoting organic farming in the country through the scheme such as Paramparagat Krishi Vikas Yojana (PKVY) and Rashtriya Vikas Yojana (RKVY).

Growth of Agricultural Output and Technology

There has been a significant increase in agricultural output and improvement in technology during the last 50 years.

- Production and yield of many crops such as rice and wheat has increased at an impressive rate. Among the other crops, the production of sugarcane, oilseeds and cotton has also increased appreciably.
- Expansion of irrigation has played a crucial role in enhancing agricultural output in the country. It provided basis for introduction of modern agricultural technology, such as high yielding varieties of seeds, chemical fertilisers, pesticides and farm machinery. The net irrigated area in the country has also increased.
- Modern agricultural technology has diffused very fast in various areas of the country. Consumption of chemical fertilizers has increased by 15 times since mid-sixties. Since the high yielding varieties are highly susceptible to pests and diseases, the use of pesticides has increased significantly since 1960s.

Farmer's Portal of India

The Farmer's Portal is a platform for farmers to seek any information related to agriculture. Detailed information on farmers' insurance, agriculture storage, crops, extension activities, seeds, pesticides, farm machineries, etc. is provided. Details of fertilizers, market prices, package and practices, programmes, welfare schemes are also given. Block level details related to soil fertility, storage, insurance, training, etc. are available in an interactive map. Users can also download farm friendly handbook, scheme guidelines, etc.

(Source: https://www.india.gov.in/farmers-portal-india-department-agriculture-and-cooperation)

Problems of Indian Agriculture

The nature of problems faced by Indian agriculture varies according to agro-ecological and historical experiences of its different regions. Hence, most of the agricultural problems in the country are region specific. Yet, there are some problems which are common and range from physical constraints to institutional hindrances. A detailed discussion on these problems follows:

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Fig. 3.12 : Roto Till Drill—A modern agricultural equipment

Dependence on Erratic Monsoon

Irrigation covers only about 33 per cent of the cultivated area in India. The crop production in rest of the cultivated land directly depends on rain. Poor performance of south-west monsoon also adversely affects the supply of canal water for irrigation. On the other hand, the rainfall in Rajasthan and other drought-prone areas is too meagre and highly unreliable. Even the areas receiving high annual rainfall experience considerable fluctuations. This makes them vulnerable to both droughts and floods. Drought is a common phenomenon in the low rainfall areas, which may also experience occasional floods. The flash floods in drylands of Maharashtra, Gujarat and Rajasthan in 2006 and 2017 are examples of this phenomenon. Droughts and floods continue to be the twin menace in Indian agriculture.

Low productivity

The yield of the crops in the country is low in comparison to the international level. Per hectare output of most of the crops such as rice, wheat, cotton and oilseeds in India is much lower than that of the U.S.A., Russia and Japan. Because of the very high pressure on the land resources, the labour productivity in Indian agriculture is also very low in comparison to international level. The vast rainfed areas of the country, particularly drylands which mostly grow coarse cereals, pulses and oilseeds have low yields.

Why is agricultural productivity low in dry regions?

Constraints of Financial Resources and Indebtedness

The inputs of modern agriculture are very expensive. This resource intensive approach has become unmanageable for marginal and small farmers as they have very meagre or no saving to invest in agriculture. To tide over these difficulties, most of such farmers have resorted to availing credit from various institutions and moneylenders. Crop failures and low returns from agriculture have forced them to fall in the trap of indebtedness.

What are the implications of severe indebtedness? Do you feel that the recent incidents of farmers' suicides in different states of the country are the result of indebtedness?

Lack of Land Reforms

Indian peasantry had been exploited for a long time as there had been unequal distribution of land. Among the three revenue systems operational during British period, i.e., Mahalwari, Ryotwari and Zamindari, the last one was most exploitative for the peasants. After Independence, land reforms were accorded priority, but these reforms were not implemented effectively due to lack of strong political will. Most of the state governments avoided taking politically tough decisions which went against strong political lobbies of landlords. Lack of implementation of land reforms has resulted in continuation of inequitous distribution of cultivable land which is detrimental to agricultural development.

Small Farm Size and Fragmentation of Landholdings

There are a large number of marginal and small farmers in the country. The average size of land holding is shrinking under increasing

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NITI, agri ministry, FAO launch investment forum for advancing climate resilient agrifood system

PTI - Last Updated: Jan 24, 2024, 06:07:00 PM IST

Synopsis

Delivering the keynote address at the inauguration, NITI Aayog Member Ramesh Chand emphasised on the need for awareness on how agriculture contributes to climate change, citing a contribution of a little more than 13 per cent of the total greenhouse gas emissions in the country.

India needs to usher in Green Revolution 2.0 to promote less water-intensive crops: GTRI

India needs to usher in Green Revolution 2.0 with a view to promote farming of less water-intensive crops such as pulses and oil seeds and discourage free power for the agri sector, economic think tank GTRI said on Thursday. There is a need to "promote less water-intensive crops like pulses, oil seeds, and vegetables that can significantly reduce water demand and the government can guarantee MSP (minimum support price) on these crops," the Global Trade Research Initiative (GTRI) report said.

In western UP, sugarcane is life

What changes have you noticed in agricultural sector and cropping pattern? Discuss in the class.

population pressure. Furthermore, in India, the land holdings are mostly fragmented. There are some states where consolidation of holding has not been carried out even once. Even the states where it has been carried out once, second consolidation is required as land holdings have fragmented again in the process of division of land among the owners of next generations. The small size fragmented landholdings are uneconomic.

Lack of Commercialisation

A large number of farmers produce crops for self-consumption. These farmers do not have enough land resources to produce more than their requirement. Most of the small and marginal farmers grow foodgrains, which are meant for their own family consumption. Modernisation and commercialisation of agriculture have, however, taken place in the irrigated areas.

Vast Underemployment

There is a massive underemployment in the agricultural sector in India, particularly in the unirrigated tracts. In these areas, there is a seasonal unemployment ranging from 4 to 8 months. Even in the cropping season, work is not available throughout as agricultural operations are not labour intensive. Hence, the people engaged in agriculture do not have the opportunity to work round the year.

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Degradation of Cultivable Land

One of the serious problems that arises out of faulty strategy of irrigation and agricultural development is degradation of land resources. This is serious because it may lead to depletion of soil fertility. The situation is particularly alarming in irrigated areas. A large tract of agricultural land has lost its fertility due to alkalisation and salinisation of soils and waterlogging. Excessive use of chemicals such as insecticides and pesticides has led to their concentration in toxic amounts in the soil profile. Leguminous crops have been displaced from the cropping pattern in the irrigated areas and duration of fallow has substantially reduced owing to multiple cropping. This has obliterated the process of natural fertilization such as nitrogen fixation. Rainfed areas in humid and semi-arid tropics also experience degradation of several types like soil erosion by water and wind erosion which are often induced by human activities.

Activity

Prepare a list of agricultural problems in your own region. How similar or different are these problems compared to the problems mentioned in this chapter?



EXERCISES

1. Choose the right answers of the following from the given options.

(i) Which one of the following is NOT a land-use category?

(a) Fallow land (c) Net Area Sown

(b) Marginal land (d) Culturable Wasteland

(ii) What one of the following is the main reason due to which share of forest has shown an increase in the last forty years?

(a) Extensive and efficient efforts of afforestation

(b) Increase in community forest land

(c) Increase in notified area allocated for forest growth

(d) Better peoples' participation in managing forest area.

(iii) Which one of the following is the main form of degradation in irrigated areas?

(a) Gully erosion (c) Salinisation of soils

(b) Wind erosion (d) Siltation of land

(iv) Which one of the following crops is not cultivated under dryland farming?

(a) Ragi (c) Groundnut

(b) Jowar (d) Sugarcane

(v) In which of the following group of countries of the world, HYVs of wheat and rice were developed?

(a) Japan and Australia (c) Mexico and Philippines

(b) U.S.A. and Japan (d) Mexico and Singapore

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Unit III

Chapter 4

WATER RESOURCES



Do you think that what exists today will continue to be so, or the future is going to be different in some respects? It can be said with some certainty that the societies will witness demographic transition, geographical shift of population, technological advancement, degradation of environment and water scarcity. Water scarcity is possibly to pose the greatest challenge on account of its increased demand coupled with shrinking supplies due to over utilisation and pollution. Water is a cyclic resource with abundant supplies on the globe. Approximately, 71 per cent of the earth's surface is covered with it but freshwater constitutes only about 3 per cent of the total water. In fact, a very small proportion of freshwater is effectively available for human use. The availability of freshwater varies over space and time. The tensions and disputes on sharing and control of this scarce resource are becoming contested issues among communities, regions, and states. The assessment, efficient use and conservation of water, therefore, become necessary to ensure development. In this chapter, we shall discuss water resources in India, its geographical distribution, sectoral utilisation, and methods of its conservation and management.

Water Resources of India

India accounts for about 2.45 per cent of the world's surface area, 4 per cent of the world's water resources and more than 17 per cent of the world's population. The total water available from precipitation in the country in a year is about 4,000 cubic km. The availability from surface water and replenishable groundwater is 1,869 cubic km. Out of this, only 60 per cent can be put to beneficial uses. Thus, the total utilisable water resource in the country is only 1,122 cubic km.

Surface Water Resources

There are four major sources of surface water. These are rivers, lakes, ponds and tanks. In the country, there are about 10,360 rivers and their tributaries longer than 1.6 km each. The mean annual flow in all the river basins

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in India is estimated to be 1,869 cubic km. However, due to topographical, hydrological and other constraints, only about 690 cubic km (32 per cent) of the available surface water can be utilised. Water flow in a river depends on size of its catchment area or river basin and rainfall within its catchment area. You have studied in your Class XI textbook *"India : Physical Environment"* that precipitation in India has very high spatial variation, and it is mainly concentrated in Monsoon season. You also have studied in the textbook that some of the rivers in the country like the Ganga, the Brahmaputra, and the Indus have huge catchment areas. Given that precipitation is relatively high in the catchment areas of the Ganga, the Brahmaputra and the Barak rivers, these rivers, although account for only about one-third of the total area in the country, have 60 per cent of the total surface water resources. Much of the annual water flow in south Indian rivers like the Godavari, the Krishna, and the Kaveri has been harnessed, but it is yet to be done in the Brahmaputra and the Ganga basins.

Groundwater Resources

The total replenishable groundwater resources in the country are about 432 cubic km. The level of groundwater utilisation is relatively high in the river basins lying in north-western region and parts of south India.

The groundwater utilisation is very high in the states of Punjab, Haryana, Rajasthan, and Tamil Nadu. However, there are States like Chhattisgarh, Odisha, Kerala, etc., which utilise only a small proportion of their groundwater potentials. States like Gujarat, Uttar Pradesh, Bihar, Tripura and Maharashtra are utilising their groundwater resources at a moderate rate. If the present trend continues, the demands for water would need the supplies. And such situation, will be detrimental to development, and can cause social upheaval and disruptions.

Lagoons and Backwaters

India has a vast coastline and the coast is very indented in some states. Due to this, a number of lagoons and lakes have formed. The States like Kerala, Odisha and West Bengal have vast surface water resources in these lagoons and lakes. Although, water is generally brackish in these water bodies, it is used for fishing and irrigating certain varieties of paddy crops, coconut, etc.

Water Demand and Utilisation

India has traditionally been an agrarian economy, and about two-third of its population have been dependent on agriculture. Hence, development of irrigation to increase agricultural production has been assigned a very high priority in the Five Year Plans, and multipurpose river valleys projects, like the Bhakra-Nangal, Hirakud, Damodar Valley, Nagarjuna Sagar, Indira Gandhi Canal Project, etc., have been taken up. In fact, India's water demand at present is dominated by irrigational needs.

Agriculture accounts for most of the surface and groundwater utilisation, it accounts for 89 per cent of the surface water and 92 per cent of the groundwater utilisation. While the share of industrial sector is limited to 2 per cent of the surface water utilisation and 5 per cent of the ground-water, the share of domestic sector is higher (9 per cent) in surface water utilisation as compared to groundwater. The share of agricultural sector in total water utilisation is much higher than other sectors. However, in future, with development, the shares of industrial and domestic sectors in the country are likely to increase.

Demand of Water for Irrigation

In agriculture, water is mainly used for irrigation. Irrigation is needed because of spatio-temporal variability in rainfall in the country. The large tracts of the country are deficient in rainfall and are drought prone. North-western India and Deccan plateau constitute such areas. Winter and summer seasons are more or less dry in most part of the country. Hence, it is difficult to practise agriculture without assured

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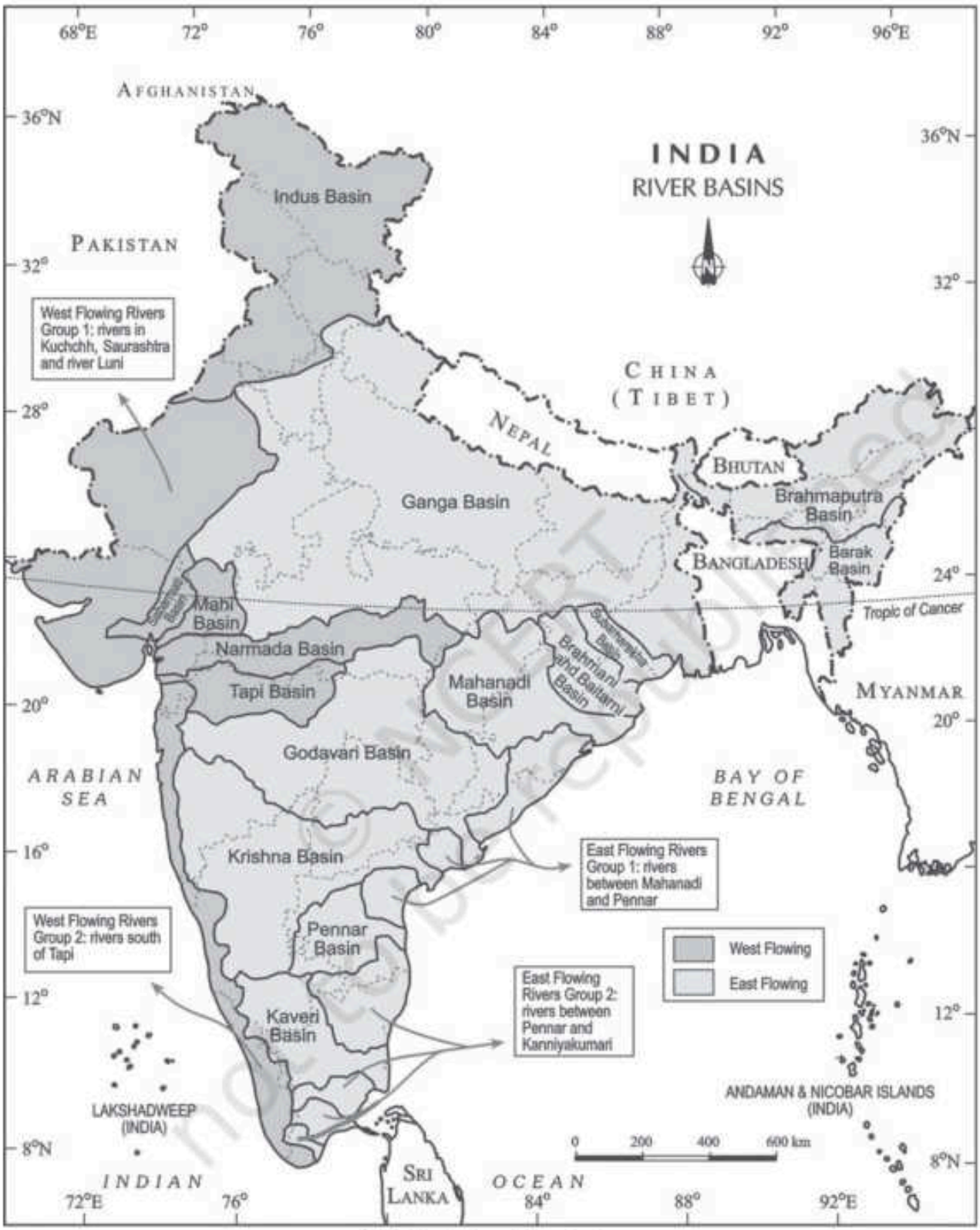


Fig. 4.1 : India – River Basins

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irrigation during dry seasons. Even in the areas of ample rainfall like West Bengal and Bihar, breaks in monsoon or its failure creates dry spells detrimental for agriculture. Water need of certain crops also makes irrigation necessary. For instance, water requirement of rice, sugarcane, jute, etc. is very high which can be met only through irrigation.

Provision of irrigation makes multiple cropping possible. It has also been found that irrigated lands have higher agricultural productivity than unirrigated land. Further, the high yielding varieties of crops need regular moisture supply, which is made possible only by a developed irrigation systems. In fact, this is why that green revolution strategy of agriculture development in the country has largely been successful in Punjab, Haryana and western Uttar Pradesh.

In Punjab, Haryana and western Uttar Pradesh, more than 85 per cent of their net sown area is under irrigation. Wheat and rice are grown mainly with the help of irrigation in these states. Of the total net irrigated area 76.1 per cent in Punjab and 51.3 per cent in Haryana are irrigated through wells and tubewells. This shows that these states utilise large proportion of their groundwater potential which has resulted in groundwater depletion in these states.

The over-use of groundwater resources has led to decline in groundwater table in these states. In fact, over withdrawals in some states, like Rajasthan and Maharashtra, has increased fluoride concentration in groundwater, and this practice has led to increase in concentration of arsenic in parts of West Bengal and Bihar.

Activity

Intensive irrigation in Punjab, Haryana and western Uttar Pradesh is increasing salinity in the soil and depletion of groundwater irrigation. Discuss its likely impacts on agriculture.

Do You Know ?

Pradhan Mantri Krishi Sinchayee Yojana (PMKSY)

Pradhan Mantri Krishi Sinchayee Yojana has been launched by the Central Government during 2015-16 with an overarching vision to ensure access to some means of protective irrigation for all agricultural farms in the country, thus bringing much desired rural prosperity. Some of the broad objectives of the this programmes are to:

- Enhance the physical access of water on the farm and expand cultivable area under assured irrigation (*Har khet ko pani*)
- Promote integration of water source, distribution and its efficient use, to make best use of water through appropriate technologies and practices.
- Improve on-farm water use efficiency to reduce wastage and increase availability both in duration and exent; irrigation and other water saving technologies (*Per drope more crop*)
- Introduce sustainable water conservation practices
- Ensure the integrated development of rain-fed areas using the waters held approach towards soil and water conservation, regeration of ground water, providing livelihood options, etc.

Emerging Water Problems

The per capita availability of water is dwindling day-by-day due to increase in population. The available water resources are also getting polluted with industrial, agricultural and domestic effluents, and this, in turn, is further limiting the availability of usable water resources.

Deterioration of Water Quality

Water quality refers to purity of water, or water without unwanted foreign substances. Water

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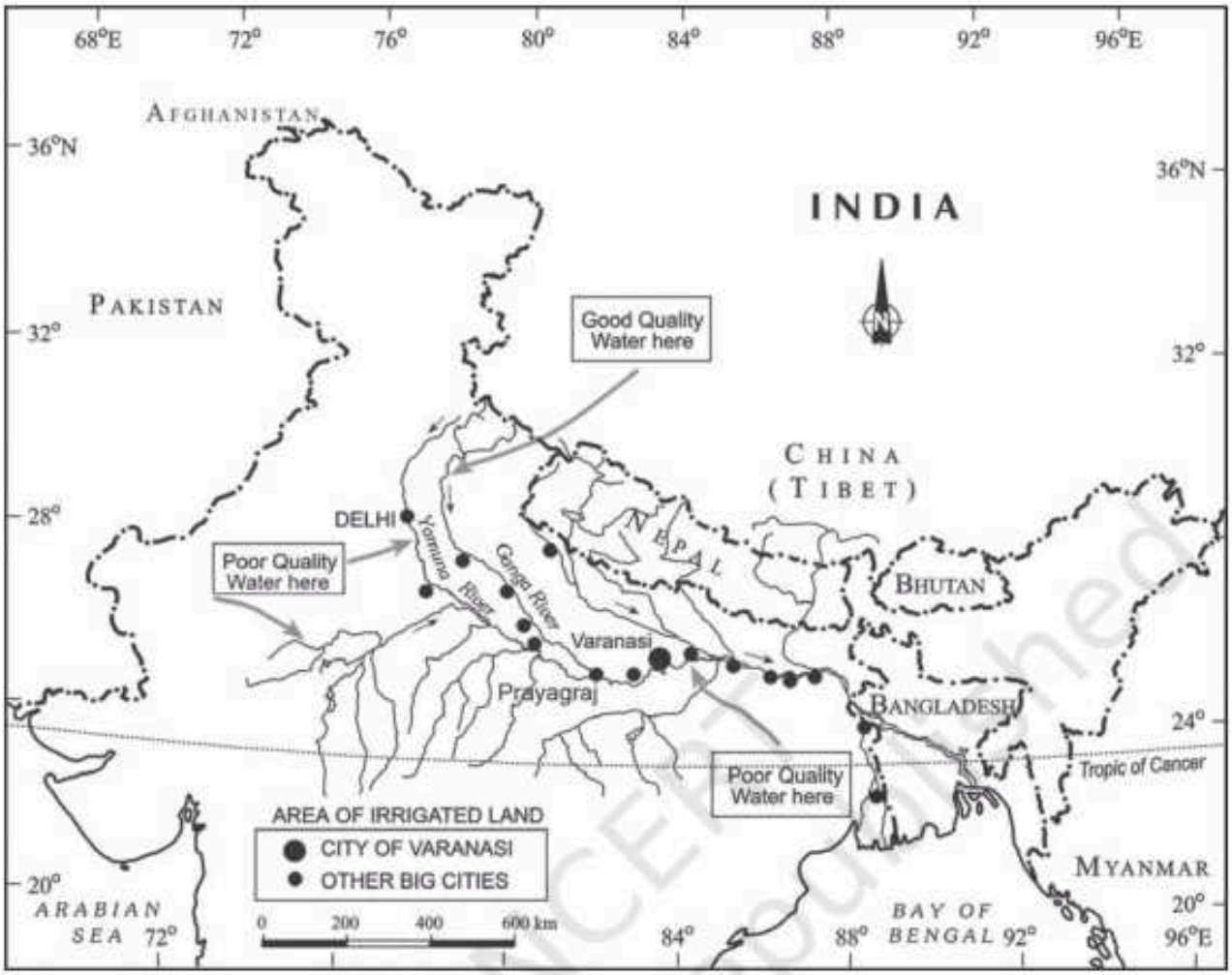


Fig. 4.2 : The Ganga and its Tributaries and Towns Located on them

gets polluted by foreign matters, such as micro-organisms, chemicals, industrial and other wastes. Such matters deteriorate the quality of water and render it unfit for human use. When toxic substances enter lakes, streams, rivers, ocean and other water bodies, they get dissolved or lie suspended in water. This results in pollution of water, whereby quality of water deteriorates affecting aquatic systems. Sometimes, these pollutants also seep down and pollute groundwater.

Activity

Find out which are the major towns/cities located on the bank of the Ganga and its tributaries and major industries they have.

Water Conservation and Management

Since there is a declining availability of freshwater and increasing demand, the need has arisen to conserve and effectively manage this precious life giving resource for sustainable development. Given that water availability from sea/ocean, due to high cost of desalinisation, is considered negligible, India has to take quick steps and make effective policies and laws, and adopt effective measures for its conservation. Besides developing water-saving technologies and methods, attempts are also to be made to prevent the pollution. There is need to encourage watershed development, rainwater harvesting, water recycling and reuse, and

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conjunctive use of water for sustaining water supply in long run.

Prevention of Water Pollution

Available water resources are degrading rapidly. The major rivers of the country generally retain better water quality in less densely populated upper stretches in hilly areas. In plains, river water is used intensively for irrigation, drinking, domestic and industrial purposes. The drains carrying agricultural (fertilizers and insecticides), domestic (solid and liquid wastes), and industrial effluents join the rivers. The concentration of pollutants in rivers, especially remains very high during the summer season when flow of water is low.

The Central Pollution Control Board (CPCB) in collaboration with State Pollution Control Boards has been monitoring water quality of national aquatic resources at 507 stations. The data obtained from these stations show that organic and bacterial contamination continues to be the main source of pollution in rivers. The Yamuna river is the most polluted river in the country between Delhi and Etawah. Other severely polluted rivers are: the Sabarmati at Ahmedabad, the Gomti at Lucknow, the Kali, the Adyar, the Cooum (entire stretches), the Vaigai at Madurai and the Musi of Hyderabad and the Ganga at Kanpur and Varanasi. Groundwater pollution has occurred due to high concentrations of heavy/toxic metals, fluoride and nitrates at different parts of the country.

The legislative provisions such as the Water (Prevention and Control of Pollution) Act 1974, and Environment Protection Act 1986 have not been implemented effectively. The result is that in 1997, 251 polluting industries were located along the rivers and lakes. The Water



Rivers of conflict...but also of peace

There have been more than 500 conflicts over water in the past century, but it's also an agent of cooperation

Rich countries poor in supply of water: WWF

Climate change? Barmer grapples with floods

AND THEY SAY IT'S A DESERT

SURVIVAL INSTINCT

A-prone Barmer?

Discuss the issues highlighted in the news items.

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Cess Act, 1977, meant to reduce pollution has also made marginal impacts. There is a strong need to generate public awareness about importance of water and impacts of water pollution. The public awareness and action can be very effective in reducing the pollutants from agricultural activities, domestic and industrial discharges.

Recycle and Reuse of Water

Another way through which we can improve fresh water availability is by recycle and reuse. Use of water of lesser quality such as reclaimed wastewater would be an attractive option for industries for cooling and fire fighting to reduce their water cost. Similarly, in urban areas water after bathing and washing utensils can be used for gardening. Water used for washing vehicle can also be used for gardening. This would conserve better quality of water for drinking purposes. Currently, recycling of water is practised on a limited scale. However, there is enormous scope for replenishing water through recycling.

Activity

Observe the quantity of water used at your home in various activities and enlist the ways in which the water can be reused and recycled in various activities.

Class teachers should organise a discussion on recycle and reuse of water.

Watershed Management

Watershed management basically refers to efficient management and conservation of surface and groundwater resources. It involves prevention of runoff and storage and recharge of groundwater through various methods like percolation tanks, recharge wells, etc. However, in broad sense watershed management includes conservation, regeneration and judicious use of all resources – natural (like land, water, plants and animals) and human with in a watershed. Watershed management aims at bringing about balance between natural resources on the one hand and society on the other. The success of watershed development largely depends upon community participation.

The Central and State Governments have initiated many watershed development and management programmes in the country. Some of these are being implemented by non-governmental organisations also. ***Haryali*** is a watershed development project sponsored by the Central Government which aims at enabling the rural population to conserve water for drinking, irrigation, fisheries and afforestation. The Project is being executed by Gram Panchayats with people's participation.

Do You Know ?

Atal Bhujal Yojana (Atal Jal) is being implemented in 8220 water stressed Gram Panchayats of 229 administrative blocks/ talukas in 80 districts of seven states,viz. Gujarat, Haryana, Karnataka, Madhya Pradesh, Maharashtra, Rajasthan, and Uttar Pradesh. The selected States account for about 37 per cent of the total number of water-stressed (over-exploited, critical and semi-critical) blocks in India. One of the key aspects of ATAL JAL is to bring in behavioural changes in the community, from the prevailing attitude of consumption to conservation and smart water management. (Source: *Annual Report, Ministry of Jal Shakti, Govt. of India, 2022-23*)

Neeru-Meeru (Water and You) programme (in Andhra Pradesh) and ***Arvary Pani Sansad*** (in Alwar, Rajasthan) have taken up constructions of various water-harvesting structures such as percolation tanks, dug out ponds (***Johad***), check dams, etc., through people's participation. Tamil Nadu has made water harvesting structures in the houses compulsory. No building can be constructed without making structures for water harvesting.

Watershed development projects in some areas have been successful in

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Watershed Development in Ralegan Siddhi, Ahmadnagar, Maharashtra: A Case Study

Ralegan Siddhi is a small village in the district of Ahmadnagar, Maharashtra. It has become an example for watershed development throughout the country.

In 1975, this village was caught in a web of poverty and illicit liquor trade. The transformation took place when a retired army personnel, settled down in the village and took up the task of watershed development. He convinced villagers about the importance of family planning and voluntary labour; preventing open grazing, felling trees, and liquor prohibition.

Voluntary labour was necessary to ensure minimum dependence on the government for financial aids. "It socialised the costs of the projects." explained the activist. Even those who were working outside the village contributed to the development by committing a month's salary every year.

Work began with the percolation tank constructed in the village. In 1975, the tank could not hold water. The embankment wall leaked. People voluntarily repaired the embankment. The seven wells below it swelled with water in summer for the first time in the living memory of the people. The people reposed their faith in him and his visions.

A youth group called Tarun Mandal was formed. The group worked to ban the dowry system, caste discrimination and untouchability. Liquor distilling units were removed and prohibition imposed. Open grazing was completely banned with a new emphasis on stall-feeding. The cultivation of water-intensive crops like sugarcane was banned. Crops such as pulses, oilseeds and certain cash crops with low water requirements were encouraged.

All elections to local bodies began to be held on the basis of consensus. "It made the community leaders complete representatives of the people." A system of Nyay Panchayats (informal courts) were also set up. Since then, no case has been referred to the police.



Ralegan Siddhi before mitigation approach

A Rs.22 lakh school building was constructed using only the resources of the village. No donations were taken. Money, if needed, was borrowed and paid back. The villagers took pride in this self-reliance. A new system of sharing labour grew out of this infusion of pride and voluntary spirit. People volunteered to help each other in agricultural operation. Landless labourers also gained employment. Today the village plans to buy land for them in adjoining villages.



Ralegan Siddhi after mitigation approach

At present, water is adequate; agriculture is flourishing, though the use of fertilisers and pesticides is very high. The prosperity also brings the question of ability of the present generation to carry on the work after the leader of the movement who declared that, "The process of Ralegan's evolution to an ideal village will not stop. With changing times, people tend to evolve new ways. In future, Ralegan might present a different model to the country."

What a mitigation approach can do? A success story.

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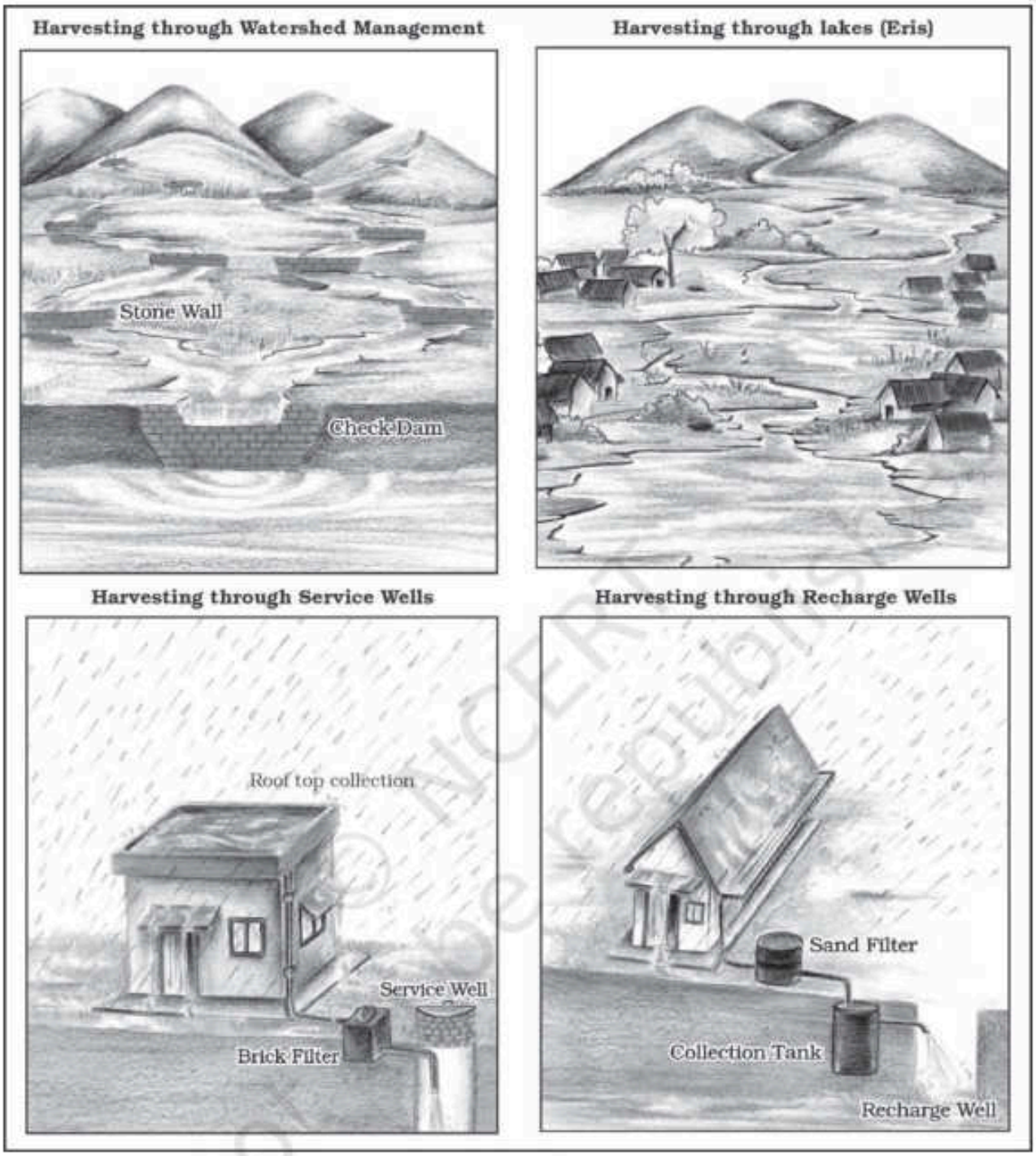


Fig. 4.3 : Various Methods of Rainwater Harvesting

rejuvenating environment and economy. However, there are only a few success stories. In majority of cases, the programme is still in its nascent stage. There is a need to generate awareness regarding benefits of watershed development and management among people in the country, and through this integrated water resource management approach water availability can be ensured on sustainable basis.

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Rainwater Harvesting

Rainwater harvesting is a method to capture and store rainwater for various uses. It is also used to recharge groundwater aquifers. It is a low cost and eco-friendly technique for preserving every drop of water by guiding the rain water to borewell, pits and wells. Rainwater harvesting increases water availability, checks the declining groundwater table, improves the quality of groundwater through dilution of contaminants, like fluoride and nitrates, prevents soil erosion, and flooding and arrests salt water intrusion in coastal areas if used to recharge aquifers.

Rainwater harvesting has been practised through various methods by different communities in the country for a long time. Traditional rainwater harvesting in rural areas is done by using surface storage bodies, like lakes, ponds, irrigation tanks, etc. In Rajasthan, rainwater harvesting structures locally known as ***Kund*** or ***Tanka*** (a covered underground tank) are constructed near or in the house or village to store harvested rainwater (see Fig. 4.3 to understand various ways of rainwater harvesting).

There is a wide scope to use rainwater harvesting technique to conserve precious water resource. It can be done by harvesting rainwater on rooftops and open spaces. Harvesting rainwater also decreases the community dependence on groundwater for domestic use. Besides bridging the demand-supply gap, it can also save energy to pump groundwater as recharge leads to rise in groundwater table. These days rainwater harvesting is being taken up on massive scale in many states in the country. Urban areas can specially benefit from rainwater harvesting as water demand has already outstripped supply in most of the cities and towns.

Apart from the above mentioned factors, the issue desalinisation of water particularly in coastal areas and brackish water in arid and semi-arid areas, transfer of water from water surplus areas to water deficit areas through inter-linking of rivers can be important remedies for solving water problem in India (read more about inter linking of rivers). However, the most important issue from the point of view of individual users, household and communities is pricing of water.

Highlights of India's National Water Policy 2012

The objective of the National Water Policy, 2012 is to assess the existing situation and to propose a framework for a plan of action with a unified national perspective. In order to achieve the objective of the Policy, a number of recommendations have been made therein for conservation, development and improved management of water resources of the country.

Some of the salient features of national water policy 2012 are:

- Emphasis on the need for a national water framework law, comprehensive legislation for optimum development of inter-State rivers and river valleys.
- Water, after meeting the pre-emptive needs for safe drinking water and sanitation, achieving food security, supporting poor people dependent on agriculture for their livelihood and high priority allocation for minimum eco-system needs, be treated as economic good so as to promote its conservation and efficient use.
- Adaptation strategies in view of climate change for designing and management of water resources structures and review of acceptability criteria has been emphasized.
- A system to evolve benchmarks for water uses for different purposes, i.e., water footprints, and water auditing be developed to ensure efficient use of water.
- Removal of large disparity in stipulations for water supply in urban areas and in rural areas has been recommended.
- Water resources projects and services should be managed with community participation.

Source: Press Information Bureau, Govt. of India, Ministry of Water Resources.

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Unit III

Chapter 5

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MINERAL AND ENERGY RESOURCES



India is endowed with a rich variety of mineral resources due to its varied geological structure. Bulk of the valuable minerals are products of pre-palaezoic age (Refer: Chapter 2 of Class XI, Textbook: *"Fundamentals of Physical Geography"* and are mainly associated with metamorphic and igneous rocks of the peninsular India. The vast alluvial plain tract of north India is devoid of minerals of economic use. The mineral resources provide the country with the necessary base for industrial development. In this chapter, we shall discuss the availability of various types of mineral and energy resources in the country.

A mineral is a natural substance of organic or inorganic origin with definite chemical and physical properties.

Types of Mineral Resources

On the basis of chemical and physical properties, minerals may be grouped under two main categories of metallics and non-metallics which may further be classified as follows :

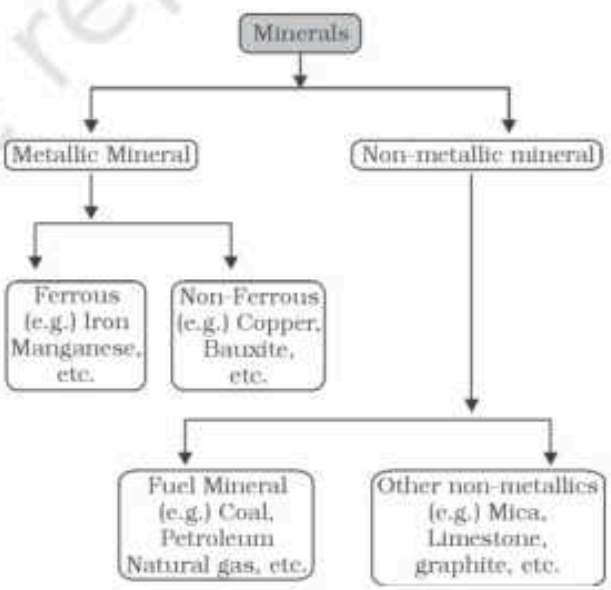


Fig. 5.1 : Classification of Minerals

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As, it is clear from the Fig. 5.1 metallic minerals are the sources of metals. Iron ore, copper, gold produce metal and are included in this category. Metallic minerals are further divided into ferrous and non-ferrous metallic minerals. Ferrous, as you know, refers to iron. All those minerals which have iron content are ferrous such as iron ore itself and those which do not have iron content are non-ferrous such as copper, bauxite, etc.

Non-metallic minerals are either organic in origin such as fossil fuels also known as mineral fuels which are derived from the buried animal and plant life such as coal and petroleum. Other type of non-metallic minerals are inorganic in origin such as mica, limestone and graphite, etc.

Minerals have certain characteristics. These are unevenly distributed over space. There is inverse relationship in quality and quantity of minerals i.e. good quality minerals are less in quantity as compared to low quality minerals. The third main characteristic is that all minerals are exhaustible over time. These take long to develop geologically and they cannot be replenished immediately at the time of need. Thus, they have to be conserved and not misused as they do not have the second crop.

Distribution of Minerals in India

Most of the metallic minerals in India occur in the peninsular plateau region in the old crystalline rocks. Over 97 per cent of coal reserves occur in the valleys of Damodar, Sone, Mahanadi and Godavari. Petroleum reserves are located in the sedimentary basins of Assam, Gujarat and Mumbai High i.e. off-shore region in the Arabian Sea. New reserves have been located in the Krishna-Godavari and Kaveri basins. Most of the major mineral resources occur to the east of a line linking Mangaluru and Kanpur.

Minerals are generally concentrated in three broad belts in India. There may be some sporadic occurrences here and there in isolated pockets. These belts are :

The North-Eastern Plateau Region

This belt covers Chhotanagpur (Jharkhand), Odisha Plateau, West Bengal and parts of Chhattisgarh. Have you ever thought about the reason of major iron and steel industry being located in this region? It has variety of minerals viz. iron ore coal, manganese, bauxite, mica.

Find out the specific region where these minerals are being extracted.

The South-Western Plateau Region

This belt extends over Karnataka, Goa and contiguous Tamil Nadu uplands and Kerala. This belt is rich in ferrous metals and bauxite. It also contains high grade iron ore, manganese and limestone. This belt lacks in coal deposits except Neyveli lignite.

This belt does not have as diversified mineral deposits as the north-eastern belt. Kerala has deposits of monazite and thorium, bauxite clay. Goa has iron ore deposits.

The North-Western Region

This belt extends along Aravali in Rajasthan and part of Gujarat and minerals are associated with Dharwar system of rocks. Copper, zinc have been major minerals. Rajasthan is rich in building stones i.e. sandstone, granite, marble. Gypsum and Fuller's earth deposits are also extensive. Dolomite and limestone provide raw materials for cement industry. Gujarat is known for its petroleum deposits. You may be knowing that Gujarat and Rajasthan both have rich sources of salt.

Why and where Dandi March was organised by Mahatma Gandhi?

The Himalayan belt is another mineral belt where copper, lead, zinc, cobalt and tungsten are known to occur. They occur on both the eastern and western parts. Assam valley has

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mineral oil deposits. Besides oil resources are also found in off-shore-areas near Mumbai Coast (Mumbai High).

In the following pages you will find the spatial pattern of some of the important minerals.

Ferrous Mineral

Ferrous minerals such as iron ore, manganese, chromite, etc., provide a strong base for the development of metallurgical industries. Our country is well-placed in respect of ferrous minerals both in reserves and production.

Iron Ore

India is endowed with fairly abundant resources of iron ore. It has the largest reserve of iron ore in Asia. The two main types of ore found in our country are *haematite* and *magnetite*. It has great demand in international market due to its superior quality. The iron ore mines occur in close proximity to the coal fields in the north-eastern plateau region of the country which adds to their advantage.

About 95 per cent of total reserves of iron ore is located in the States of Odisha, Jharkhand, Chhattisgarh, Karnataka, Goa, Telangana, Andhra Pradesh and Tamil Nadu. In Odisha, iron ore occurs in a series of hill ranges in Sundergarh, Mayurbhanj and Jhar. The important mines are Gurumahisani, Sulaipet, Badampahar (Mayurbhaj), Kiruburu (Kendujhar) and Bonai (Sundergarh). Similar hill ranges, Jharkhand has some of the oldest iron ore mines and most of the iron and steel plants are located around them. Most of the important mines such as Noamundi and Gua are located in Poorbi and Pashchimi Singhbhum districts. This belt further extends to Durg, Dantewara and Bailadila. Dalli, and Rajhara in Durg are the important mines of iron ore in the country. In Karnataka, iron ore deposits occur in Sandur-Hospet area of Ballari district, Baba Budan hills and Kudremukh in Chikkamagaluru district and


Iron ore mining gets a boost

The iron ore mining industry in India is attracting several new players, both large and small

Can you find out its reason?

parts of Shivamogga, Chitradurg and Tumakuru districts. The districts of Chandrapur, Bhandara and Ratnagiri in Maharashtra, Karimnagar and Warangal district of Telangana, Kurnool, Cuddapah and Anantapur districts of Andhra Pradesh, Salem and Nilgiris districts of Tamil Nadu are other iron mining regions. Goa has also emerged as an important producer of iron ore.

Manganese

Manganese is an important raw material for smelting of iron ore and also used for manufacturing ferro alloys. Manganese deposits are found in almost all geological formations, however, it is mainly associated with Dharwar system.

Madhya Pradesh and Odisha are the leading producers of Manganese. Major mines in Odisha are located in the central part of the iron ore belt of India, particularly in Bonai, Kendujhar, Sundergarh, Gangpur, Koraput, Kalahandi and Bolangir.

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Fig. 5.2 : India – Metallic Minerals (Ferrous)

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The manganese belt of Madhya Pradesh extends in a belt in Balaghat-Chhindwara-Nimar-Mandla and Jhabua districts. Karnataka is another major producer and here the mines are located in Dharwar, Ballari, Belagavi, North Canara, Chikkmagaluru, Shivamogga, Chitradurg and Tumakuru. Maharashtra is also an important producer of manganese, which is mined in Nagpur, Bhandara and Ratnagiri districts. The disadvantage to these mines is that they are located far from steel plants.

Telangana, Goa, and Jharkhand are other minor producers of manganese.

Non-Ferrous Minerals

India is poorly endowed with non-ferrous metallic minerals except bauxite.

Bauxite

Bauxite is the ore, which is used in manufacturing of aluminium. Bauxite is found mainly in tertiary deposits and is associated with laterite rocks occurring extensively either on the plateau or hill ranges of peninsular India and also in the coastal tracts of the country.

Odisha happens to be the largest producer of Bauxite. Kalahandi and Sambalpur are the leading producers. The other two areas which have been increasing their production are Bolangir and Koraput. The patlands of Lohardaga in Jharkhand have rich deposits. Gujarat, Chhattisgarh, Madhya Pradesh and Maharashtra are other major producers. Bhavanagar, and Jamnagar in Gujarat have the major deposits. Chhattisgarh has bauxite deposits in Amarkantak plateau while Katni-Jabalpur area and Balaghat in M.P. have important deposits of bauxite. Kolaba, Thane, Ratnagiri, Satara, Pune and Kolhapur in Maharashtra are important producers. Tamil Nadu, Karnataka and Goa are minor producers of bauxite.

Copper

Copper is an indispensable metal in the electrical industry for making wires, electric motors, transformers and generators. It is alloyable, malleable and ductile. It is also mixed with gold to provide strength to jewellery.

The Copper deposits mainly occur in Singhbhum district in Jharkhand, Balaghat district in Madhya Pradesh and Jhunjhunu and Alwar districts in Rajasthan.

Minor producers of Copper are Agnigundala in Guntur District (Andhra Pradesh), Chitradurg and Hasan districts (Karnataka) and South Arcot district (Tamil Nadu).

Non-metallic Minerals

Among the non-metallic minerals produced in India, mica is the important one. The other minerals extracted for local consumption are limestone, dolomite and phosphate.

Mica

Mica is mainly used in the electrical and electronic industries. It can be split into very thin sheets which are tough and flexible. Mica in India is produced in Jharkhand, Andhra Pradesh, Telanganga and Rajasthan followed by Tamil Nadu, West Bengal and Madhya Pradesh. In Jharkhand, high quality mica is obtained in a belt extending over a distance of about 150 km, in length and about 22 km, in width in lower Hazaribagh plateau. In Andhra Pradesh, Nellore district produces the best quality mica. In Rajasthan, mica belt extends for about 320 kms from Jaipur to Bhilwara and around Udaipur. Mica deposits also occur in Mysuru and Hasan districts of Karanataka, Coimbatore, Tiruchirapalli, Madurai and Kanniyakumari in Tamil Nadu, Alleppey in Kerala, Ratnagiri in Maharashtra, Purulia and Bankura in West Bengal.

Energy Resources

Mineral fuels are essential for generation of power, required by agriculture, industry, transport and other sectors of the economy. Mineral fuels like coal, petroleum and natural gas (known as fossil fuels), nuclear energy minerals, are the conventional sources of energy. These conventional sources are exhaustible resources.



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Fig. 5.3 : India – Minerals (Non-Ferrous)

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Coal

Coal is a one of the important minerals which is mainly used in the generation of thermal power and smelting of iron ore. Coal occurs in rock sequences mainly of two geological ages, namely Gondwana and tertiary deposits.

About 80 per cent of the coal deposits in India is of bituminous type and is of non-coking grade. The most important Gondwana coal fields of India are located in Damodar Valley. They lie in Jharkhand-Bengal coal belt and the important coal fields in this region are Raniganj, Jharia, Bokaro, Giridih, Karanpura.

Jharia is the largest coal field followed by Raniganj. The other river valleys associated with coal are Godavari, Mahanadi and Sone. The most important coal mining centres are Singrauli in Madhya Pradesh (part of Singrauli coal field lies in Uttar Pradesh), Korba in Chhattisgarh, Talcher and Rampur in Odisha, Chanda–Wardha, Kamptee and Bander in Maharashtra and Singareni in Telangana and Pandur in Andhra Pradesh.

Tertiary coals occur in Assam, Arunachal Pradesh, Meghalaya and Nagaland. It is extracted from Darangiri, Cherrapunji, Mewlong and Langrin (Meghalaya); Makum, Jaipur and Nazira in upper Assam, Namchik – Namphuk (Arunachal Pradesh) and Kalakot (Jammu and Kashmir).

Besides, the brown coal or lignite occur in the coastal areas of Tamil Nadu, Puducherry, Gujarat and Jammu and Kashmir.

Petroleum

Crude petroleum consists of hydrocarbons of liquid and gaseous states varying in chemical composition, colour and specific gravity. It is an essential source of energy for all internal combustion engines in automobiles, railways and aircraft. Its numerous by-products are processed in petrochemical industries, such as fertiliser, synthetic rubber, synthetic fibre, medicines, vaseline, lubricants, wax, soap and cosmetics.

DO YOU KNOW ?

Petroleum is referred to as liquid gold because of its scarcity and diversified uses.

Crude petroleum occurs in sedimentary rocks of the tertiary period. Oil exploration and production was systematically taken up after the Oil and Natural Gas Commission was set up in 1956. Till then, Digboi in Assam was the only oil producing region but the scenario changed after 1956. In recent years, new oil deposits have been found at the extreme western and eastern parts of the country. In Assam, Digboi, Naharkatiya and Moran are important oil producing areas. The major oilfields of Gujarat are Ankaleshwar, Kalol, Mehsana, Nawagam, Kosamba and Lunej. Mumbai High which lies 160 km off Mumbai was discovered in 1973 and production commenced in 1976. Oil and natural gas have been found in exploratory wells in Krishna-Godavari and Kaveri basin on the east coast.

Oil extracted from the wells is crude oil and contains many impurities. It cannot be used directly. It needs to be refined. There are two types of refineries in India: (a) field-based and (b) market-based. Digboi is an example of field-based and Barauni is an example of market-based refinery.

Natural Gas

Natural Gas is found with petroleum deposits and is released when crude oil is brought to

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Fig. 5.4 : India – Conventional Energy Resources

Activity: Collect information about cross country natural gas pipelines laid by GAIL (India) under 'One Nation One Grid'.

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the surface. It can be used as a domestic and industrial fuel. It is used as fuel in power sector to generate electricity, for heating purpose in industries, as raw material in chemical, petrochemical and fertiliser industries. With the expansion of gas infrastructure and local city gas distribution (COD) networks, natural gas is also emerging as a preferred transport fuel (CNG) and cooking fuel (PNG) at homes. India's major gas reserves are found in the Mumbai High and allied fields along the west coast which are supplemented by finds in the Cambay basin. Along the East Coast, new reserves of natural gas have been discovered in the Krishna-Godavari basin.

Non-Conventional Energy Sources

Fossil fuel sources, such as coal, petroleum, natural gas and nuclear energy use exhaustible raw materials. Sustainable energy resources are only the renewable energy sources like solar, wind, hydro-geothermal and biomass. These energy sources are more equitably distributed and environment-friendly. The non-conventional energy sources will provide more sustained, eco-friendly cheaper energy after the initial cost is taken care of.

Nuclear Energy Resources

Nuclear energy has emerged as a viable source in recent times. Important minerals used for the generation of nuclear energy are uranium and thorium. Uranium deposits occur in the Dharwar rocks. Geographically, uranium ores are known to occur in several locations along the Singbhum Copper belt. It is also found in Udaipur, Alwar and Jhunjhunu districts of Rajasthan, Durg district of Chhattisgarh, Bhandara district of Maharashtra and Kullu district of Himachal Pradesh. Thorium is mainly obtained from monazite and ilmenite in the beach sands along the coast of Kerala and Tamil Nadu. World's richest monazite deposits occur in Palakkad and Kollam districts of Kerala, near Vishakhapatnam in Andhra Pradesh and Mahanadi river delta in Odisha.

Atomic Energy Commission was established in 1948, progress could be made only after the establishment of the Atomic Energy Institute at Trombay in 1954 which was renamed as the Bhabha Atomic Research Centre in 1967. The important nuclear power projects are Tarapur (Maharashtra), Rawatbhata near Kota (Rajasthan), Kalpakkam (Tamil Nadu), Narora (Uttar Pradesh), Kaiga (Karnataka) and Kakarapara (Gujarat).

Solar Energy

Sun rays tapped in photovoltaic cells can be converted into energy, known as solar energy. The two effective processes considered to be very effective to tap solar energy are photovoltaics and solar thermal technology. Solar thermal technology has some relative advantages over all other non-renewable energy sources. It is cost competitive, environment friendly and easy to construct. Solar energy is 7 per cent more effective than coal or oil based plants and 10 per cent more effective than nuclear plants. It is generally used more in appliances like heaters, crop dryers, cookers, etc. The western part of India has greater potential for the development of solar energy in Gujarat and Rajasthan.

Wind Energy

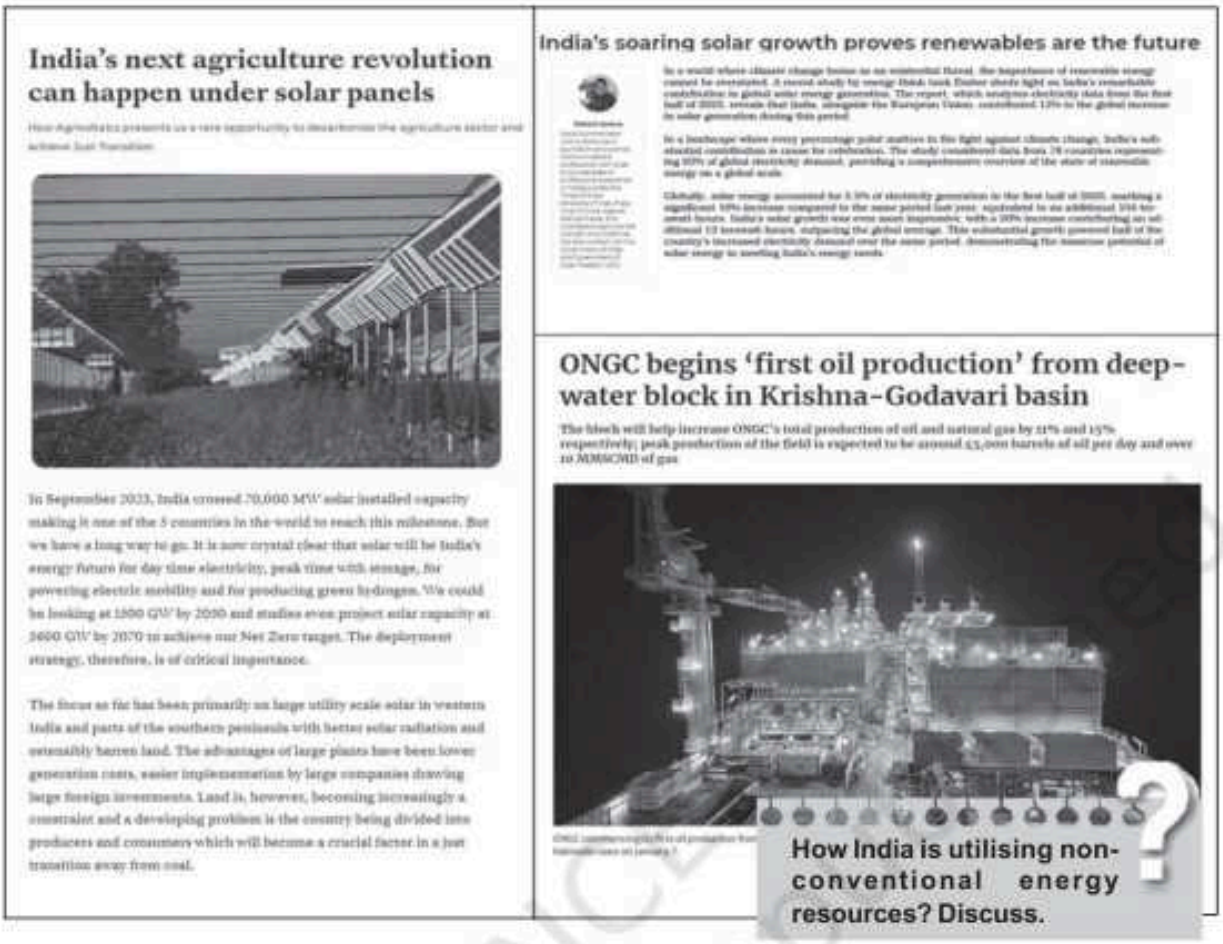
Wind energy is absolutely pollution free, inexhaustible source of energy. The mechanism of energy conversion from blowing wind is simple. The kinetic energy of wind, through turbines is converted into electrical energy. The

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Fig. 5.5 : India – Oil Refineries

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India's next agriculture revolution can happen under solar panels

How Agrivoltaics presents us a rare opportunity to decarbonise the agriculture sector and achieve Just Transition

In September 2023, India crossed 70,000 MW solar installed capacity making it one of the 5 countries in the world to reach this milestone. But we have a long way to go. It is now crystal clear that solar will be India's energy future for day time electricity, peak time with storage, for powering electric mobility and for producing green hydrogen. We could be looking at 1000 GW by 2030 and studies even project solar capacity at 3600 GW by 2070 to achieve our Net Zero target. The deployment strategy, therefore, is of critical importance.

The focus so far has been primarily on large utility scale solar in western India and parts of the southern peninsula with better solar radiation and ostensibly barren land. The advantages of large plants have been lower generation costs, easier implementation by large companies drawing large foreign investments. Land is, however, becoming increasingly a constraint and a developing problem in the country being divided into producers and consumers which will become a crucial factor in a just transition away from coal.

India's soaring solar growth proves renewables are the future

In a world where climate change looms as an existential threat, the importance of renewable energy cannot be overstated. A recent study by energy think tank Ember sheds light on India's remarkable contribution to global solar energy generation. The report, which analyses electricity data from the first half of 2023, reveals that India, alongside the European Union, contributed 13% to the global increase in solar generation during this period.

In a landscape where every percentage point matters in the fight against climate change, India's substantial contribution is cause for celebration. The study considered data from 78 countries representing 93% of global electricity demand, providing a comprehensive overview of the state of renewable energy on a global scale.

Globally, solar energy accounted for 5.9% of electricity generation in the first half of 2023, marking a significant 19% increase compared to the same period last year, equivalent to an additional 334 terawatt-hours. India's solar growth was even more impressive, with a 30% increase contributing an additional 13 terawatt-hours, outpacing the global average. This substantial growth powered half of the country's increased electricity demand over the same period, demonstrating the immense potential of solar energy in meeting India's energy needs.

ONGC begins 'first oil production' from deep-water block in Krishna-Godavari basin

The block will help increase ONGC's total production of oil and natural gas by 11% and 15% respectively; peak production of the field is expected to be around 45,000 barrels of oil per day and over 10 MMSCMD of gas

How India is utilising non-conventional energy resources? Discuss.

permanent wind systems such the trade winds, westerlies and seasonal wind like monsoon have been used as source of energy. Besides these, local winds, land and sea breezes can also be used to produce electricity.

India, already has started generating wind energy. In Rajasthan, Gujarat, Maharashtra and Karnataka, favourable conditions for wind energy exist.

Tidal and Wave Energy

Ocean currents are the store-house of infinite energy. Since the beginning of seventeenth and eighteenth century, persistent efforts were made to create a more efficient energy system from the ceaseless tidal waves and ocean current.

Large tidal waves are known to occur along the west coast of India. Hence, India has great potential for the development of tidal energy along the coasts but so far these have not yet been utilised.

Geothermal Energy

When the magma from the interior of earth, comes out on the surface, tremendous heat is released. This heat energy can successfully be tapped and converted to electrical energy. Apart from this, the hot water that gushes out through the geyser wells is also used in the generation of thermal energy. It is popularly known as Geothermal energy. This energy is now considered to be one of the key energy sources which can be developed as an alternate source. The hot springs and geysers are being used since medieval period. In India, a geothermal energy plant has been commissioned at Manikaran in Himachal Pradesh.

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The first successful (1890) attempt to tap the underground heat was made in the city of Boise, Idaho (U.S.A.), where a hot water pipe network was built to give heat to the surrounding buildings. This plant is still working.

Bio-energy

Bio-energy refers to energy derived from biological products which includes agricultural residues, municipal, industrial and other wastes. Bio-energy is a potential source of energy conversion. It can be converted into electrical energy, heat energy or gas for cooking. It will also process the waste and garbage and produce energy. This will improve economic life of rural areas in developing countries, reduce environmental pollution, enhance self-reliance and reduce pressure on fuel wood. One such project converting municipal waste into energy is Okhla in Delhi.

Conservation of Mineral Resources

The challenge of sustainable development requires integration of quest for economic development with environmental concerns. Traditional methods of resource use result into generating enormous quantity of waste as well as create other environmental problems. Hence, for sustainable development calls for the protection of resources for the future generations. There is an urgent need to conserve the resources. The alternative energy sources like solar power, wind, wave, geothermal energy are inexhaustible resource. These should be developed to replace the exhaustible resources. In case of metallic minerals, use of scrap metals will enable recycling of metals. Use of scrap is specially significant in metals like copper, lead and zinc in which India's reserves are meagre. Use of substitutes for scarce metals may also reduce their consumption. Export of strategic and scarce minerals must be reduced, so that the existing reserve may be used for a longer period.



EXERCISES

1. Choose the right answers of the following from the given options.

(i) In which one of the following States are the major oil fields located?

(a) Assam (c) Rajasthan

(b) Bihar (d) Tamil Nadu

(ii) At which one of the following places was the first atomic power station started?

(a) Kalpakkam (c) Rana Pratap Sagar

(b) Narora (d) Tarapur

(iii) Which one of the following is non-renewable source of energy?

(a) Hydel (c) Thermal

(b) Solar (d) Wind power

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beneficiaries of infrastructural development. The remote villages in Tundah and Kugti areas still do not have sufficient infrastructure.

The social benefits derived from ITDP include tremendous increase in literacy rate, improvement in sex ratio and decline in child marriage. The female literacy rate in the region increased from 1.88 per cent in 1971 to 65 per cent in 2011. The difference between males and females in literacy level i.e. gender inequality, has also declined. Traditionally, the *Gaddis* had subsistence agricultural-cum-pastoral economy having emphasis on foodgrains and livestock production. But during the last three decades of twentieth century, the cultivation of pulses and other cash crops has increased in Bharmaur region. But the crop cultivation is still done with traditional technology. The declining importance of pastoralism in the economy of the region can be gauged from the fact that at present only about one-tenth of the total households practise transhumance. But the *Gaddis* are still very mobile as a sizeable section of them migrate to Kangra and surrounding areas during winter to earn their livings from wage labour.

Sustainable Development

The term development is generally used to describe the state of particular societies and the process of changes experienced by them. During a fairly large period of human history, the state of the societies has largely been determined by the interaction processes between human societies and their bio-physical environment. The processes of human-environment interaction depend upon the level of technology and institutions nurtured by a society. While the technology and institutions have helped in increasing the pace of human-environment interaction, the momentum thus, generated in return has accelerated technological progress and transformation and creation of institutions. Hence, development is a multi-dimensional concept and signifies the positive, irreversible transformation of the economy, society and environment.

The concept of development is dynamic and has evolved during the second half of twentieth century. In the post World War II era, the concept of development was synonymous to economic growth which is measured in terms of temporal increase in gross national product (GNP) and per capita income/per capita consumption. But, even the countries having high economic growth, experienced speedy rise in poverty because of its unequal distribution. So, in 1970s, the phrases such as *redistribution with growth* and *growth and equity* were incorporated in the definition of development. While dealing with the questions related to redistribution and equity, it was realised that the concept of development cannot be restricted to the economic sphere alone. It also includes the issues such as improving the well-being and living standard of people, availing of the health, education and equality of opportunity and ensuring political and civil rights. By 1980s, development emerged as a concept encapsulating wide-spread improvement in social as well as material well-being of all in a society.

The notion of sustainable development emerged in the wake of general rise in the awareness of environmental issues in the late 1960s in Western World. It reflected the concern of people about undesirable effects of industrial development on the environment. The publication of '***The Population Bomb***' by *Ehrlich* in 1968 and '***The Limits to Growth***' by *Meadows* and others in 1972 further raised the level of fear among environmentalists in particular and people in general. This sets the scenario for the emergence of new models of development under a broad phrase '*sustainable development.*'

Concerned with the growing opinion of world community on the environmental issues, the United Nations established a *World Commission on Environment and Development* (WCED) headed by the Norwegian Prime Minister Gro Harlem Brundtland. The Commission gave its report (also known as *Brundtland Report*) entitled '*Our Common Future*' in 1987. The report defines sustainable development as a "*development that meets the needs of the present without compromising the ability of future generations to meet their own needs.*"

Sustainable development takes care of ecological, social and economic aspects of development during the present times and pleads

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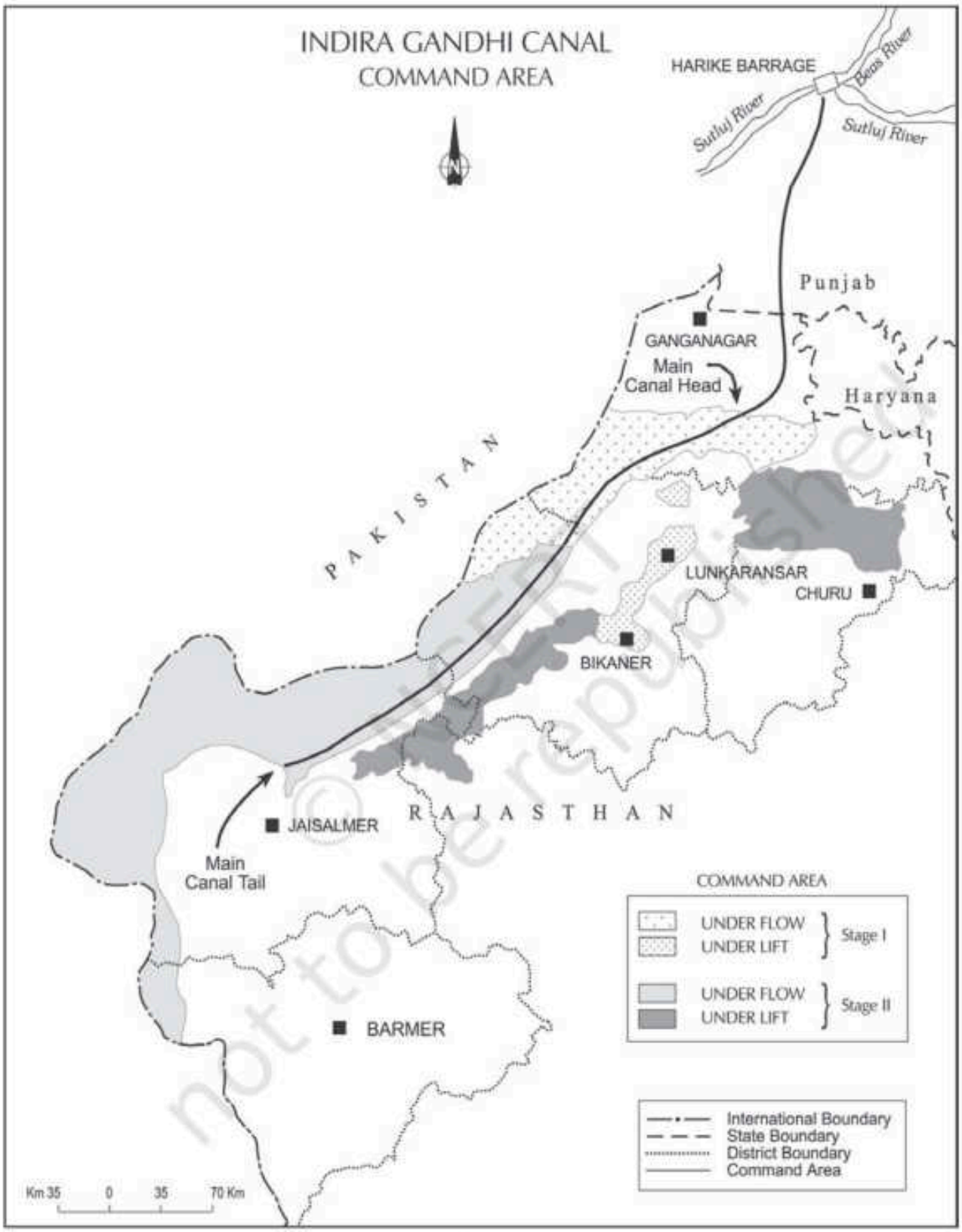


Fig. 6.3

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Fig. 6.4: Indira Gandhi Canal



Fig. 6.5 : Indira Gandhi Canal and its adjoining areas

for conservation of resources to enable the future generations to use these resources. It takes into account the development of whole human kind which have common future.

Case Study

Indira Gandhi Canal (Nahar) Command Area

Indira Gandhi Canal, previously known as the Rajasthan Canal, is one of the largest canal systems in India. Conceived by Kanwar Sain in 1948, the canal project was launched on 31 March, 1958. The canal originates at Harike barrage in Punjab and runs parallel to Pakistan border at an average distance of 40 km in Thar Desert (Marusthali) of Rajasthan. The total planned length of the system is 9,060 km catering to the irrigation needs of a total culturable command area of 19.63 lakh hectares. Out of the total command area, about 70 per cent was envisaged to be irrigated by flow system and the rest by lift system. The construction work of the canal system has been carried out through two stages. The command area of Stage-I lies in Ganganagar, Hanumangarh and northern part of Bikaner districts. It has a gently undulating topography and its culturable command area is 5.53 lakh hectares. The command area of Stage-II is spread over Bikaner, Jaisalmer, Barmer, Jodhpur, Nagaur and Churu districts covering culturable command area of 14.10 lakh ha. It comprises desert land dotted with shifting sand dunes and temperature soaring to 50°C in summers. In the lift canal, the water is lifted up to make it to flow against the slope of the land. All the lift canals of Indira Gandhi Canal system originate at the left bank of main canal while all the canals on the right bank of main canal are flow channels.

Irrigation in Stage-I command area of the canal was introduced in early 1960s, whereas, the command area of Stage-II began receiving irrigation in mid-1980s. The introduction of canal irrigation in this dry land has transformed its ecology, economy and society. It has influenced the environmental conditions of the region both positively as well as negatively. The availability of soil moisture for a longer period of time and various afforestation and pasture development programmes under CAD have resulted in greening the land. This has also helped in reducing wind erosion and siltation of canal systems. But the intensive irrigation and excessive use of water has led to the emergence of twin environmental problems of waterlogging and soil salinity.

Introduction of canal irrigation has brought about a perceptible transformation in the agricultural economy of the region. Soil moisture has been a limiting factor in successful growing of crops in this area. Spread of canal irrigation has led to increase in cultivated area and intensity of cropping. The traditional crops sown in the area, gram, bajra and jowar have been replaced by wheat, cotton, groundnut and rice. This is the result of intensive irrigation. This intensive irrigation, no doubt, initially has led to tremendous increase in agricultural and livestock productivity. This has also caused waterlogging

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and soil salinity, and thus, in the long run, it hampers the sustainability of agriculture.

Measures for Promotion of Sustainable Development

The ecological sustainability of Indira Gandhi Canal Project has been questioned by various scholars. Their point of view has also largely been validated by the course of development this region has taken during the last four decades, which has resulted in degradation of physical environment. It is a hard fact that attaining sustainable development in the command area requires major thrust upon the measures to achieve ecological sustainability. Hence, five of the seven measures proposed to promote sustainable development in the command area are meant to restore ecological balance.

(i) The first requirement is strict implementation of water management policy. The canal project envisages protective irrigation in Stage-I and extensive irrigation of crops and pasture development in Stage-II.

(ii) In general, the cropping pattern shall not include water intensive crops. It shall be adhered to and people shall be encouraged to grow plantation crops such as citrus fruits.

(iii) The CAD programmes such as lining of water courses, land development and levelling and *warabandi* system (equal distribution of canal water in the command area of outlet) shall be effectively implemented to reduce the conveyance loss of water.

(iv) The areas affected by water logging and soil salinity shall be reclaimed.

(v) The eco-development through afforestation, shelterbelt plantation and pasture development is necessary particularly in the fragile environment of Stage-II.

(vi) The social sustainability in the region can be achieved only if the land allottees having poor economic background are provided adequate financial and institutional support for cultivation of land.

(vii) The economic sustainability in the region cannot be attained only through development of agriculture and animal husbandry. The agricultural and allied activities have to develop alongwith other sectors of economy. This shall lead to diversification of economic base and establishment of functional linkages between basic villages, agro-service centres and market centres.

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people from one place to another. Ropeways, cableways and pipelines were devised to cater to the demands of transporting specific goods under special circumstances.

Road Transport

India has one of the second largest road networks in the world with a total length of about 62.16 lakh km (morth.nic.in, Annual Report 2020-21).

the princely states and British India. After Independence, twenty-year road plan (1961) was introduced to improve the conditions of roads in India. However, roads continue to concentrate in and around urban centres. Rural and remote areas had the least connectivity by road.

For the purpose of construction and maintenance, roads are classified as National Highways (NH), State Highways(SH), Major District Roads and Rural Roads.



Rain-soaked : Nomads go about their routine during an early morning downpour in Srinagar. Traffic on the 300-km Srinagar-Jammu and 434-km Srinagar-Leh National Highways is suspended as upper reaches of J&K saw heavy snowfall while rain lashed the plains.

A view of traffic flow in Delhi

Fig. 7.1

About 85 per cent of passenger and 70 per cent of freight traffic are carried by roads every year. Road transport is relatively suitable for shorter distance travel.

DO YOU KNOW ?

Sher Shah Suri built the *Shahi* (Royal) road to strengthen and consolidate his empire from the Indus Valley to the Sonar Valley in Bengal. This road was renamed the Grand Trunk (GT) road during the British period, connecting Calcutta and Peshawar. At present, it extends from Amritsar to Kolkata.

Collect information about National Highway number (old and new) from the website morth.nic.in/national-highway-details.

Road transport in modern sense was very limited in India before World War-II. The first serious attempt was made in 1943 when 'Nagpur Plan' was drawn. This plan could not be implemented due to lack of coordination among

National Highways

The main roads which are constructed and maintained by the Central Government are known as the National Highways. These roads are meant for inter-state transport and movement of defence men and material in strategic areas. These also connect the state capitals, major cities, important ports, railway junctions, etc. The length of the National Highways has increased from 19,700 km in 1951 to 1,36,440 km in 2020. The National Highways constitute only about 2 per cent of the total road length but carry 40 per cent of the road traffic.

The National Highways Authority of India (NHAI) was operationalised in 1995. It is an autonomous body under the Ministry of Surface Transport. It is entrusted with the responsibility of development, maintenance and operation of National Highways. This is also the apex body to improve the quality of the roads designated as National Highways.

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Table 7.1 : India Road Network 2020

Serial No.	Road Category	Length in Km
1.	National Highways	136440
2.	State Highways	176818
3.	Other Roads	5902539
	Total	**6215797**

Source: *Ministry of Road Transport and Highways Annual Report 2020-21. For latest data see website morth.nic.in*

National Highways Development Projects

NHAI has taken up some major projects in the country under different phases :

Golden Quadrilateral : It comprises construction of 5,846-km long 4/6 lane, high density traffic corridor, to connect India's four big metro cities of Delhi-Mumbai-Chennai-Kolkata. With the construction of Golden Quadrilateral, the time, distance and cost of movement among the mega cities of India will be considerably minimised.

North-South and East-West Corridors : North-South corridor aims at connecting Srinagar in Jammu and Kashmir with Kanniyakumari in Tamil Nadu (including Cochin-Salem Spur) with 4,076-km long road. The East-West Corridor has been planned to connect Silchar in Assam with the port town of Porbandar in Gujarat with 3,640-km of road length.

State Highways

These are constructed and maintained by state governments. They join the state capitals with district headquarters and other important towns. These roads are connected to the National Highways.

District Roads

These roads are the connecting link between District Headquarters and the other important nodes in the district.

Rural Roads

These roads are vital for providing links in the rural areas. About 80 per cent of the total road length in India are categorised as rural roads. There is regional variation in the density of rural road because these are influenced by the nature of the terrain.



Fig. 7.2 : Road constructed under the Pradhan Mantri Gram Sadak Yojna

Why are the rural roads' density very low in hilly, plateau and forested areas? Why does the quality of rural roads deteriorate away from the urban centres?

Other Roads

Other roads include Border Roads and International Highways. The Border Road Organisation (BRO) was established in May 1960 for accelerating economic development and strengthening defence preparedness through rapid and coordinated improvement of strategically important roads along the

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northern and north-eastern boundary of the country. It is a premier multifaceted construction agency. It has constructed roads in high altitude mountainous terrain joining Chandigarh with Manali (Himachal Pradesh) and Leh (Ladakh). This road runs at an average altitude of 4,270 metres above the mean sea level.

Apart from the construction and maintenance of roads in strategically sensitive areas, the BRO also undertakes snow clearance in high altitude areas. The international highways are meant to promote the harmonious relationship with the neighbouring countries by providing effective links with India. (Fig. 7.4 and 7.5)



Fig. 7.3 : Khardung La Pass in Ladakh

Do You Know ?

The World's longest Highway tunnel — **Atal Tunnel** (9.02 Km) has been built by Border Road Organisation. This tunnel connects Manali to Lahaul-Spiti valley throughout the year. Earlier the valley was cut off for about 6 months each year owing to heavy snowfall. The Tunnel is built with ultra-modern specifications in the Pir Panjal range of Himalayas at an altitude of 3000 metres from the Mean Sea Level (MSL).

Source: http://www.bro.gov.in/pagefimg.asp?imid=144, And PIB Delhi 03 October 2020

Delhi-Lahore Bus



Fig. 7.4 : A Bus from Lahore to Delhi at Wagah Border



Fig. 7.5 : Aman Setu between Baramula and Muzaffarabad

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Activity

Why have Bengaluru and Hyderabad in the South and Delhi, Kanpur and Patna in north India have emerged as important nodes?

Bharatmala Pariyojana – 'Road' to country's infrastructure development

- The Bharatmala Pariyojna envisages development of about 26,000 km length of Economic Corridors, which along with Golden Quadrilateral (GQ) and North–South and East–West (NS-EW) Corridors are expected to carry majority of freight trafiic on roads.
- The Programme focuses on development of ring roads, bypasses and elevated corridores to decongest the traffic passing through cities and enhance logistic efficiency.

[Source: Press Information Bureau (Reseach Unit), Ministry of Information and Broadcasting, Government of India, December 13, 2021]

Rail Transport

Indian Railways, network is one of the longest in the world. It facilitates the movement of both freight and passengers and contributes to the growth of the economy. Mahatma Gandhi said, the Indian railways *"...brought people of diverse cultures together to contribute to India's freedom struggle"*.

Indian Railway was introduced in 1853, when a line was constructed from Bombay to Thane covering a distance of 34 km.

Indian Railways is the largest government undertaking in the country. The length of Indian Railways network was 67,956 km (Railway yearbook 2019-20). Its very large size puts a lot of pressure on a centralised railway management system. Thus, in India, the railway system has been divided into 17 zones.

Table 7.2 : Indian Railways: Railway Zones and Headquarters

Railway Zone	Headquarters
Central	Mumbai CST
Eastern	Kolkata
East Central	Hajipur
East Coast	Bhubaneswar
Northern	New Delhi
North Central	Prayagraj
North Eastern	Gorakhpur
North East Frontier	Maligaon (Guwahati)
North Western	Jaipur
Southern	Chennai
South Central	Secunderabad
South Eastern	Kolkata
South East Central	Bilaspur
South Western	Hubli
Western	Mumbai (Church Gate)
West Central	Jabalpur
Metro Railway	Kolkata

Do You Know ?

On the basis of the width of track of the Indian Railways, three categories have been made:

Broad gauge: The distance between rails in broad gauge is 1.676 metre. The total length of broad gauge lines was 63950 km (2019-20).

Metre gauge: The distance between rails is one metre. Its total length was 2402 km (2019-20).

Narrow gauge: The distance between the rails in this case is 0.762 metre or 0.610 metre. The total length of narrow guage was 1604 km (2019-20). It is generally confined to hilly areas.

Indian Railways has launched extensive programme to convert the metre and narrow gauges to broad gauge. Moreover, steam

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engines have been replaced by diesel and electric engines. This step has increased the speed, as well as, the haulage capacity.

The replacement of steam engines run by coal has also improved the environment of the stations.

Metro rail has revolutionlised the urban transport system in India. Replacement of diesel buses by CNG-run vehicles along with the introduction of metro is a welcome step towards controlling the air pollution in urban centres.

Areas around towns, raw material producing areas and of plantations and other commercial crops, hill stations and cantonment towns were well-connected by railways from the British colonial era. These were mostly developed for the exploitation of resources. After the Independence of the country, railway routes have been extended to other areas too. The most significant development has been the development of Konkan Railway along the western coast providing a direct link between Mumbai and Mangaluru.

Railway continues to remain the main means of transport for the masses. Railway network is relatively less dense in the hill states, north eastern states, central parts of India and Rajasthan.

Which cities of India have Metro Rail Facility? Collect information about it and discuss in the classroom.

Konkan Railway

One of the important achievements of the Indian Railways has been the construction of Konkan Railway in 1998. It is 760-km long rail route connecting Roha in Maharashtra to Mangalore in Karnataka. It is considered an engineering marvel. It crosses 146 rivers, streams, nearly 2000 bridges and 91 tunnels. Asia's largest tunnel which is nearly 6.5 km long, also lies on this route. The states of Maharashtra, Goa and Karnataka are partners in this undertaking.

Water Transport

Waterways is an important mode of transport for both passenger and cargo traffic in India. It is the cheapest means of transport and is most suitable for carrying heavy and bulky material. It is a fuel-efficient and eco-friendly mode of transport. The water transport is of two types– (a) inland waterways, and (b) oceanic waterways.

Inland Waterways

It was the chief mode of transport before the advent of railways. It, however, faced tough competition from road and railway transport. Moreover, diversion of river water for irrigation purposes made them non-navigable in large parts of their courses. India has 14,500 km of navigable waterways, contributing about 1% to the country's transportation. It comprises rivers, canals, backwaters, creeks, etc.



Fig. 7.6 : River navigation in the North-east

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For the development, maintenance and regulation of national waterways in the country, the Inland Waterways Authority was set up in 1986. In order to create wide waterways network and to promote inland water transport in the country as an economical, environment friendly supplementary mode of transport to rail and road, 111 inland waterways (including 5 national waterways declared earlier, Table 7.3) were declared as National Waterways (NWs) by the National Waterways Act, 2016. *(Source: Annual Report, Ministry of Ports, Shipping & Waterways, Govt. of India, 2022-23)*

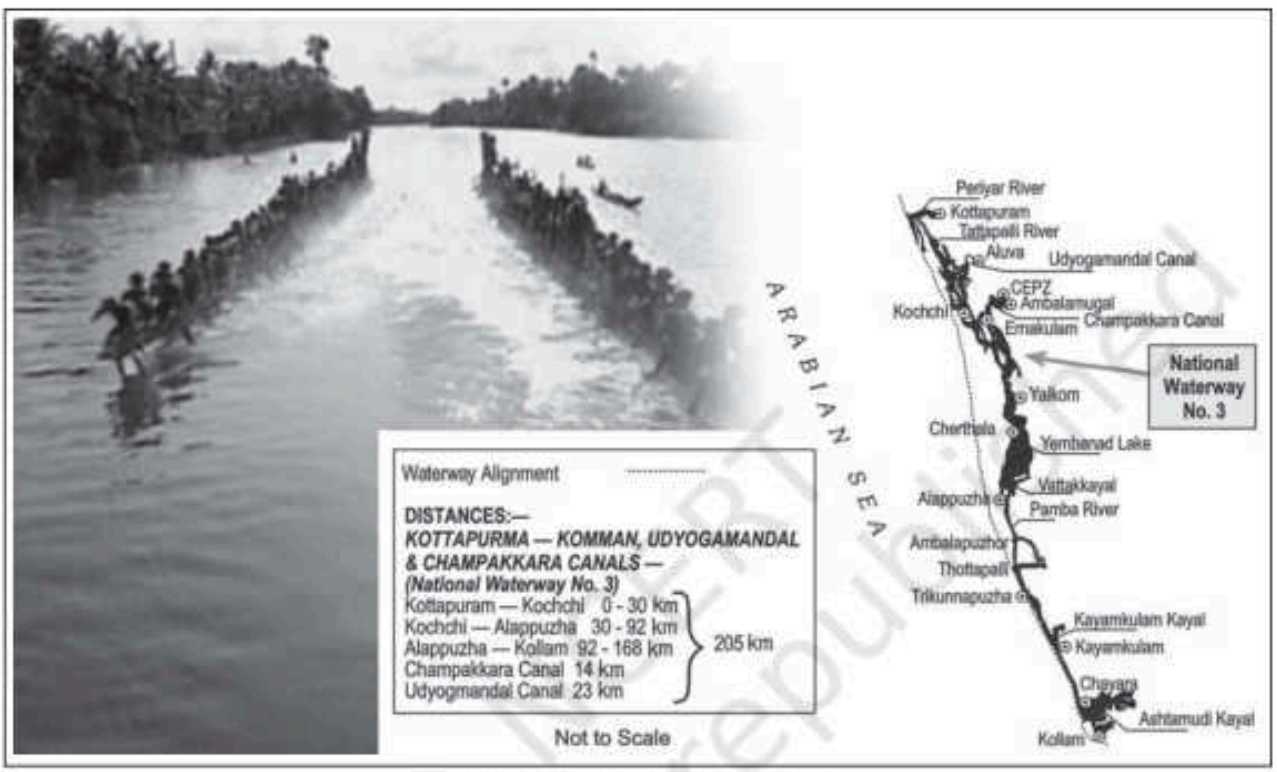


Fig. 7.7 : Natinal Waterway No.3

Table 7.3 National Waterways of India

Waterways	Stretch	Specification
NW 1	Prayagraj–Haldia stretch (1,620 km)	It is one of the most important waterways in India, which is navigable by mechanical boats up to Patna and by ordinary boats up to Haridwar. It is divided into three parts for developmental purposes – (i) Haldia–Farakka (560 km), (ii) Farakka–Patna (460 km), (iii) Patna–Prayagraj (600 km).
NW 2	Sadiya-Dhubri stretch (891 km)	Brahmaputra is navigable by steamers up to Dibrugarh (1,384 km) which is shared by India and Bangladesh
NW 3	Kottapuram-Kollam stretch (205 km)	It includes 168 km of west coast canal along with Champakara canal (14 km) and Udyogmandal canal (23 km).
NW 4	Specified streches of Godavari and Krishna rivers along with Kakinada Puducherry stretch of canals (1078 km)	
NW 5	Specified stretches of river Brahmani along with Matai river, delta channels of Mahanadi and Brahmani rivers and East Coast canals (588km).	

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The Inland Waterways Authority has also identified 10 other inland waterways, which could be upgraded. The backwaters (Kayal) of Kerala has special significance in Inland Waterway. Apart from providing cheap means of transport, they are also attracting a large number of tourists in Kerala. The famous Nehru Trophy Boat Race (VALLAMKALI) is also held in the backwaters.

Oceanic Routes

India has a vast coastline of approximate 7,517 km, including islands. Twelve major and 200 minor ports provide infrastructural support to these routes. Oceanic routes play an important role in the transport sector of India's economy. Approximately 95 per cent of India's foreign trade by volume and 70 per cent by value moves through ocean routes. Apart from international trade, these are also used for the purpose of transportation between the islands and the rest of the country.

Air Transportation

Air transport is the fastest means of movement from one place to the other. It has reduced distances by minimising the travel time. It is essential for a vast country like India, where distances are large and the terrain and climatic conditions are diverse.

Air transport in India made a beginning in 1911 when airmail operation commenced over a little distance of 10 km between Prayagraj and Naini. But its real development took place in post-Independent period. The Airport Authority of India is responsible for providing safe, efficient air traffic and aeronautical communication services in the Indian Air Space.

Do You Know ?

UDAN (*Ude Desh ka Aam Nagrik*) is a first-of-its kind scheme globally, designed to jump- start the regional aviation market. Regional Connectivity Scheme (RCS) -UDAN was conceived by the Ministry of Civil Aviation (MoCA) Govt. of India, to promote regional connectivity by making fly affordable for the common citizen. The central idea of the scheme is to encourage airlines to operate flights on regional and remote routes through enabling policies and extending incentives.

Pawan Hans is the helicopter service operating in hilly areas and is widely used by tourists in north-eastern sector.

In addition, Pawan Hans Limited mainly provides helicopter services to petroleum sector and for tourism.

Oil and Gas Pipelines

Pipelines are the most convenient and efficient mode of transporting liquids and gases over long distances. Even solids can also be transported by pipelines after converting them into slurry. Oil India Limited (OIL) under the administrative set up of the Ministry of Petroleum and Natural Gas is engaged in the exploration, production and transportation of crude oil and natural gas. It was incorporated in 1959 as a company. Asia's first cross country pipeline covering a distance of 1,157 km was constructed by OIL from Naharkatiya oilfield in Assam to Barauni refinery in Bihar. It was further extended up to Kanpur in 1966. GAIL (India) Ltd. was set up in 1984 as a public sector undertaking to transport, process and market natural gas for its economic use. The first 1,700 km long Hazira-Vijaipur-Jagdishpur (HVJ) cross country gas pipeline, constructed by GAIL (India), linked Mumbai

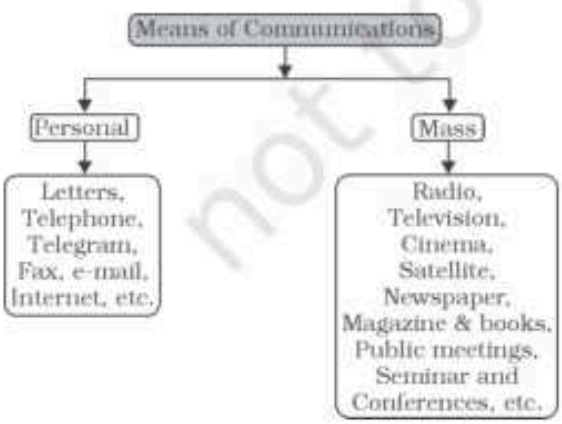
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High and Bassein gas fields with various fertiliser, power and industrial complexes in western and northern India. This artery provided impetus to Indian gas market development. Overall, India's gas infrastructure has expanded over ten times from 1,700 km to 18,500 km of cross-country pipelines and is expected to soon reach over 34,000 km as Gas Grid by linking all the gas sources and consuming markets across the country including North Eastern States.

Communication Networks

Human beings have evolved different methods of communication over time. In earlier times, the messages were delivered by beating the drum or hollow tree trunks, giving indications through smoke or fire or with the help of fast runners. Horses, camels, dogs, birds and other animals were also used to send messages. Initially, the means of communication were also the means of transportation. Invention of post-office, telegraph, printing press, telephone, satellite, etc has made the communication much faster and easier. Development in the field of science and technology has significantly contributed in bringing about revolution in the field of communication.

People use different modes of communication to convey the messages. On the basis of scale and quality, the mode of communication can be divided into following categories :



Personal Communication System

Among all the personal communication system internet is the most effective and advanced one. It is widely used in urban areas. It enables the user to establish direct contact through e-mail to get access to the world of knowledge and information. It is increasingly used for e-commerce and carrying out money transactions. The internet is like a huge central warehouse of data, with detailed information on various items. The network through internet and e-mail provides an efficient access to information at a comparatively low cost. It enables us with the basic facilities of direct communication.

Mass Communication System

Radio

Radio broadcasting started in India in 1923 by the Radio Club of Bombay. Since then, it gained immense popularity and changed the socio-cultural life of people. Within no time, it made a place in every household of the country. Government took this opportunity and brought this popular mode of communication under its control in 1930 under the Indian Broadcasting System. It was changed to All India Radio in 1936 and to Akashwani in 1957.

All India Radio broadcasts a variety of programmes related to information, education and entertainment. Special news bulletins are also broadcast at specific occasions like session of parliament and state legislatures.

Television (T.V.)

Television broadcasting has emerged as the most effective audio-visual medium for disseminating information and educating masses. Initially, the T.V. services were limited only to the National Capital where it began in 1959. After 1972, several other centres became operational. In 1976, TV was delinked from All India Radio (AIR) and got a separate identity

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Table 8.5 : Import of Some Principal Commodities
(in crore rupees)

Commodities	2021-22
Fertilisers and fertiliser manufacturing	105796
Edible oils	141532
Pulp and waste paper	11934
Non-ferrous metals	499766
Iron and steel	94053
Petroleum, oil and lubricants	1207803
Pearls, precious and semi-precious stones	231279
Medicinal and Pharma products	67545
Chemical products	308882

Source : Economic Survey 2022-23

Direction of Trade

India has trade relations with most of the countries and major trading blocks of the world.

Region-wise and sub-region-wise trade during the period 2021-22 has been given in Table 8.6.

Table 8.6 Direction of India's Import trade
(in crore rupees)

Region	Imports	
	2016-17	2021-22
Europe	403972	640577
Africa	193327	368156
North America	195332	378041
Latin America	115762	161995
Asia and ASEAN	1544520	2918577

Source : Economic Survey 2016-17 and 2022-23

India aims to double its share in the international trade within the next five years. It has already started adopting suitable measures such as import liberalisation, reduction in import duties, delicensing and change from process to product patents.

Activity

Draw a multiple bar diagram to represent the major trading partners.

Most of India's foreign trade is carried through sea and air routes. However, a small portion is also carried through land route to neighbouring countries like Nepal, Bhutan, Bangladesh and Pakistan.

Sea Ports as Gateways of International Trade

India is surrounded by sea from three sides and is bestowed with a long coastline. Water provides a smooth surface for very cheap transport provided there is no turbulence. India has a long tradition of sea faring and developed many ports with place name suffixed with *pattan* meaning port. An interesting fact about ports in India is that its west coast has more ports than its east coast.



Fig. 8.3 : Unloading of goods on port

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Can you find out the reasons for the variations in the location of ports along the two coasts?

Though ports have been in use since ancient times, the emergence of ports as gateways of international trade became important after the coming of the European traders and colonisation of the country by the British. This led to the variation in the size and quality of ports. There are some ports which have very vast area of influence and some have limited area of influence. At present, India has 12 major ports and 200 minor or intermediate ports. In case of the major ports, the central government decides the policy and plays regulatory functions. The minor ports are there whose policy and functions are regulated by state governments. The major ports handle larger share of the total traffic.

The British used the ports as suction points of the resources from their hinterlands. The extension of railways towards the interior facilitated the linking of the local markets to regional markets, regional markets to national markets and national markets to the international markets. This trend continued till 1947. It was expected that the country's Independence will reverse the process, but the partition of the country snatched away two very important ports, i.e., Karachi port went to Pakistan and Chittagong port to the erstwhile east-Pakistan and now Bangladesh. To compensate the losses, many new ports, like the Kandla in the west and the Diamond Harbour near Kolkata on river Hugli in the east were developed.

Despite this major setback, Indian ports continued to grow after the Independence. Today, Indian ports are handling large volumes of domestic, as well as, overseas trade. Most of the ports are equipped with modern infrastructure. Previously, the development and modernisation was the responsibility of the government agencies, but considering the increase in function and need to bring these ports at par with the international ports, private entrepreneurs have been invited for the modernisation of ports in India.

The capacity of Indian ports increased from 20 million tonnes of cargo handling in 1951 to more than 837 million tonnes in 2016.

Some of the Indian ports along with their hinterlands are as follows :

Deendayal Port (Kandla port) situated at the head of Gulf of Kuchchh has been developed as a major port to cater to the needs of western and north western parts of the country and also to reduce the pressure at Mumbai port. The port is specially designed to receive large quantities of petroleum and petroleum products and fertiliser. The offshore terminal at Vadinar has been developed to reduce the pressure at the port.

Demarcation of the boundary of the hinterland would be difficult as it is not fixed over space. In most of the cases, hinterland of one port may overlap with that of the other.

Mumbai is a natural harbour and the biggest port of the country. The port is situated closer to the general routes from the countries of Middle East, Mediterranean countries, North Africa, North America and Europe where the major share of country's overseas trade is carried out. The port is 20 km long and 6-10 km wide with 54 berths and has the country's largest oil terminal. M.P., Maharashtra, Gujarat, U.P. and parts of Rajasthan constitute the main hinterlands of Mumbai ports.

Jawaharlal Nehru Port at Nhava Sheva was developed as a satellite port to relieve the pressure at the Mumbai port. It is the largest container port in India.

Mormugao Port, situated at the entrance of the Zuari estuary, is a natural harbour in Goa. It gained significance after its remodelling in 1961 to handle iron-ore exports to Japan. Construction of Konkan railway has considerably extended the hinterland of this port. Karnataka, Goa, Southern Maharashtra constitute its hinterland.

New Mangalore Port is located in the state of Karnataka and caters to the needs of the export of iron-ore and iron-concentrates. It also handles fertilisers, petroleum products, edible

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Fig. 8.4 : India – Major Ports and Sea Routes

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oils, coffee, tea, wood pulp, yarn, granite stone, molasses, etc. Karnataka is the major hinterland for this port.

Cochin Port, situated at the head of Vembanad Kayal, popularly known as the *'Queen of the Arabian Sea'*, is also a natural harbour. This port has an advantageous location being close to the Suez-Colombo route. It caters to the needs of Kerala, southern-Karnataka and south western Tamil Nadu.

Shyama Prasad Mookerjee, Kolkata Port is located on the Hugli river, 128 km inland from the Bay of Bengal. Like the Mumbai port, this port was also developed by the British. Kolkata had the initial advantage of being the capital of British India. The port has lost its significance considerably on account of the diversion of exports to the other ports such as Vishakhapatnam, Paradwip and its satellite port, Haldia.

This port is also confronted with the problem of silt accumulation in the Hugli river which provides a link to the sea. Its hinterland covers U.P., Bihar, Jharkhand, West Bengal, Sikkim and the north-eastern states. Apart from this, it also extends ports facilities to our neighbouring land-locked countries such as Nepal and Bhutan.

Haldia Port is located 105 km downstream from Kolkata. It has been constructed to reduce the congestion at Kolkata port. It handles bulk cargo like iron ore, coal, petroleum, petroleum products and fertilisers, jute, jute products, cotton and cotton yarn, etc.

Paradwip Port is situated in the Mahanadi delta, about 100 km from Cuttack. It has the deepest harbour specially suited to handle very large vessels. It has been developed mainly to handle large-scale export of iron-ore. Odisha, Chhattisgarh and Jharkhand are the parts of its hinterland.

Visakhapatnam Port in Andhra Pradesh is a land-locked harbour, connected to the sea by a channel cut through solid rock and sand. An outer harbour has been developed for handling iron-ore, petroleum and general cargo. Andhra Pradesh and Telangana are the main hinterland for this port.

Chennai Port is one of the oldest ports on the eastern coast. It is an artificial harbour built in 1859. It is not much suitable for large ships because of the shallow waters near the coast. Tamil Nadu and Puducherry are its hinterland.

Kamarajar (Ennore), a newly developed port in Tamil Nadu, has been constructed 25 km north of Chennai to relieve the pressure at Chennai port.

V.O. Chidambaranar (Tuticorin) Port was also developed to relieve the pressure of Chennai port. It deals with a variety of cargo, including coal, salt, food grains, edible oils, sugar, chemicals and petroleum products.

Airports

Air transport plays an important role in the international trade. It has the advantage of taking the least time for carriage and handling high value or perishable goods over long distances. It is very costly and unsuitable for carrying heavy and bulky commodities. This ultimately reduces the participation of this sector in the international trade as compared to the oceanic routes.

There were 25 major airports functioning in the country (Annual Report 2016-17). They are Ahmedabad, Bengaluru, Chennai, Delhi, Goa, Guwahati, Hyderabad, Kolkata, Mumbai, Thiruvananthapuram, Srinagar, Jaipur, Calicut, Nagpur, Coimbatore, Cochin, Lucknow, Pune, Chandigarh, Mangaluru, Vishakhapatnam, Indore, Patna, Bhubaneswar and Kannur. Since 2017, under the UDAN scheme, a total number of 73 unserved/underserved airports including 9 Heliports and 2 water aerodromes have been operationalized (*Source: PIB, Ministry of Civil Aviation, Govt. of India, 2023*)

You have already studied about the air transport in the previous chapter. You consult the chapter on transport to find out the main features of air transport in India.

Activity

Name the nearest domestic and international airports from your place. Identify the state with maximum number of domestic airports.

Identify four cities where maximum number of air routes converge and also give reasons for this.

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Table 9.1 : Types and Sources of Pollution

Pollution Types	Pollution Involved	Sources of Pollution
Air Pollution	Oxides of sulphur (SO_2, SO_3), Oxides of nitrogen, carbon monoxide, hydro-carbon, ammonia, lead, aldehydes asbestos and beryllium.	Combustion of coal, petrol and diesel, industrial processes, solid waste disposal, sewage disposal, etc.
Water Pollution	Odour, dissolved and suspended solids, ammonia and urea, nitrate and nitrites, chloride, fluoride, carbonates, oil and grease, insecticide and pesticide residue, tannin, coliform MPM (bacterial count) sulphates and sulphides, heavy metals e.g. lead, aresenic, mercury, manganese, etc., radioactive substances.	Sewage disposal, urban run-off, toxic effluents from industries, run-off over cultivated lands and nuclear power plants.
Land Pollution	Human and animal excreta viruses and bacteria, garbage and vectors therein, pesticides and fertiliser-residue alkalinity, fluorides, radio-active substances.	Improper human activities, disposal of untreated industrial waste, use of pesticides and fertilisers.
Noise Pollution	High level of noise above tolerance level.	Aircrafts, automobiles, trains, industrial processing and advertising media.

Industries produce several undesirable products including industrial wastes, polluted waste water, poisonous gases, chemical residuals, numerous heavy metals, dust, smoke, etc. Most of the industrial wastes are disposed off in running water or lakes. Consequently, poisonous elements reach the reservoirs, rivers and other water bodies, which destroy the bio-system of these waters. Major water polluting industries are leather, pulp and paper, textiles and chemicals.

Various types of chemicals used in modern agriculture such as inorganic fertilisers, pesticides and herbicides are also pollution generating components. These chemicals are washed down to rivers, lakes and tanks. These chemicals also infiltrate the soil to reach the ground water. Fertiliser induces an increase in the nitrate content of surface waters. Cultural activities such as pilgrimage, religious fairs, tourism, etc. also cause water pollution. In India, almost all

Table 9.2 : Sources of Pollution in the Ganga and the Yamuna Rivers

River and State	Polluted Stretches	Nature of Pollution	Main Polluters
Ganga (Uttar Pradesh) Bihar and West Bengal	(a) Downstream of Kanpur (b) Downstream of Varanasi (c) Farrakka Barrage	1. Industrial pollution from towns like Kanpur 2. Domestic wastes from urban centres 3. Dumping of carcasses in the river	Cities of Kanpur, Prayagraj, Varanasi, Patna and Kolkata release domestic waste into the river
Yamuna (Delhi) and (Uttar Pradesh)	(a) Delhi to confluence with Chambal (b) Mathura and Agra	1. Extraction of water by Haryana and Uttar Pradesh for irrigation 2. Agricultural run off resulting in high levels of micro-pollutants in the Yamuna 3. Domestic and industrial waste of Delhi flowing into the river	Delhi dumping its domestic waste

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surface water sources are contaminated and unfit for human consumption.

Water pollution is a source of various water- borne diseases. The diseases commonly caused due to contaminated water are diarrhoea, intestinal worms, hepatitis, etc. The World Health Organization shows that about one-fourth of the communicable diseases in India are water-borne. Though river pollution is common to all rivers, yet pollution of river Ganga flowing through one of the most populous regions of India has caused great concerns among all. To imporve the condition of the river, National Mission for Clean Ganga was initiated. The Namami Gange Programme has been launched for the same.

Namami Gange Programme

Ganga, as a river, has national importance but the river requires cleaning by effectively controlling the pollution for its water. The Union Government has launched the '*Namami Gange Programme*' with the following objectives:

- developing sewerage treatment systems in towns,
- monitoring of industrial effluents,
- development of river front,
- afforestation along the bank to increase biodiversity,
- cleaning of the river surface,
- development of 'Ganga Grams' in Uttarakhand, UP, Bihar, Jharkhand and West Bengal, and
- creating public awareness to avoid adding pollutants into the river even in the form of rituals.

Air Pollution

Air pollution is taken as addition of contaminants, like dust, fumes, gas, fog, odour, smoke or vapour to the air in substantial proportion and duration that may be harmful to flora and fauna and to property. With increasing use of varieties of fuels as the source of energy, there is a marked increase in emission of toxic gases into the atmosphere resulting in the pollution of air. Combustion of fossil fuels, mining and industries are the main sources of air pollution. These processes release oxides of sulphur and nitrogen, hydrocarbons, carbon dioxide, carbon monoxide, lead and asbestos.

Air pollution causes various diseases related to respiratory, nervous and circulatory systems.



Smog in Mumbai

Air pollution is world's top environmental health risk, WHO says

Sushmi Dey / TNN / Updated: Jun 2, 2015, 03:19 IST

Air pollution is the world's biggest environmental health risk, causing at least one in eight deaths around the globe, the World Health Organization has said.

Air pollution biggest killer in Southeast Asia, says WHO

A smoky haze that shrouded parts of Southeast Asia this month, forcing schools and businesses to close, is just one element of an air pollution problem that kills hundreds of thousands of people in the region annually, the World Health Organization said.

Air pollution in major Southeast Asian and Chinese cities ranks among the worst in the world and contributes to the deaths of about 500,000 people each year, said Michal Krzyzanowski, an air quality specialist at the WHO's European Centre for Environment and Health in Bonn.

Drifting smoke from purposely set forest fires in Indonesia caused Malaysia to declare a state of emergency last week in two areas outside Kuala Lumpur. Parts of Thailand were also blanketed in the haze.

Malaysia said hospitals reported a 150% increase in breathing problems and seven people who had a history of respiratory problems reportedly died. The government could not confirm the smoky air was to blame.

Worldwide, air pollution contributes to some 800,000 deaths each year. The emergency in Malaysia was lifted after two days, but meteorologists are predicting a new cloud will hover over parts of Malaysia and possibly Singapore.

The haze, blamed on illegal dry-season burning to clear land on Sumatra island, is a annual problem

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Smoky fog over cities called as urban *smog* is caused by atmospheric pollution. It proves very harmful to human health. Air pollution can also cause acid rains. Rainwater analysis of urban environment has indicated that pH value of the first rain after summer is always lower than the subsequent rains.

Noise Pollution

Noise pollution refers to the state of unbearable and uncomfortable to human beings which is caused by noise from different sources. This matter has become a serious concern only in recent years due to a variety of technological innovations.

The main sources of noise pollution are various factories, mechanised construction and demolition works, automobiles and aircraft, etc. There may be added periodical but polluting noise from sirens, loudspeakers used in various festivals, programmes



Fig. 9.2 : Noise monitoring at Panchpatmalai Bauxite Mine

associated with community activities. The level of steady noise is measured by sound level expressed in terms of decibels (dB).

Of all these sources, the biggest nuisance is the noise produced by traffic, because its intensity and nature depend upon factors, such as the type of aircraft, vehicle, train and the condition of road, as well as, that of vehicle (in case of automobiles). In sea traffic, the noise pollution is confined to the harbour due to loading and unloading activities being carried. Industries cause noise pollution but with varying intensity depending upon the type of industry.

Noise pollution is location specific and its intensity declines with increase in distance from the source of pollution, i.e. industrial areas, arteries of transportation, airport, etc. Noise pollution is hazardous in many metropolitan and big cities in India.

Do You Know ?

Oceans 10 times noisier today than 40 years ago

A study by Scripps Institute of Oceanography has revealed that Ocean Noise has increased tenfold since the 1960s. Oceanologists Sean Wiggins, John Hildebrand from Scripps and Mark McDonald from WhaleAcoustics, Colorado, studied declassified US Navy documents and came to the conclusion that global shipping has contributed a lot to increased undersea noise pollution. They said with populations increasing around the globe in recent decades, the underwater world had also become a noisier place, adding that the effects of greater noise on marine life was still unknown. Findings revealed a tenfold increase in underwater ocean noise as compared with the 1960s. They said the noise levels in 2003-2004 were about 10 to 12 decibels higher than in 1964-1966. The reasons could be due to the vast increase in the global shipping trade, the number of ships plying the oceans and higher speed of vessels.

Urban Waste Disposal

Urban areas are generally marked by overcrowding, congestion, inadequate facilities to support the fast growing population and consequent poor sanitary conditions and foul air. Environmental pollution by solid wastes has now got significance because of enormous growth in the quantity of wastes generated from various sources. Solid waste refers to a variety of old and used articles, for example stained small pieces of metals, broken glassware, plastic containers, polythene bags, ash, floppies, CDs, etc., dumped at different places. These discarded materials are also termed as refuse, garbage and rubbish, etc., and are disposed of from two sources : (i) household or domestic establishments, and (ii) industrial or commercial establishments. The household wastes are disposed off either on public lands or on private contractors' sites,

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whereas the solid wastes of industrial units are collected and disposed off through public (municipal) facilities at low lying public grounds (landfill areas). The huge turn out of ashes and debris from industries, thermal power houses and building constructions or demolitions have posed problems of serious consequences. Solid wastes cause health hazard through creation of obnoxious smell, and harbouring of flies and rodents, which act as carriers of diseases like typhoid, diphtheria, diarrhoea, malaria and cholera, etc. These wastes cause frequent nuisance as and when these are carelessly handled, spread by wind and splittered through rain water.

Concentration of industrial units in and around urban centres gives rise to disposal of industrial wastes. The dumping of industrial waste into rivers leads to water pollution. River pollution from city-based industries and untreated sewage leads to serious health problems downstream.

Urban waste disposal is a serious problem in India. In metropolitan cities like Mumbai, Kolkata, Chennai, Bengaluru, etc., about 90 per cent of the solid waste is collected and disposed. But in most of other cities and towns in the country, about 30 to 50 per cent of the waste generated are left uncollected which accumulate on streets, in open spaces between houses and in wastelands leading to serious



I moved into this second floor from the first to get a view of the sea and the garbage has piled up to this level obstructing the view.

Case Study : A Role Model to Restore the Ecology and Safeguard Human Health in Daurala

Based on the universal law "Polluter pays", effort to restore the ecology and safeguard the human health with people's participation has taken place in Daurala near Meerut. These efforts are now bearing fruits after a span of three years when Meerut based NGO had developed a model for ecological restoration. The meeting of the Daurala Industries officials, NGOs, Government officials and other stakeholders at Meerut has brought out results. The powerful logics, authentic studies and the pressure of people have brought a new lease of life to the twelve thousand residents of this village. It was in the year 2003 that the pitiable condition of Dauralaites drew the attention of the civil society. The groundwater of this village was contaminated with heavy metals. The reason was that the untreated wastewater of Daurala industries was leaching to the groundwater table. The NGO conducted a door to door survey of the health status of the residents and came out with a report. The organisation, the village community and people's representatives sat together to find out sustainable solutions to the health problem. The industrialists showed a keen interest towards checking the deteriorating ecology. The overhead water tank's capacity in the village was enhanced and a 900m extra pipeline was laid to supply potable water to the community. The silted pond of the village was cleaned and recharged by desilting it. Large quantity of silt was removed paving way to large quantity of water so that it recharged the aquifers. Rainwater harvesting structures have been constructed at different places which has helped in diluting the contaminants of the groundwater after the monsoons. 1000 trees have also been planted which have improved the environment.

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health hazards. These wastes should be treated as resource and utilised for generating energy and compost. Untreated wastes ferment slowly and release toxic biogas to the atmosphere, including methane.

Activity

What do we throw away? Why?

Where does our waste end up?

Why do ragpickers sort out rubbish dumps? Does it have some value?

Is our urban waste worth anything?



Fig. 9.3 : A view of urban waste in Mahim, Mumbai

Rural–Urban Migration

Population flow from rural to urban areas is caused by many factors, like high demand for labour in urban areas, low job opportunities in rural areas and unbalanced pattern of development between urban and rural areas. In India, population in cities is rapidly increasing. Due to low opportunities in smaller and medium cities, the poor people generally bypass these small cities and directly come to the mega cities for their livelihood.

A case study given below to have better understanding of the subject. Read it carefully and try to comprehend the process of rural urban migration.

A Case Study

Ramesh has been working in contract as a welder on construction site in Talcher (coal region of Odisha) for the last two years. He moved with the contractor to various places like Surat, Mumbai, Gandhi Nagar, Bharuch, Jamnagar and so on. He remits Rs. 20,000 per year to his father in his native village. The remittances have been mainly used for daily consumption, healthcare, schooling of children, etc. Part of the money is also used in agriculture, purchasing of land and building of houses, etc. The standard of living of Ramesh's family improved significantly.

Fifteen years ago, the situation was not the same. The family was passing through very tough times. Three of his brothers and their families had to survive on three acres of land. The family was highly in debt. Ramesh had to discontinue his studies after ninth standard. He was further hard pressed when he got married.

Do You Know ?

At present, 55 per cent of the world's population lives in cities and more will join them in near future. This proportion is estimated to go up to 68 per cent by 2050. That will put pressure on governments to make urban areas better places to live with optimum infrastructure facilities for desirable quality of life.

Urban population grows as a result of natural increase (when birth rate exceeds death rate), net in-migration (when people move in than out), and sometimes reclassification of urban areas to encompass formerly rural population settlements. In India, it is estimated that after 1961 around 60 per cent of the urban growth has been attributed and about 29 per cent of them from rural areas to urban migration.

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Simultaneously, he was also impressed by some successful out-migrants of his village who had been working in Ludhiana and supporting their families in village by sending money and some consumer goods. Thus, due to abject poverty in the family and perceived job promises at Ludhiana, he made a move to Punjab with his friend. He worked there in a woolen factory for six months at the rate of only Rs. 20 per day in 1988. Apart from the crisis of managing his personal expenditure from this meagre income, he was also facing difficulty in assimilation to the new culture and environment. Then he decided to change his place of work from Ludhiana to Surat under the guidance of his friend. He learnt the skills of welding in Surat and after that he has been moving to different places with the same contractor. Though the economic condition of Ramesh's family at village improved, he is bearing the pain of separation of his near and dear ones. He cannot shift them with him, as the job is temporary and transferable.

Comments

In developing countries, poor, semi-illiterate and the unskilled like Ramesh migrating from rural areas frequently end up performing menial jobs at low wages in informal sector in urban areas. Since wages are very low to support the family at the place of destination, the spouses are left behind in rural areas to look after children and elderly people. Thus, the rural-urban migration stream is dominated by the males.

Problems of Slums

The concept "Urban or Urban Centre" is defined in settlement geography to differentiate it from the "Rural" about which you have learnt in some previous chapters of this book. You have also learnt in the book entitled *"Fundamentals of Human Geography"* that this concept is defined differently in different countries.

Both urban and rural settlements are different in their functions, sometimes,



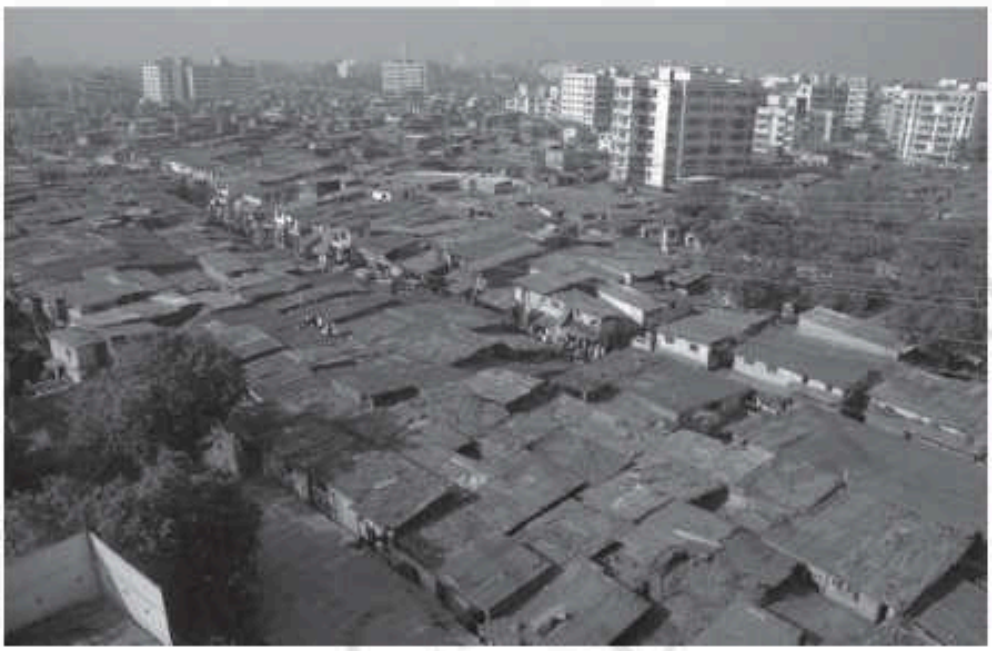
Have you visited a slum? Visit a slum in your city, and write about the problems faced by slum dwellers.

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Dharavi–Asia's Largest Slum

"…. Buses merely skirt the periphery. Autorickshaws cannot go there, Dharavi is part of central Bombay where three wheelers are banned.

Only one main road traverses the slum, the miscalled 'ninety-foot road', which has been reduced to less than half of that for most of its length. Some of the side alleys and lanes are so narrow that not



even a bicycle can pass. The whole neighbourhood consists of temporary buildings, two or three storeyed high with rusty iron stairways to the upper part, where a single room is rented by a whole family, sometimes accommodating twelve or more people; it is a kind of tropical version of the industrial dwelling of Victorian London's East End.

But Dharavi is a keeper of more sombre secrets than the revulsion it inspires in the rich; a revulsion, moreover, that is, in direct proportion to the role it serves in the creation of the wealth of Bombay. In this place of shadowless, treeless sunlight, uncollected garbage, stagnant pools of foul water, where the only non-human creatures are the shining black crows and long grey rats, some of the most beautiful, valuable and useful articles in India are made. From Dharavi come delicate ceramics and pottery, exquisite embroidery and zari work, sophisticated leather goods, high-fashion garments, finely-wrought metalwork, delicate jewellery settings, wood carvings and furniture that would find its way into the richest houses, both in India and abroad…

Dharavi was an arm of the sea, that was filled by waste, largely produced by the people who have come to live there: Scheduled Castes and poor Muslims. It comprises rambling buildings of corrugated metal, 20 metres high in places, used for the treatment of hides and tanning. There are pleasant parts, but rotting garbage is everywhere…"

(Seabrook, 1996, pp. 50, 51-52)

complementing each other. Apart from these, rural and urban areas have also emerged into two separate cultural, social, political, economic and technological divide.

India, which has a predominance of rural population (approximately 69 per cent of the total population in 2011) and where villages were considered the ideal republics by

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Mahatma Gandhi, most of the rural areas are still poor performing primary activities. Here most of the villages exist as appendix to the core urban centre forming its hinterland.

This may give an impression that urban centres exist as undifferentiated homogeneous entities in opposition to the rural areas. On the contrary, urban centres in India are more differentiated in terms of the socio-economic, politico-cultural and other indicators of development than any other areas. At the top, there are farm houses and high income group localities characterised by well-developed urban infrastructures, like wide roads, streetlights, water and sanitation facilities, lawns, well-developed green belt, parks, playgrounds and provisions for individual security and right to privacy. At the other extreme of it are the *slums*, *jhuggi-jhopari* clusters and colonies of shanty structures. These are inhabited by those people who were forced to migrate from the rural areas to these urban centres in search of livelihood but could not afford proper housing due to high rent and high costs of land. They occupy environmentally incompatible and degraded areas.

Slums are residential areas of the least choice, dilapidated houses, poor hygienic conditions, poor ventilation, lack of basic amenities, like drinking water, light and toilet facilities, etc. Open defecation, unregulated drainage system and overcrowded narrow street patterns are serious health and socio-environmental hazards.

The *Swachh Bharat Mission* (SBM) is part of the urban renewal mission launched by the Government of India to improve the quality of life in urban slums.

Moreover, most of the slum population works in low-paid, high risk-prone, unorganised sectors of the urban economy. Consequently, they are the undernourished, prone to different types of diseases and illness and can not afford to give proper education to their children. The poverty makes them vulnerable to drug abuse, alcoholism, crime, vandalism, escapism, apathy and ultimately social exclusion.

Why are the children of slum-dwellers deprived of school education?

Land Degradation

The pressure on agricultural land increases not only due to the limited availability but also by deterioration of quality of agricultural land. Soil erosion, waterlogging, salinisation and alkalinisation of land lead to land degradation. What happens if land is consistently used without managing its fertility? Land is degraded and productivity declines. Land degradation is generally understood either as a temporary or a permanent decline in productive capacity of the land.

Though all degraded land may not be wasteland, but unchecked process of degradation may lead to the conversion to wasteland.

There are two processes that induce land degradation. These are natural and created by human beings. National Remote Sensing Centre (NRSC) has classified wastelands by using remote sensing techniques and it is possible to categorise these wastelands according to the processes that have created them. There are a few types of wastelands such as gullied /ravinous land, desertic or coastal sands, barren rocky areas, steep sloping land, and glacial areas, which are primarily caused by *natural* agents. There are other types of degraded lands such as waterlogged and marshy areas, land affected by salinity and alkalinity and land with or without scrub, which have largely been caused by *natural as well as human factors*. There are some other types of wastelands such as degraded shifting cultivation area, degraded land under plantation crops, degraded forests, degraded pastures, and mining and industrial wastelands,

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THEME ONE

BRICKS, BEADS AND BONES
THE HARAPPAN CIVILISATION

The Harappan seal (Fig.1.1) is possibly the most distinctive artefact of the Harappan or Indus valley civilisation. Made of a stone called steatite, seals like this one often contain animal motifs and signs from a script that remains undeciphered. Yet we know a great deal about the lives of the people who lived in the region from what they left behind – their houses, pots, ornaments, tools and seals – in other words, from archaeological evidence. Let us see *what* we know about the Harappan civilisation, and *how* we know about it. We will explore *how* archaeological material is interpreted and *how* interpretations sometimes change. Of course, there are some aspects of the civilisation that are as yet unknown and may even remain so.



Fig. 1.1
A Harappan seal

Terminologies, Places and Time

"The Harappan Civilisation" is also known as the Indus Valley Civilisation. The term Harappa is derived from the place where the civilization was identified. The total time span of the civilisation ranges from 6000 BCE to 1300 BCE. The early phase, termed as Early Harappan (6000 BCE-2600 BCE) is a formative phase of the civilisation. The urban phase of the civilisation, termed as Mature Harappan (2600 BCE-1900 BCE) is the most prosperous phase. The decline of the civilisation starts around 1900 BCE. This decadent phase (1900 BCE-1300 BCE) is termed as Late Harappan.

The distinctive Harappan pottery, bricks (baked and unbaked) seals, weights, beads, copper and bronze articles, belonging to the Harappan civilisation have been found in the region, as apart as Afghanistan, Baluchistan, Sind and Punjab provinces of Pakistan and the Indian States of Jammu and Kashmir, Punjab, Haryana, Rajasthan, Uttar Pradesh, Gujarat and Maharashtra (Map 1).



Fig. 1.2
Beads, weights, blades

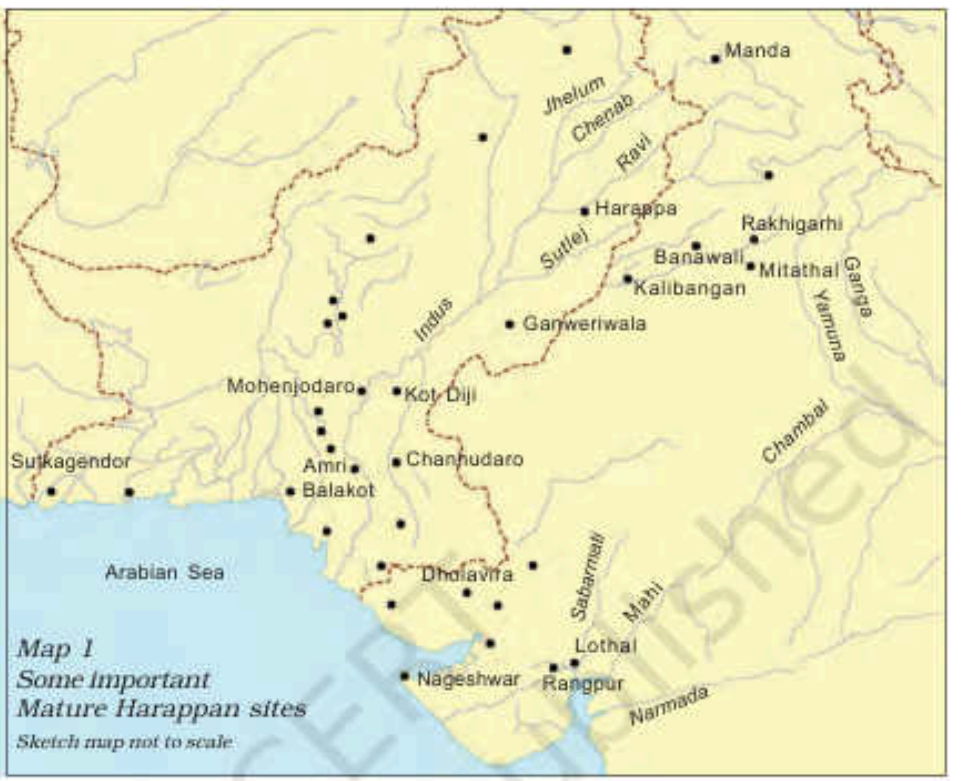
You will find certain abbreviations, related to dates, in this book.

BP stands for Before Present

BCE stands for Before Common Era

CE stands for the Common Era. The present year is 2015 according to this dating system.

c. stands for the Latin word circa and means "approximate."



Map 1
Some important Mature Harappan sites
Sketch map not to scale

Harappan Settlements

So far, more than 2000 Harappan archaeological sites have been discovered in the Indian sub-continent. Most of the sites are found located between The Indus and The Saraswati river basins. Nearly two-thirds of these settlements are in the Saraswati basin, suggesting its immense importance for the Harappan Civilisation. Among these, five major cities have been identified. These are Rakhigarhi, Mohenjodaro, Harappa, Dholavira and Ganweriwala. The remaining sites come under different categories such as Regional Centres, agricultural villages, ports and manufacturing centres.

1. BEGINNINGS

There were several archaeological cultures in the region prior to the Mature Harappan. These cultures were associated with distinctive pottery, evidence of agriculture and pastoralism, and some crafts. Settlements were generally small, and there were virtually no large buildings. The recent archaeological evidences also indicate the emergence of Harappan culture from early farming communities dated around 7000 BCE. The urban phase (Mature Harappan) was a consequent result of the gradual transformation and internal development of the civilisation during the early Harappan phase.

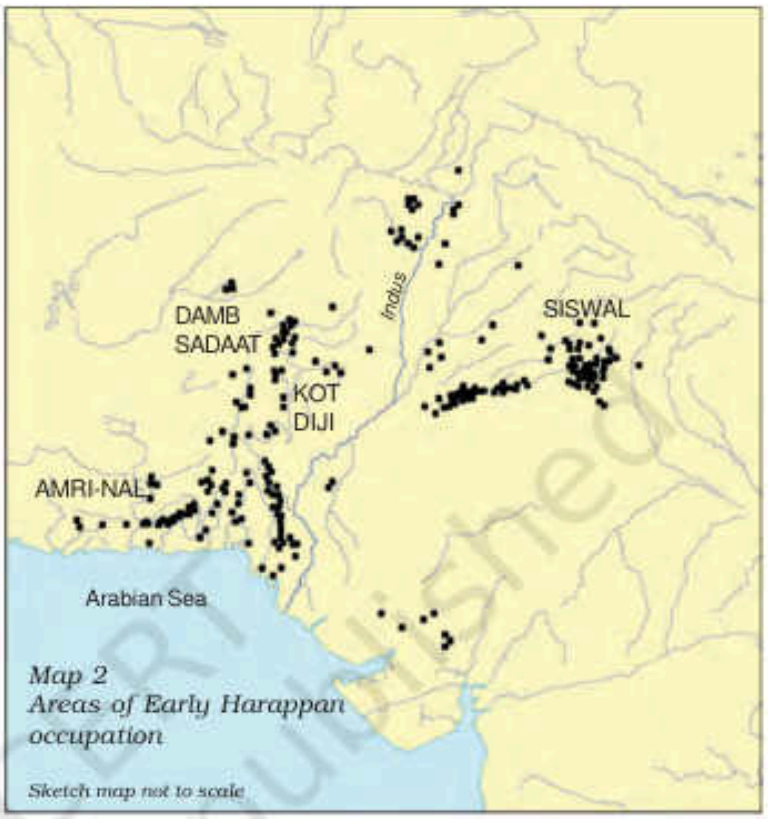
2. SUBSISTENCE STRATEGIES

If you look at Maps 1 and 2 you will notice that the Mature Harappan culture developed in some of the areas occupied by the Early Harappan cultures. These cultures also shared certain common elements including subsistence strategies. The Harappans ate a wide range of plant and animal products, including fish. Archaeologists have been able to reconstruct dietary practices from finds of charred grains and seeds. These are studied by archaeo-botanists, who are specialists in ancient plant remains. Grains

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found at Harappan sites include wheat, barley, lentil, chickpea and sesame. Millets are found from sites in Gujarat. Finds of rice are relatively rare.

Animal bones found at Harappan sites include those of cattle, sheep, goat, buffalo and pig. Studies done by archaeo-zoologists or zoo-archaeologists indicate that these animals were domesticated. Bones of wild species such as boar, deer and *gharial* are also found. We do not know whether the Harappans hunted these animals themselves or obtained meat from other hunting communities. Bones of fish and fowl are also found.



Map 2
Areas of Early Harappan occupation
Sketch map not to scale

2.1 Agricultural technologies

While the prevalence of agriculture is indicated by finds of grain, it is more difficult to reconstruct actual agricultural practices. Were seeds broadcast (scattered) on ploughed lands? Representations on seals and terracotta sculpture indicate that the bull was known, and archaeologists extrapolate from this that oxen were used for ploughing. Moreover, terracotta models of the plough have been found at sites in Cholistan and at Banawali (Haryana). Archaeologists have also found evidence of a ploughed field at Kalibangan (Rajasthan), associated with Early Harappan levels (see p. 20). The field had two sets of furrows at right angles to each other, suggesting that two different crops were grown together.

Fig. 1.3
A terracotta bull



Archaeologists have also tried to identify the tools used for harvesting. Did the Harappans use stone blades set in wooden handles or did they use metal tools?

Most Harappan sites are located in semi-arid lands, where irrigation was probably required for agriculture. Traces of canals have been found at the Harappan site of Shortughai in Afghanistan, but not in Punjab or Sind. It is possible that ancient

➲ Discuss...

Are there any similarities or differences in the distribution of settlements shown on Maps 1 and 2?

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Fig. 1.4
Copper tools

➲ Do you think these tools could have been used for harvesting?

Fig. 1.5
Reservoir at Dholavira
Note the masonry work.



➲ Discuss...

What is the evidence used by archaeologists to reconstruct dietary practices?

canals silted up long ago. It is also likely that water drawn from wells was used for irrigation. Besides, water reservoirs found in Dholavira (Gujarat) may have been used to store water for agriculture.

Source 1

How artefacts are identified

Processing of food required grinding equipment as well as vessels for mixing, blending and cooking. These were made of stone, metal and terracotta. This is an excerpt from one of the earliest reports on excavations at Mohenjodaro, the best-known Harappan site:

> Saddle querns ... are found in considerable numbers ... and they seem to have been the only means in use for grinding cereals. As a rule, they were roughly made of hard, gritty, igneous rock or sandstone and mostly show signs of hard usage. As their bases are usually convex, they must have been set in the earth or in mud to prevent their rocking. Two main types have been found: those on which another smaller stone was pushed or rolled to and fro, and others with which a second stone was used as a pounder, eventually making a large cavity in the nether stone. Querns of the former type were probably used solely for grain; the second type possibly only for pounding herbs and spices for making curries. In fact, stones of this latter type are dubbed "curry stones" by our workmen and our cook asked for the loan of one from the museum for use in the kitchen.

FROM ERNEST MACKAY, *Further Excavations at Mohenjodaro*, 1937.



Fig. 1.6
Saddle quern

➲ Archaeologists use present-day analogies to try and understand what ancient artefacts were used for. Mackay was comparing present-day querns with what he found. Is this a useful strategy?

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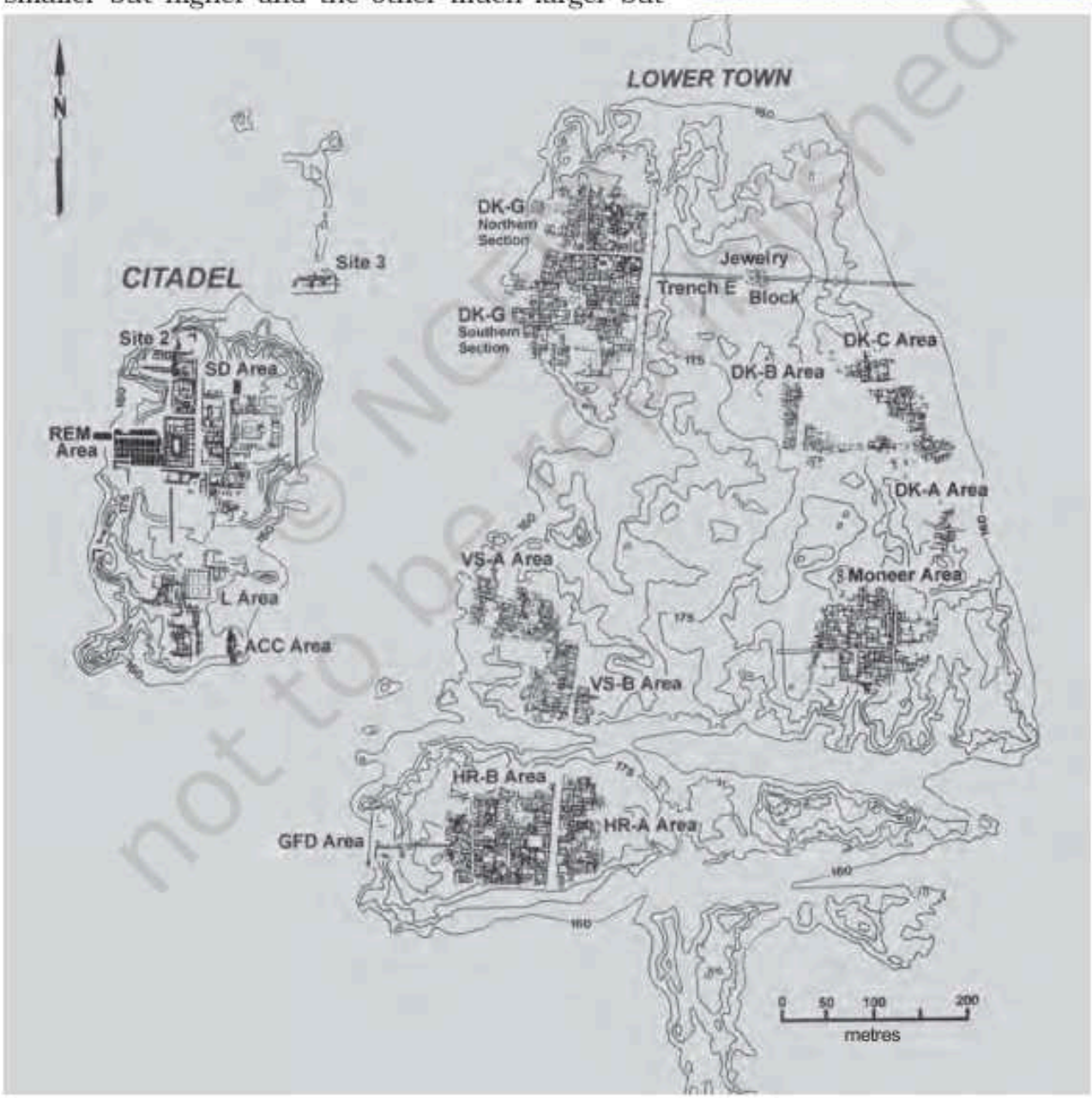
3. MOHANJODARO
A PLANNED URBAN CENTRE

Perhaps the most unique feature of the Harappan civilisation was the development of urban centres. Let us look at one such centre, Mohenjodaro, more closely. Although Mohenjodaro is the most well-known site, the first site to be discovered was Harappa.

The settlement is divided into two sections, one smaller but higher and the other much larger but

Fig. 1.7
Layout of Mohenjodaro

➲ How is the Lower Town different from the Citadel?



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The plight of Harappa

Although Harappa was the first site to be discovered, it was badly destroyed by brick robbers. As early as 1875, Alexander Cunningham, the first Director-General of the Archaeological Survey of India (ASI), often called the father of Indian archaeology, noted that the amount of brick taken from the ancient site was enough to lay bricks for "about 100 miles" of the railway line between Lahore and Multan. Thus, many of the ancient structures at the site were damaged. In contrast, Mohenjodaro was far better preserved.

lower. Archaeologists designate these as the Citadel and the Lower Town respectively. The Citadel owes its height to the fact that buildings were constructed on mud brick platforms. It was walled, which meant that it was physically separated from the Lower Town.

The Lower Town was also walled. Several buildings were built on platforms, which served as foundations. It has been calculated that if one labourer moved roughly a cubic metre of earth daily, just to put the foundations in place it would have required four million person-days, in other words, mobilising labour on a very large scale.

Consider something else. Once the platforms were in place, all building activity within the city was restricted to a fixed area on the platforms. So it seems that the settlement was first planned and then implemented accordingly. Other signs of planning include bricks, which, whether sun-dried or baked, were of a standardised ratio, where the length and breadth were four times and twice the height respectively. Such bricks were used at all Harappan settlements.

3.1 Laying out drains

One of the most distinctive features of Harappan cities was the carefully planned drainage system. If you look at the plan of the Lower Town you will notice that roads and streets were laid out along an approximate "grid" pattern, intersecting at right angles. It seems that streets with drains were laid out first and then houses built along them. If domestic waste water had to flow into the street drains, every house needed to have at least one wall along a street.

Fig. 1.8
A drain in Mohenjodaro
Notice the huge opening of the drain.



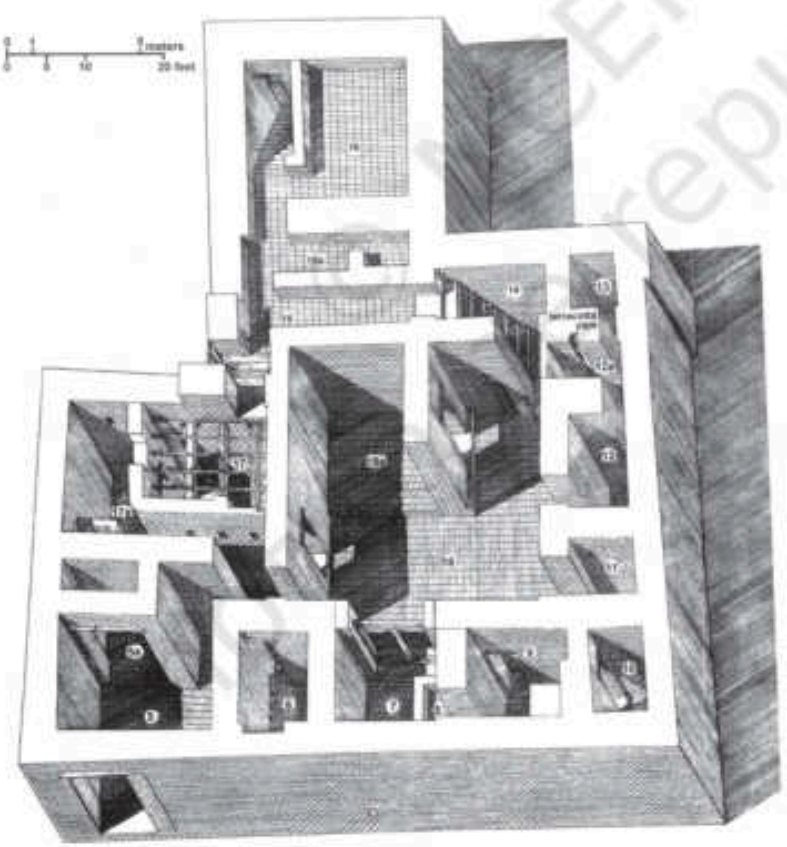
Citadels

While most Harappan settlements have a small high western part and a larger lower eastern section, there are variations. At sites such as Dholavira and Lothal (Gujarat), the entire settlement was fortified, and sections within the town were also separated by walls. The Citadel within Lothal was not walled off, but was built at a height.

3.2 Domestic architecture

The Lower Town at Mohenjodaro provides examples of residential buildings. Many were centred on a courtyard, with rooms on all sides. The courtyard was probably the centre of activities such as cooking and weaving, particularly during hot and dry weather. What is also interesting is an apparent concern for privacy: there are no windows in the walls along the ground level. Besides, the main entrance does not give a direct view of the interior or the courtyard.

Every house had its own bathroom paved with bricks, with drains connected through the wall to the street drains. Some houses have remains of staircases to reach a second storey or the roof. Many houses had wells, often in a room that could be reached from the outside and perhaps used by passers-by. Scholars have estimated that the total number of wells in Mohenjodaro was about 700.



➲ Where is the courtyard? Where are the two staircases? What is the entrance to the house like?

Fig. 1.9
This is an isometric drawing of a large house in Mohenjodaro. There was a well in room no 6.

Source 2

The most ancient system yet discovered

About the drains, Mackay noted: "It is certainly the most complete ancient system as yet discovered." Every house was connected to the street drains. The main channels were made of bricks set in mortar and were covered with loose bricks that could be removed for cleaning. In some cases, limestone was used for the covers. House drains first emptied into a sump or cesspit into which solid matter settled while waste water flowed out into the street drains. Very long drainage channels were provided at intervals with sumps for cleaning. It is a wonder of archaeology that "little heaps of material, mostly sand, have frequently been found lying alongside drainage channels, which shows ... that the debris was not always carted away when the drain was cleared".

FROM ERNEST MACKAY, *Early Indus Civilisation*, 1948.

Drainage systems were not unique to the larger cities, but were found in smaller settlements as well. At Lothal for example, while houses were built of mud bricks, drains were made of burnt bricks.

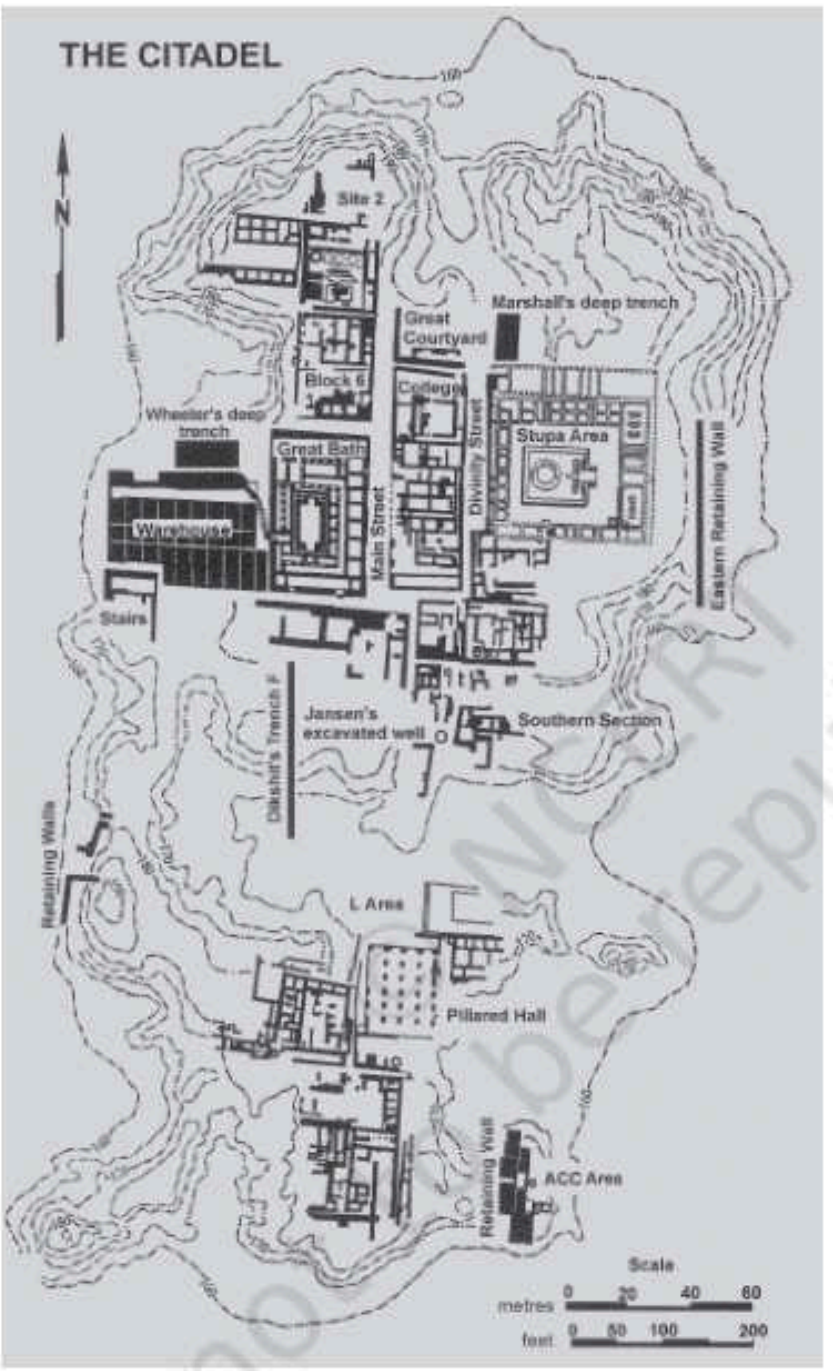


Fig. 1.10
Plan of the Citadel

3.3 The Citadel

It is on the Citadel that we find evidence of structures that were probably used for special public purposes. These include the warehouse – a massive structure of which the lower brick portions remain, while the upper portions, probably of wood, decayed long ago – and the Great Bath.

The Great Bath was a large rectangular tank in a courtyard surrounded by a corridor on all four sides. There were two flights of steps on the north and south leading into the tank, which was made watertight by setting bricks on edge and using a mortar of gypsum. There were rooms on three sides, in one of which was a large well. Water from the tank flowed into a huge drain. Across a lane to the north lay a smaller building with eight bathrooms, four on each side of a corridor, with drains from each bathroom connecting to a drain that ran along the corridor. The uniqueness of the structure, as well as the *context* in which it was found (the Citadel, with several distinctive buildings), has led scholars to suggest that it was meant for some kind of a special ritual bath.

➲ Discuss...
Which of the architectural features of Mohenjodaro indicate planning?

➲ Are there other structures on the Citadel apart from the warehouse and the Great Bath?

4. TRACKING SOCIAL DIFFERENCES

4.1 Burials

Archaeologists generally use certain strategies to find out whether there were social or economic differences amongst people living within a particular culture. These include studying burials. You are probably familiar with the massive pyramids of Egypt, some of which were contemporaneous with the Harappan civilisation. Many of these pyramids were royal burials, where enormous quantities of wealth was buried.

At burials in Harappan sites the dead were generally laid in pits. Sometimes, there were differences in the way the burial pit was made – in some instances, the hollowed-out spaces were lined with bricks. Could these variations be an indication of social differences? We are not sure.

Some graves contain pottery and ornaments, perhaps indicating a belief that these could be used in the afterlife. Jewellery has been found in burials of both men and women. In fact, in the excavations at the cemetery in Harappa in the mid-1980s, an ornament consisting of three shell rings, a jasper (a kind of semi-precious stone) bead and hundreds of micro beads was found near the skull of a male. In some instances the dead were buried with copper mirrors. But on the whole, it appears that the Harappans did not believe in burying precious things with the dead.



Fig. 1.11
A copper mirror

4.2 Looking for "luxuries"

Another strategy to identify social differences is to study artefacts, which archaeologists broadly classify as utilitarian and luxuries. The first category includes objects of daily use made fairly easily out of ordinary materials such as stone or clay. These include querns, pottery, needles, flesh-rubbers (body scrubbers), etc., and are usually found distributed throughout settlements. Archaeologists assume objects were luxuries if they are rare or made from costly, non-local materials or with complicated technologies. Thus, little pots of faience (a material made of ground sand or silica mixed with colour and a gum and then fired) were probably considered precious because they were difficult to make.

The situation becomes more complicated when we find what seem to be articles of daily

Fig. 1.12
A faience pot



Hoards are objects kept carefully by people, often inside containers such as pots. Such hoards can be of jewellery or metal objects saved for reuse by metalworkers. If for some reason the original owners do not retrieve them, they remain where they are left till some archaeologist finds them.

➲ Discuss...

What are the modes of disposal of the dead prevalent at present? To what extent do these represent social differences?

use, such as spindle whorls made of rare materials such as faience. Do we classify these as utilitarian or luxuries?

If we study the distribution of such artefacts, we find that rare objects made of valuable materials are generally concentrated in large settlements like Mohenjodaro and Harappa and are rarely found in the smaller settlements. For example, miniature pots of faience, perhaps used as perfume bottles, are found mostly in Mohenjodaro and Harappa, and there are none from small settlements like Kalibangan. Gold too was rare, and as at present, probably precious – all the gold jewellery found at Harappan sites was recovered from hoards.

5. Finding Out About Craft Production

Locate Chanhudaro on Map 1. This is a tiny settlement (less than 7 hectares) as compared to Mohenjodaro (125 hectares), almost exclusively devoted to craft production, including bead-making, shell-cutting, metal-working, seal-making and weight-making.

The variety of materials used to make beads is remarkable: stones like carnelian (of a beautiful red colour), jasper, crystal, quartz and steatite; metals like copper, bronze and gold; and shell, faience and terracotta or burnt clay. Some beads were made of two or more stones, cemented together, some of stone with gold caps. The shapes were numerous – disc-shaped, cylindrical, spherical, barrel-shaped, segmented. Some were decorated by incising or painting, and some had designs etched onto them.



Fig. 1.13
A tool and beads

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Techniques for making beads differed according to the material. Steatite, a very soft stone, was easily worked. Some beads were moulded out of a paste made with steatite powder. This permitted making a variety of shapes, unlike the geometrical forms made out of harder stones. How the steatite micro bead was made remains a puzzle for archaeologists studying ancient technology.

Archaeologists' experiments have revealed that the red colour of carnelian was obtained by firing the yellowish raw material and beads at various stages of production. Nodules were chipped into rough shapes, and then finely flaked into the final form. Grinding, polishing and drilling completed the process. Specialised drills have been found at Chanhudaro, Lothal and more recently at Dholavira.

If you locate Nageshwar and Balakot on Map 1, you will notice that both settlements are near the coast. These were specialised centres for making shell objects – including bangles, ladles and inlay – which were taken to other settlements. Similarly, it is likely that finished products (such as beads) from Chanhudaro and Lothal were taken to the large urban centres such as Mohenjodaro and Harappa.



Fig. 1.15
A terracotta figurine

5.1 Identifying centres of production

In order to identify centres of craft production, archaeologists usually look for the following: raw material such as stone nodules, whole shells, copper ore; tools; unfinished objects; rejects and waste material. In fact, waste is one of the best indicators of craft work. For instance, if shell or stone is cut to make objects, then pieces of these materials will be discarded as waste at the place of production.



Fig. 1.14
Pottery
Some of these can be seen in the National Museum, Delhi or in the site museum at Lothal.

➲ **Discuss...**

Should the stone artefacts illustrated in the chapter be considered as utilitarian objects or as luxuries? Are there any that may fall into both categories?



Fig.1.16
Copper and bronze vessels

Sometimes, larger waste pieces were used up to make smaller objects, but minuscule bits were usually left in the work area. These traces suggest that apart from small, specialised centres, craft production was also undertaken in large cities such as Mohenjodaro and Harappa.

6. STRATEGIES FOR PROCURING MATERIALS

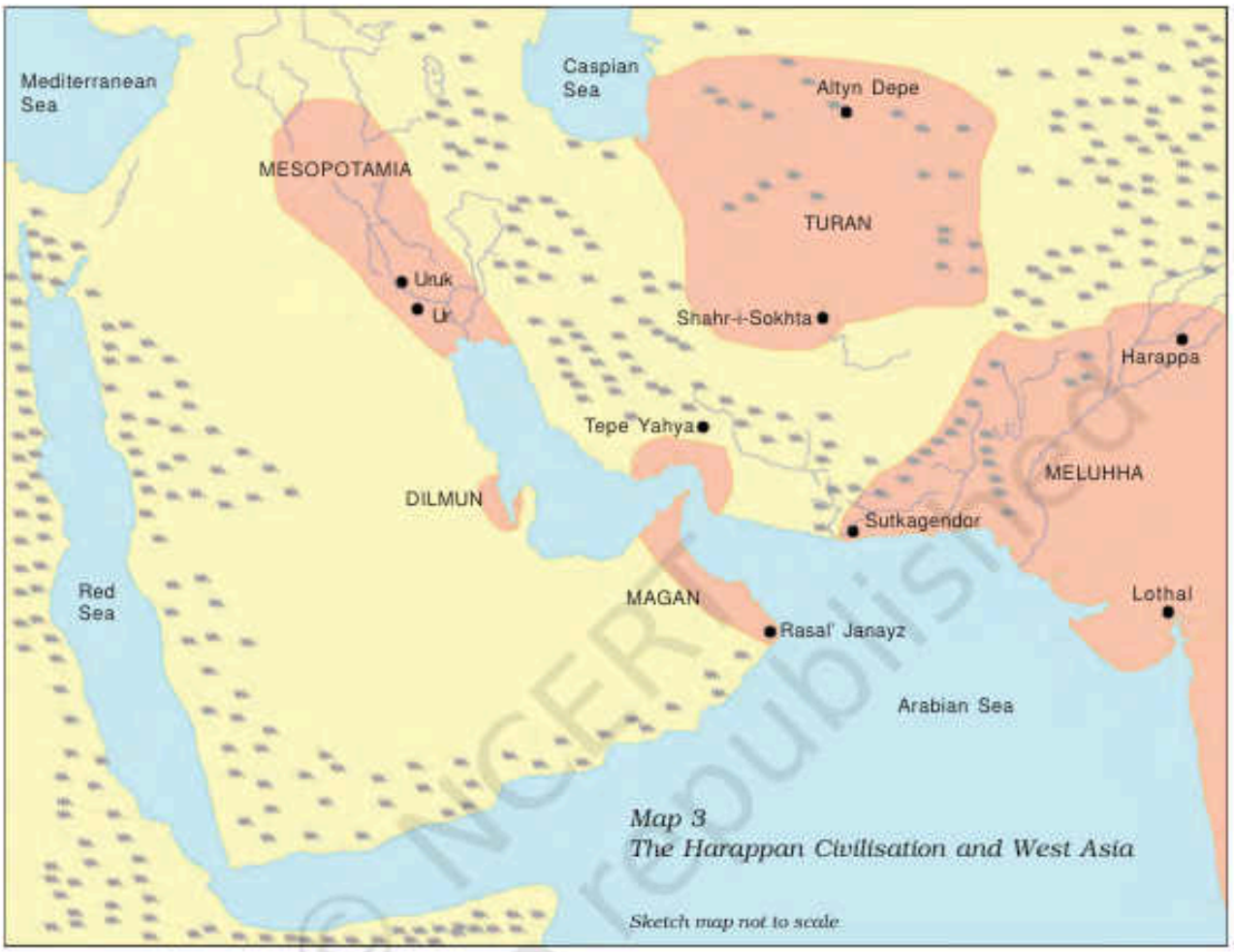
As is obvious, a variety of materials was used for craft production. While some such as clay were locally available, many such as stone, timber and metal had to be procured from outside the alluvial plain. Terracotta toy models of bullock carts suggest that this was one important means of transporting goods and people across land routes. Riverine routes along the Indus and its tributaries, as well as coastal routes were also probably used.

6.1 Materials from the subcontinent and beyond

The Harappans procured materials for craft production in various ways. For instance, they established settlements such as Nageshwar and Balakot in areas where shell was available. Other such sites were Shortughai, in far-off Afghanistan, near the best source of lapis lazuli, a blue stone that was apparently very highly valued, and Lothal which was near sources of carnelian (from Bharuch in Gujarat), steatite (from south Rajasthan and north Gujarat) and metal (from Rajasthan).

Another strategy for procuring raw materials may have been to send expeditions to areas such as the Khetri region of Rajasthan (for copper) and south India (for gold). These expeditions established communication with local communities. Occasional finds of Harappan artefacts such as steatite micro beads in these areas are indications of such contact. There is evidence in the Khetri area for what archaeologists call the Ganeshwar-Jodhpura culture, with its distinctive non-Harappan pottery and an unusual wealth of copper objects. It is possible that the inhabitants of this region supplied copper to the Harappans.

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6.2 Contact with distant lands

Recent archaeological finds suggest that copper was also probably brought from Oman, on the south-eastern tip of the Arabian peninsula. Chemical analyses have shown that both the Omani copper and Harappan artefacts have traces of nickel, suggesting a common origin. There are other traces of contact as well. A distinctive type of vessel, a large Harappan jar coated with a thick layer of black clay has been found at Omani sites. Such thick coatings prevent the percolation of liquids. We do not know what was carried in these vessels, but it is possible that the Harappans exchanged the contents of these vessels for Omani copper.

Mesopotamian texts datable to the third millennium BCE refer to copper coming from a region called Magan, perhaps a name for Oman, and interestingly enough copper found at

Fig. 1.17
A Harappan jar found in Oman



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Fig. 1.18
This is a cylinder seal, typical of Mesopotamia, but the humped bull motif on it appears to be derived from the Indus region.

Mesopotamian sites also contains traces of nickel. Other archaeological finds suggestive of long-distance contacts include Harappan seals, weights, dice and beads. In this context, it is worth noting that Mesopotamian texts mention contact with regions named Dilmun (probably the island of Bahrain), Magan and Meluhha, possibly the Harappan region. They mention the products from Meluhha: carnelian, lapis lazuli, copper, gold, and varieties of wood. A Mesopotamian myth says of Meluhha: "May your bird be the *haja*-bird, may its call be heard in the royal palace." Some archaeologists think the *haja*-bird was the peacock. Did it get this name from its call? It is likely that communication with Oman, Bahrain or Mesopotamia was by sea. Mesopotamian texts refer to Meluhha as a land of seafarers. Besides, we find depictions of ships and boats on seals.



Fig. 1.19
The round "Persian Gulf" seal found in Bahrain sometimes carries Harappan motifs. Interestingly, local "Dilmun" weights followed the Harappan standard.

Fig. 1.20
Seal depicting a boat



➲ Discuss...

What were the possible routes from the Harappan region to Oman, Dilmun and Mesopotamia?

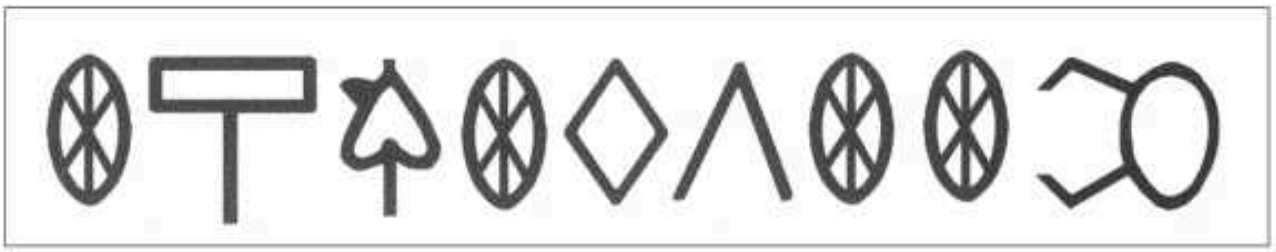


Fig. 1.21
Symbols, Letters on what is considered a signboard found at Dholavira.

7. SEALS, SCRIPT, WEIGHTS

7.1 Seals and sealings

Seals and sealings were used to facilitate long-distance communication. Imagine a bag of goods being sent from one place to another. Its mouth was tied with rope and on the knot was affixed some wet clay on which one or more seals were pressed, leaving an impression. If the bag reached with its sealing intact, it meant that it had not been tampered with. The sealing also conveyed the identity of the sender.

7.2 An enigmatic script

Harappan seals usually have a line of writing, probably containing the name and title of the owner. Scholars have also suggested that the motif (generally an animal) conveyed a meaning to those who could not read.

Most inscriptions are short, the longest containing about 26 signs. Although the script remains undeciphered to date, it was evidently not alphabetical (where each sign stands for a vowel or a consonant) as it has just too many signs – somewhere between 375 and 400. It is apparent that the script was written from right to left as some seals show a wider spacing on the right and cramping on the left, as if the engraver began working from the right and then ran out of space.

Consider the variety of objects on which writing has been found: seals, copper tools, rims of jars, copper and terracotta tablets, jewellery, bone rods, even an ancient signboard! Remember, there may have been writing on perishable materials too. Could this mean that literacy was widespread?

7.3 Weights

Exchanges were regulated by a precise system of weights, usually made of a stone called chert and generally cubical (Fig. 1.2), with no markings. The



Fig. 1.22
A sealing from Ropar

➲ How many seals are impressed on this piece of clay?

➲ Discuss...

What are some of the present-day methods used for long-distance exchange of goods? What are their advantages and problems?

lower denominations of weights were binary (1, 2, 4, 8, 16, 32), while the higher denominations followed the decimal system (160, 200, 320, 640). The smaller weights were probably used for weighing jewellery and beads. Metal scale-pans have also been found.

8. ANCIENT AUTHORITY

There are indications of complex decisions being taken and implemented in Harappan society. Take for instance, the extraordinary uniformity of Harappan artefacts as evident in pottery (Fig. 1.14), seals, weights and bricks. Notably, bricks, though obviously not produced in any single centre, were of a uniform ratio throughout the region, from Jammu to Gujarat. We have also seen that settlements were strategically set up in specific locations for various reasons. Besides, labour was mobilised for making bricks and for the construction of massive walls and platforms.

Who organised these activities?



Fig. 1.23
A "priest-king"

8.1 Palaces and kings

If we look for a centre of power or for depictions of people in power, archaeological records provide no immediate answers. A large building found at Mohenjodaro was labelled as a palace by archaeologists but no spectacular finds were associated with it. A stone statue was labelled and continues to be known as the "priest-king". This is because archaeologists were familiar with Mesopotamian history and its "priest-kings" and have found parallels in the Indus region. But as we will see (p.23), the ritual practices of the Harappan civilisation are not well understood yet nor are there any means of knowing whether those who performed them also held political power.

Some archaeologists are of the opinion that Harappan society had no rulers, and that everybody enjoyed equal status. Others feel there was no single ruler but several, that Mohenjodaro had a separate ruler, Harappa another, and so forth. Yet others argue that there was a single state, given the similarity in artefacts, the evidence for planned settlements, the standardised ratio of brick size, and the establishment of settlements near sources of raw

➲ Discuss...
Could everybody in Harappan society have been equal?

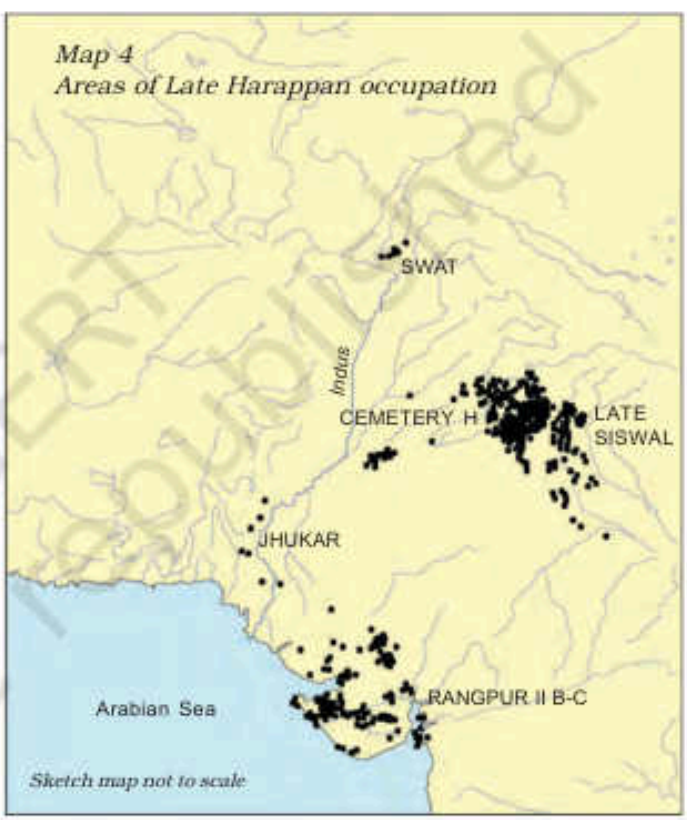
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material. It is likely that the Harappans practiced some sort of democratic system. Some of the structures with their given features were created apparently for utilitarian purposes.

9. THE END OF THE CIVILISATION

There is evidence that by *c.* 1800 BCE most of the Mature Harappan sites in regions such as Cholistan had been abandoned. Simultaneously, there was an expansion of population into new settlements in Gujarat, Haryana and western Uttar Pradesh.

In the few Harappan sites that continued to be occupied after 1900 BCE there appears to have been a transformation of material culture, marked by the disappearance of the distinctive artefacts of the civilisation – weights, seals, special beads. Writing, long-distance trade, and craft specialisation also disappeared. In general, far fewer materials were used to make far fewer things. House construction techniques deteriorated and large public structures were no longer produced. Overall, artefacts and settlements indicate a rural way of life in what are called "Late Harappan" or "successor cultures".



What brought about these changes? Several explanations have been put forward. These range from climatic change, deforestation, excessive floods, the shifting and/or drying up of rivers, to overuse of the landscape. Some of these "causes" may hold for certain settlements, but they do not explain the collapse of the entire civilisation.

It appears that a strong unifying element, perhaps the Harappan state, came to an end. This is evidenced by the disappearance of seals, the script, distinctive beads and pottery, the shift from a standardised weight system to the use of local weights; and the decline and abandonment of cities. The subcontinent would have to wait for over a millennium for new cities to develop in a completely different region.

➲ Discuss...

What are the similarities and differences between Maps 1, 2 and 4?

Source 3

Evidence of an "invasion"

Deadman Lane is a narrow alley, varying from 3 to 6 feet in width ... At the point where the lane turns westward, part of a skull and the bones of the thorax and upper arm of an adult were discovered, all in very friable condition, at a depth of 4 ft 2 in. The body lay on its back diagonally across the lane. Fifteen inches to the west were a few fragments of a tiny skull. It is to these remains that the lane owes its name.

From John Marshall, Mohenjodaro and the Indus Civilisation, 1931.

Sixteen skeletons of people with the ornaments that they were wearing when they died were found from the same part of Mohenjodaro in 1925.

Much later, in 1947, R.E.M. Wheeler, then Director-General of the ASI, tried to correlate this archaeological evidence with that of the *Rigveda*, the earliest known text in the subcontinent. He wrote:

The *Rigveda* mentions *pur*, meaning rampart, fort or stronghold. Indra, the Aryan war-god is called *puramdara*, the fort-destroyer.

Where are – or were – these citadels? It has in the past been supposed that they were mythical ... The recent excavation of Harappa may be thought to have changed the picture. Here we have a highly evolved civilisation of essentially non-Aryan type, now known to have employed massive fortifications ... What destroyed this firmly settled civilisation? Climatic, economic or political deterioration may have weakened it, but its ultimate extinction is more likely to have been completed by deliberate and large-scale destruction. It may be no mere chance that at a late period of Mohenjodaro men, women, and children, appear to have been massacred there. On circumstantial evidence, Indra stands accused.

From R.E.M. Wheeler, "Harappa 1946", Ancient India, 1947.

In the 1960s, the evidence of a massacre in Mohenjodaro was questioned by an archaeologist named George Dales. He demonstrated that the skeletons found at the site did not belong to the same period:

Whereas a couple of them definitely seem to indicate a slaughter, ... the bulk of the bones were found in contexts suggesting burials of the sloppiest and most irreverent nature. There is no destruction level covering the latest period of the city, no sign of extensive burning, no bodies of warriors clad in armour and surrounded by the weapons of war. The citadel, the only fortified part of the city, yielded no evidence of a final defence.

From G.F. Dales, "The Mythical Massacre at Mohenjodaro", Expediton, 1964.

As you can see, a careful re-examination of the data can sometimes lead to a reversal of earlier interpretations. In this respect, it is necessary to mention the recent **archaeogenetic** research carried out at the site of Rakhigarhi. The site, located in Hisar district of Haryana, is the biggest Harappan city, spread over an area of 550 Hectare. In order to study the genetic history of the Harappans, DNA was extracted from the skeletal remains excavated at Rakhigarhi. This research was carried out by The Deccan College Deemed University, Pune in collaboration with Centre for Cellular and Molecular Biology, Hyderabad and The Harvard Medical College. The analysis of the data indicates that the Harappans are the indigenous people of this region.



Harappan skeletal remains from Rakhigarhi - DNA was extracted from this skeleton.

The genetic roots of the Harappans go back to 10,000 BCE. The DNA of the Harappans has continued till today and a majority of the South Asian population appears to be their descendents. Due to trade and cultural contacts of the Harappans with distant regions there is a mixture of genes in small quantity. The continuity without any break in genetic history as well as cultural history rules out large scale immigration of the so-called Aryans. This research also indicates that the people coming from bordering areas and distant regions were absorbed in the Indian society. At no stage, the genetic history of the Indians was either discontinued or broken. As the Harappans started moving towards Iran and Central Asia, their genes also gradually spread in these regions.

The same team of researchers reconstructed 3D features of the Harappan people. The reconstructed facial features of the Harappans, male and female, show remarkable similarity with the modern population of Haryana. Thus, this study has indicated unbroken continuity for 5000 years in this region.

From Vasant Shinde, et.al., An ancient Harrapan genome lacks ancestry from steppe pastoralists or Iranian farmers. Cell, 179 (3), 2019.



Reconstructed 3D facial features of Harappan male and female respectively.

10. DISCOVERING THE HARAPPAN CIVILISATION

So far, we have examined facets of the Harappan civilisation in the context of how archaeologists have used evidence from material remains to piece together parts of a fascinating history. However, there is another story as well – about how archaeologists "discovered" the civilisation.

When Harappan cities fell into ruin, people gradually forgot all about them. When men and women began living in the area millennia later, they did not know what to make of the strange artefacts that occasionally surfaced, washed by floods or exposed by soil erosion, or turned up while ploughing a field, or digging for treasure.

Archaeogenetics is the study of DNA of ancient population. It involves various methods of molecular genetics to understand population history.

10.1 Cunningham's confusion

When Cunningham, the first Director-General of the ASI, began archaeological excavations in the mid-nineteenth century, archaeologists preferred to use the written word (texts and inscriptions) as a guide to investigations. In fact, Cunningham's main interest was in the archaeology of the Early Historic (*c.* sixth century BCE-fourth century CE) and later periods. He used the accounts left by Chinese Buddhist pilgrims who had visited the subcontinent between the fourth and seventh centuries CE to locate early settlements. Cunningham also collected, documented and translated inscriptions found during his surveys. When he excavated sites he tended to recover artefacts that he thought had cultural value.

A site like Harappa, which was not part of the itinerary of the Chinese pilgrims and was not known as an Early Historic city, did not fit very neatly within his framework of investigation. So, although Harappan artefacts were found fairly often during the nineteenth century and some of these reached Cunningham, he did not realise how old these were.

A Harappan seal was given to Cunningham by an Englishman. He noted the object, but unsuccessfully tried to place it within the time-frame with which he was familiar. This was because he, like many others, thought that Indian history began with the first cities in the Ganga valley (see Chapter 2). Given his specific focus, it is not surprising that he missed the significance of Harappa.



Fig. 1.24
Cunningham's sketch of the first-known seal from Harappa

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Sites, mounds, layers

Archaeological *sites* are formed through the production, use and discarding of materials and structures. When people continue to live in the same place, their constant use and reuse of the landscape results in the build up of occupational debris, called a *mound*. Brief or permanent abandonment results in alteration of the landscape by wind or water activity and erosion. Occupations are detected by traces of ancient materials found in *layers*, which differ from one another in colour, texture and the artefacts that are found in them. Abandonment or desertions, what are called "sterile layers", can be identified by the absence of such traces.

Generally, the lowest layers are the oldest and the highest are the most recent. The study of these layers is called stratigraphy. Artefacts found in layers can be assigned to specific *cultural periods* and can thus provide the *cultural sequence* for a site.

10.2 A new old civilisation

Subsequently, seals were discovered at Harappa by archaeologists such as Daya Ram Sahni in the early decades of the twentieth century, in layers that were definitely much older than Early Historic levels. It was then that their significance began to be realised. Another archaeologist, Rakhal Das Banerji found similar seals at Mohenjodaro, leading to the conjecture that these sites were part of a single archaeological culture. Based on these finds, in 1924, John Marshall, Director-General of the ASI, announced the discovery of a new civilisation in the Indus valley to the world. As S.N. Roy noted in *The Story of Indian Archaeology*, "Marshall left India three thousand years older than he had found her." This was because similar, till-then-unidentified seals were found at excavations at Mesopotamian sites. It was then that the world knew not only of a new civilisation, but also of one contemporaneous with Mesopotamia.

In fact, John Marshall's stint as Director-General of the ASI marked a major change in Indian archaeology. He was the first professional archaeologist to work in India, and brought his experience of working in Greece and Crete to the field. More importantly, though like Cunningham he too was interested in spectacular finds, he was equally keen to look for patterns of everyday life.

Marshall tended to excavate along regular horizontal units, measured uniformly throughout the mound, ignoring the stratigraphy of the site. This meant that all the artefacts recovered from the same unit were grouped together, even if they were found at different stratigraphic layers. As a result, valuable

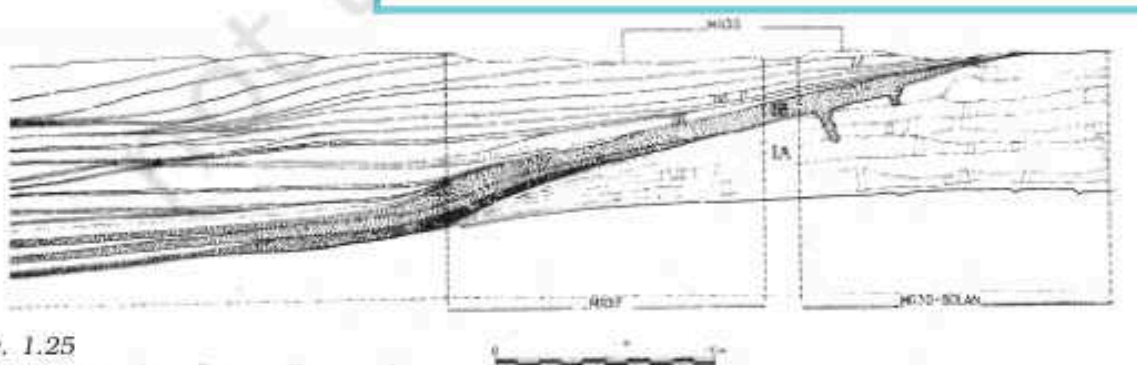


Fig. 1.25
The stratigraphy of a small mound
Notice that the layers are not exactly horizontal.

information about the context of these finds was irretrievably lost.

10.3 New techniques and questions

It was R.E.M. Wheeler, after he took over as Director-General of the ASI in 1944, who rectified this problem. Wheeler recognised that it was necessary to follow the stratigraphy of the mound rather than dig mechanically along uniform horizontal lines. Moreover, as an ex-army brigadier, he brought with him a military precision to the practice of archaeology.

The frontiers of the Harappan civilisation have little or no connection with present-day national boundaries. However, with the partition of the subcontinent and the creation of Pakistan, the major known sites became part of Pakistan. This has spurred Indian archaeologists to try and locate sites in India. An extensive survey in Kutch has revealed a number of Harappan settlements and explorations in Punjab and Haryana have added to the list of Harappan sites. While Kalibangan, Lothal, Rakhi Garhi and most recently Dholavira have been discovered, explored and excavated as part of these efforts, fresh explorations continue.

Over the decades, new issues have assumed importance. Where some archaeologists are often keen to obtain a cultural sequence, others try to understand the logic underlying the location of specific sites. They also grapple with the wealth of artefacts, trying to figure out the functions these may have served.

Since the 1980s, there has also been growing international interest in Harappan archaeology. Specialists from the subcontinent and abroad have been jointly working at both Harappa and Mohenjodaro. They are using modern scientific techniques including surface exploration to recover traces of clay, stone, metal and plant and animal remains as well as to minutely analyse every scrap of available evidence. These explorations promise to yield interesting results in the future.

➲ **Discuss...**

Which of the themes in this chapter would have interested Cunningham? Which are the issues that have been of interest since 1947?

Wheeler at Harappa

Early archaeologists were often driven by a sense of adventure. This is what Wheeler wrote about his experience at Harappa:

> It was, I recall, on a warm May night in 1944 that a four miles' *tonga*-ride brought me as the newly appointed Director General of the Archaeological Survey with my local Muslim officer from a little railway-station labelled "Harappa" along a deep sand track to a small rest-house beside the moonlit mounds of the ancient site. Warned by my anxious colleague that we must start our inspection at 5.30 next morning and finish by 7.30 "after which it would be too hot", we turned in with the dark figure of the *punka-walla* crouched patiently in the entrance and the night air rent by innumerable jackals in the neighbouring wilderness.
>
> Next morning, punctually at 5.30, our little procession started out towards the sandy heaps. Within ten minutes I stopped and rubbed my eyes as I gazed upon the tallest mound, scarcely trusting my vision. Six hours later my embarrassed staff and I were still toiling with picks and knives under the blazing sun, the mad sahib (I am afraid) setting a relentless pace.

FROM R.E.M. WHEELER, *My Archaeological Mission to India and Pakistan*, 1976.

11. PROBLEMS OF PIECING TOGETHER THE PAST

As we have seen, it is not the Harappan script that helps in understanding the ancient civilisation. Rather, it is material evidence that allows archaeologists to better reconstruct Harappan life. This material could be pottery, tools, ornaments, household objects, etc. Organic materials such as cloth, leather, wood and reeds generally decompose, especially in tropical regions. What survive are stone, burnt clay (or terracotta), metal, etc.

It is also important to remember that only broken or useless objects would have been thrown away. Other things would probably have been recycled. Consequently, valuable artefacts that are found intact were either lost in the past or hoarded and never retrieved. In other words, such finds are accidental rather than typical.

11.1 Classifying finds

Recovering artefacts is just the beginning of the archaeological enterprise. Archaeologists then classify their finds. One simple principle of classification is in terms of material, such as stone, clay, metal, bone, ivory, etc. The second, and more complicated, is in terms of function: archaeologists have to decide whether, for instance, an artefact is a tool or an ornament, or both, or something meant for ritual use.

An understanding of the function of an artefact is often shaped by its resemblance with present-day things – beads, querns, stone blades and pots are obvious examples. Archaeologists also try to identify the function of an artefact by investigating the context in which it was found: was it found in a house, in a drain, in a grave, in a kiln?

Sometimes, archaeologists have to take recourse to indirect evidence. For instance, though there are traces of cotton at some Harappan sites, to find out about clothing we have to depend on indirect evidence including depictions in sculpture.

Archaeologists have to develop frames of reference. We have seen that the first Harappan seal that was found could not be understood till archaeologists had a context in which to place it – both in terms of the cultural sequence in which it was found, and in terms of a comparison with finds in Mesopotamia.

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11.2 Problems of interpretation

The problems of archaeological interpretation are perhaps most evident in attempts to reconstruct religious practices. Early archaeologists thought that certain objects which seemed unusual or unfamiliar may have had a religious significance. These included terracotta figurines of women, heavily jewelled, some with elaborate head-dresses. These were regarded as mother goddesses. Rare stone statuary of men in an almost standardised posture, seated with one hand on the knee – such as the "priest-king" – was also similarly classified. In other instances, structures have been assigned ritual significance. These include the Great Bath and fire altars found at Kalibangan and Lothal.

Attempts have also been made to reconstruct religious beliefs and practices by examining seals, some of which seem to depict ritual scenes. Others, with plant motifs, are thought to indicate nature worship. Some animals – such as the one-horned animal, often called the "unicorn" – depicted on seals seem to be mythical, composite creatures. In some seals, a figure shown seated cross-legged in a "yogic" posture, sometimes surrounded by animals, has been regarded as a depiction of "proto-Shiva", that is, an early form of one of the major deities of Hinduism. Besides, conical stone objects have been classified as *lingas*.

Many reconstructions of Harappan religion are made on the assumption that later traditions provide parallels with earlier ones. This is because archaeologists often move from the known to the unknown, that is, from the present to the past. While this is plausible in the case of stone querns and pots, it becomes more speculative when we extend it to "religious" symbols.

Let us look, for instance, at the "proto-Shiva" seals. The earliest religious text, the *Rigveda* (compiled *c.* 1500-1000 BCE) mentions a god named Rudra, which is a name used for Shiva in later Puranic traditions (in the first millennium CE; see also Chapter 4). However, unlike Shiva, Rudra in the *Rigveda* is neither depicted as Pashupati (lord of animals in general and cattle in particular), nor as a yogi. In other words, this depiction does not match the description of Rudra in the *Rigveda*. Is this, then, possibly a shaman as some scholars have suggested?



Fig. 1.26
Was this a mother goddess?



Fig. 1.27
A "proto-Shiva" seal

A *linga* is a polished stone that is worshipped as a symbol of Shiva.

Shamans are men and women who claim magical and healing powers, as well as an ability to communicate with the other world.

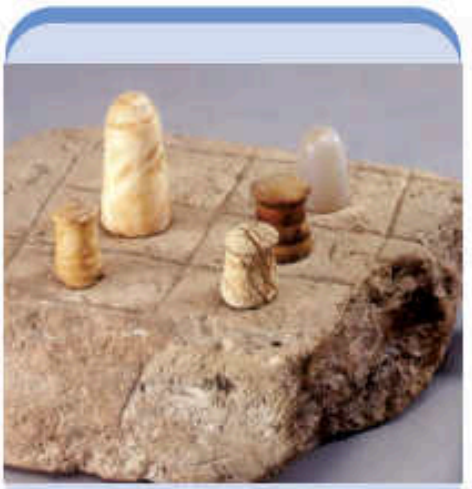


Fig. 1.28
Gamesmen or lingas*?*

This is what Mackay, one of the earliest excavators, had to say about these stones:

> Various small cones made of lapis lazuli, jasper, chalcedony, and other stones, most beautifully cut and finished, and less than two inches in height, are also thought to be *lingas* ... on the other hand, it is just as possible that they were used in the board-games ...
>
> FROM ERNEST MACKAY, *Early Indus Civilisation*, 1948.

➲ Discuss...

What are the aspects of Harappan economy that have been reconstructed from archaeological evidence?

What has been achieved after so many decades of archaeological work? We have a fairly good idea of the Harappan economy. We have been able to tease out social differences and we have some idea of how the civilisation functioned. It is really not clear how much more we would know if the script were to be deciphered. If a bilingual inscription is found, questions about the languages spoken by the Harappans could perhaps be put to rest.

Several reconstructions remain speculative at present. Was the Great Bath a ritual structure? How widespread was literacy? Why do Harappan cemeteries show little social differentiation? Also unanswered are questions on gender – did women make pottery or did they only paint pots (as at present)? What about other craftspersons? What were the terracotta female figurines used for? Very few scholars have investigated issues of gender in the context of the Harappan civilisation and this is a whole new area for future work.

More research is also required on the relationship between the Harappans and the Vedic people as some scholars have argued that the authors of the Harappans civilisation and the Vedic people were the same.



Fig. 1.29
A terracotta cart

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TIMELINE 1
MAJOR PERIODS IN EARLY INDIAN ARCHAEOLOGY

2 million BP (BEFORE PRESENT)	Lower Palaeolithic
80,000	Middle Palaeolithic
35,000	Upper Palaeolithic
12,000	Mesolithic
10,000	Neolithic (early agriculturists and pastoralists)
6,000	Chalcolithic (first use of copper)
2600 BCE	Harappan civilisation
1000 BCE	Early iron, megalithic burials
600 BCE-400 CE	Early Historic

(Note: All dates are approximate. Besides, there are wide variations in developments in different parts of the subcontinent. Dates indicated are for the earliest evidence of each phase.)

TIMELINE 2
MAJOR DEVELOPMENTS IN HARAPPAN ARCHAEOLOGY

Nineteenth century	
1875	Report of Alexander Cunningham on Harappan seal
Twentieth century	
1921	Daya Ram Sahni begins excavations at Harappa
1922	Excavations begin at Mohenjodaro
1946	R.E.M. Wheeler excavates at Harappa
1955	S.R. Rao begins excavations at Lothal
1960	B.B. Lal and B.K. Thapar begin excavations at Kalibangan
1974	M.R. Mughal begins explorations in Bahawalpur
1980	A team of German and Italian archaeologists begins surface explorations at Mohenjodaro
1986	American team begins excavations at Harappa
1990	R.S. Bisht begins excavations at Dholavira
1997	Amrendra Nath starts excavations at Rakhigarhi
2013	Vasant Shinde begins archaeogenetic research at Rakhigarhi

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Fig. 1.30
A Harappan burial

ANSWER IN 100-150 WORDS

1. List the items of food available to people in Harappan cities. Identify the groups who would have provided these.
2. How do archaeologists trace socio-economic differences in Harappan society? What are the differences that they notice?
3. Would you agree that the drainage system in Harappan cities indicates town planning? Give reasons for your answer.
4. List the materials used to make beads in the Harappan civilisation. Describe the process by which any one kind of bead was made.
5. Look at Fig. 1.30 and describe what you see. How is the body placed? What are the objects placed near it? Are there any artefacts on the body? Do these indicate the sex of the skeleton?

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WRITE A SHORT ESSAY (ABOUT 500 WORDS) ON THE FOLLOWING:

6. Describe some of the distinctive features of Mohenjodaro.
7. List the raw materials required for craft production in the Harappan civilisation and discuss how these might have been obtained.
8. Discuss how archaeologists reconstruct the past.
9. Discuss the functions that may have been performed by rulers in Harappan society.

MAP WORK

10. On Map 1, use a pencil to circle sites where evidence of agriculture has been recovered. Mark an X against sites where there is evidence of craft production and R against sites where raw materials were found.

PROJECT (ANY ONE)

11. Find out if there are any museums in your town. Visit one of them and write a report on any ten items, describing how old they are, where they were found, and why you think they are on display.
12. Collect illustrations of ten things made of stone, metal and clay produced and used at present. Compare these with the pictures of the Harappan civilisation in this chapter, and discuss the similarities and differences that you find.

If you would like to know more, read:

Raymond and Bridget Allchin. 1997. *Origins of a Civilization.* Viking, New Delhi.

G.L. Possehl. 2003. *The Indus Civilization.* Vistaar, New Delhi.

Shereen Ratnagar. 2001. *Understanding Harappa.* Tulika, New Delhi.

For more information, you could visit:

http://www.harappa.com/har/harres0.html

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